



TM Forum Technical Report

Digital Services Reference Architecture Guide

TR274

Release 17.5.1

May 2018

Latest Update: TM Forum Release 17.5.1	TM Forum Approved
Version 8.0.2	IPR Mode: RAND

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1. Digital Services Reference Architecture Guide

The Digital Services Reference Architecture (DSRA) is the TM Forum's proposal to enable richer and yet open, distributed digital collaboration with the essential tools to support increasingly demanding technical and commercial models.



2. Introduction to DSRA R17.5

- [Defining a Digital Ecosystem R17.5](#)
- [Enabling a Digital Ecosystem R17.5](#)
- [The DSRA Approach R17.5](#)
- [Related TM Forum Initiatives R17.5](#)
- [Who Should Get Involved R17.5](#)

2.1. Defining a Digital Ecosystem R17.5

What is a Digital Ecosystem?

The rise of e-commerce and the advent of the digital economy are profoundly changing how businesses think and operate. In the digital environment, products are increasingly assembled in cross-vertical collaborations, with building-block web services contributed from and exposed by myriad partners and hence are demanding management on an end-to-end basis. These complex products are increasingly critical to organizations and to customers, and require more complex commercial and revenue sharing models among diverse parties. These high value, sometimes technically demanding and cross-industry products have outgrown today's collaboration infrastructure and are placing extreme pressure on traditional operational models. Today's state of the art is either "pair-wise" collaboration, or "Over the top" loose coupling without effective end-to-end management.

In addition, businesses are shifting to platform businesses, whereby the business model is based on a platform. The platform creates value by connecting consumers and producers, and is owned by a curator.

For example, Airbnb applies the platform business model to the accommodation business whereby they don't own any rooms. Instead, the company created and maintains a platform enabling individual partners in their ecosystem to offer rooms directly to customers. In return Airbnb takes a transaction fee for every rental arranged through their platform.



[Platform Revolution](#), book by Geoffrey Parker, Marshall Van Alstyne and Sangeet Paul Choudary



[Airbnb](#)

The TM Forum is defining a [Digital Reference Platform Architecture \(DPRA\)](#), as a business architecture, a set of best practices and technical standards, metrics and open APIs that will be needed for the communications industry to collectively build the platform required to capture the global opportunities.

 [IG1157 DPRA Concepts & Principles](#)

A digital ecosystem is an open socio-technical digital environment with properties of self-organization, aimed at building digital services to create value and revenue streams amongst networking partners. It provides access to resources, and supports collaboration, knowledge sharing and development of open and adaptive technologies and evolutionary business models, in an open environment which can span across different enterprise boundaries and different parties (operators, social networks). It clearly extends the traditional value chain beyond the evolutionary horizontal & vertical value chain paradigms to a full [digital value fabric](#).

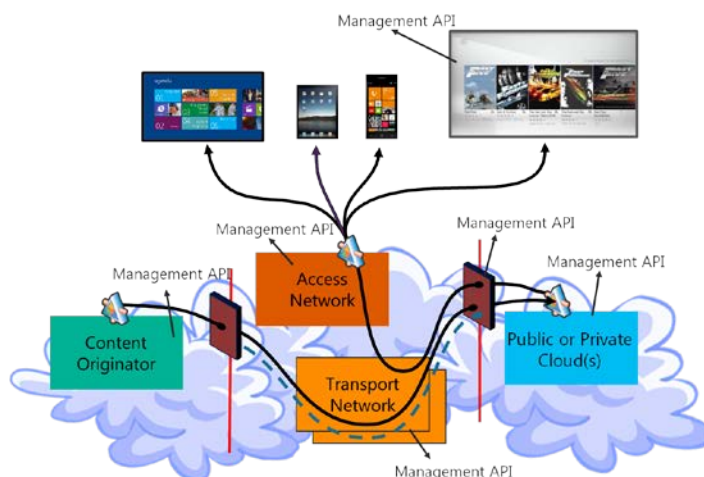
 [Digital Value Fabric](#)

This digital value fabric is also a characteristic of the complex relationships with producers, consumers and the platform, as businesses shift to the platform structure.

A digital ecosystem is an open socio-technical digital environment with properties of self-organization, aimed at building digital services to create value and revenue streams amongst networking partners.

The [Digital Services Reference Architecture \(DSRA\)](#) is the TM Forum's proposal to enable richer and yet open, distributed digital collaboration with the essential tools to support increasingly demanding technical and commercial models.

 [Digital Services Reference Architecture \(DSRA\)](#)



As more new products are assembled out of building blocks that are provided by a range of digital service providers, there is great opportunity, and growing complexity to be managed. The opportunity comes from placing a wide range of capabilities at a product designer's fingertips. The complexity comes from the distributed nature of those very capabilities – for example, my product may rely on the performance of your building block, building blocks from a 3rd enterprise delivered over a set of network capabilities provided by yet another enterprise – a network provider.

This digital economy needs a set of standards and functions that allow product composers to learn about these capabilities, configure and fulfill them, monitor them, understand dependencies, and execute what may be multi-dimensional commercial and revenue sharing models among end users, component service suppliers, advertisers, etc.

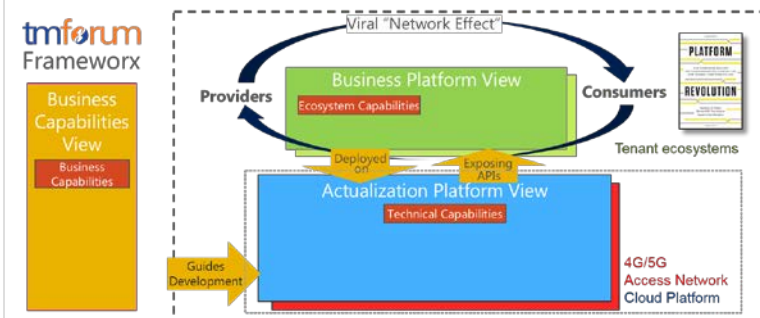
The DSRA aims to make this process simple, capable and standardized to the degree necessary. To this end, our proposed DSRA will enable collaboration and product composition from a broad set of building-block services spanning multiple companies, industries, clouds, networks and devices. And it will allow this composition to be accomplished with (if desired) commercial grade quality, accountability, business models and operational efficiencies. Finally, it will be as simple as possible, maximizing “loose coupling” and minimizing impacts on existing investments. It aims to enable, not replace.



The TM Forum's [Open Digital Program](#) (ODP) assumes that there will be not just one, but many ecosystems within the broader digital economy. These ecosystems must coexist, using some of the same enabling services but solving different problems; for instance, ecosystems focused on digital content and Machine-to-Machine (M2M) / Internet of Things (IoT), each leveraging similar sets of underlying cloud and network resources.

 [Open Digital Program](#)

The Digital Service Reference Architecture maps to the Actualization Platform view described in the [Digital Platform Reference Architecture](#).



 [IG1157 DPRA Concepts & Principles](#)

2.2. Enabling a Digital Ecosystem R17.5

What is the enabler for a Digital Ecosystem?

A Digital Ecosystem is enabled by a federation of platforms that implement the [Digital Service Reference Architecture](#) (DSRA), and delivers three key value propositions:

A Great User Experience – Users can access digital products, assembled from diverse building blocks, in the manner they expect.

Robust Operations – Ecosystem participants, including partners, service providers and enterprises can conduct rich commerce as well as create and deploy those products and manage end-to-end performance.

A Robust Developer Environment – Developers – both providers of component services and end-to-end product developers – have the tools and infrastructure to support both above goals.

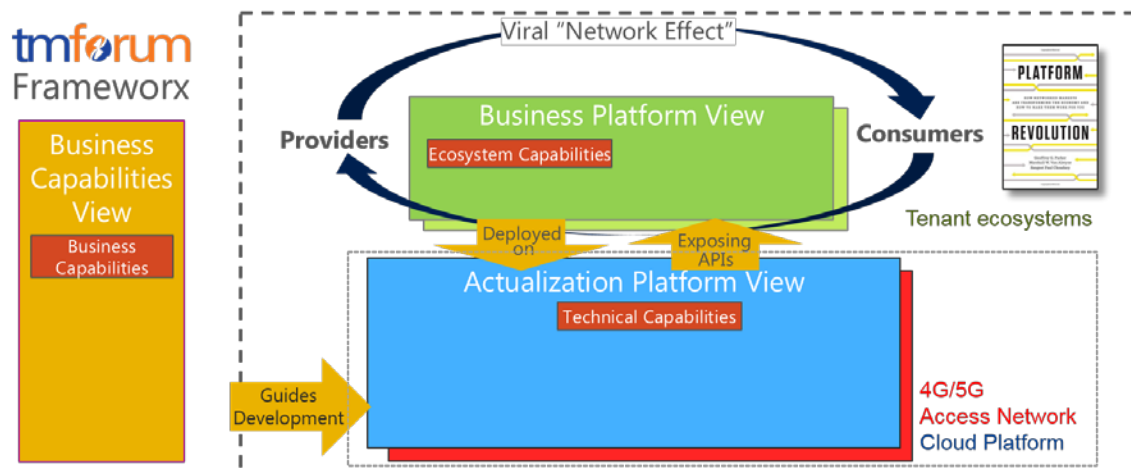
A Digital Ecosystem is enabled by a federation of platforms that implement the Digital Service Reference Architecture

The Business Capabilities required to achieve the above goals are described in the [DPRA Business Capabilities](#) view that maps to the [Business Process Framework eTOM](#).

 [IG1157 DPRA Concepts & Principles](#)

 [Business Process Framework](#)

The implementation of these Business Capabilities, is the DSRA enabling the ecosystem. This is described as the DPRA Actualization Platform View that maps to the Application Framework TAM.	 Application Framework
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2.3. The DSRA Approach R17.5

What is the DSRA approach?

DSRA is the name for the architecture that supports these collaborations, and specifies a set of best practices from leading edge ecosystems today, plus a set of new management capabilities designed to enable the three objectives above. The DSRA must meet two additional objectives, which at first glance may appear to be in conflict:

1. it must be distributed, spanning and encompassing many service providers and thus many platforms, and
2. it must support a logically unified approach so that the three objectives (above) may be met.

This implies:

1. a degree of federation,
2. some common supporting platform infrastructure and
3. a standardized method for abstraction and exposure of management functionality as end-to-end products are assembled from component services.

Note, this “assembly” of component services into products is expected to be a recursive process.

In the spirit of innovation and loose coupling, the DSRA assumes that most component web services are created, abstracted and exposed by various ecosystem partners, building a rich and changing library over time. In addition, the TM Forum believes that there exists a small set of platform capabilities that are necessary to support these ecosystems and should exist in each DSRA platform implementation.

In this document we distinguish between two kinds of platform capabilities:

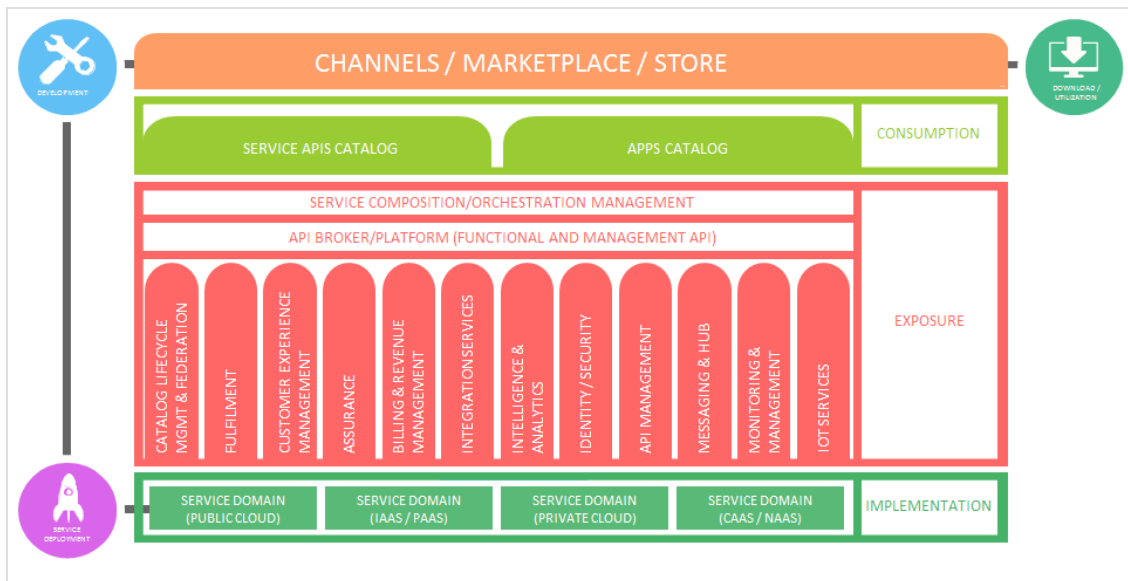
1. “Platform Capabilities” based on “Best Practices” culled from today’s leading-edge cloud ecosystems, that the TM Forum believes are essential to the digital ecosystem and represents the best of current practices. TM Forum is making no effort to standardize such a capability, but rather to point out the importance of the availability of such a capability to the platform. Hence, these are referenced in the DSRA but not specified by the TM Forum.
2. “Platform Capabilities” that are required to meet the objectives set out in this document (e.g. Catalog Management). These represent the main work effort of the DSRA and often apply ongoing work of the TM Forum.

There exists a small set of platform capabilities that are necessary to support these ecosystems and should exist in each DSRA platform implementation.

Any platform that intends to form part of future digital ecosystems should implement both the best practices and support these new “platform capabilities”. The most important are shown in the diagram hereinafter, with more information available in [IG1157 DPRA Concepts & Principles](#) and on the [ODE web site](#).

 [IG1157 DPRA Concepts & Principles](#)

 [Open Digital Ecosystem website](#)



Any platform that intends to form part of future digital ecosystems should implement both the best practices and support these new “platform capabilities”.

The main DSRA work is an effort to describe each of these “platform capabilities”, how they contribute to digital services management, and how to expose them to support distributed digital ecosystems. The following “platform capabilities” should be provided by the platform infrastructure that exposes services into one or more ecosystems. Note, each service provider (SP) *could* have its own DSRA exposure environment, but many smaller players may wish to use exposure environments provided by larger SPs or aggregators.

Each of the following “platform capabilities” is intended to support an environment where N services, potentially from M different service providers, and even resident on K platforms, are composed (“mashed up”) into a single product offering. They address the many questions across multiple parties about how one discovers, assembles, configures, assures, and charges for that composite product offering. This context is essential – it demands that information be exposed into a loosely coupled environment where the ultimate use and even composition environment is often un-known. The management services should not be confused with the functional services they support. The focus of DSRA is not about the functional services themselves but about their management. It is entirely possible that a digital service owner will offer a digital service – at which point it is blank or null, yet the service exists.

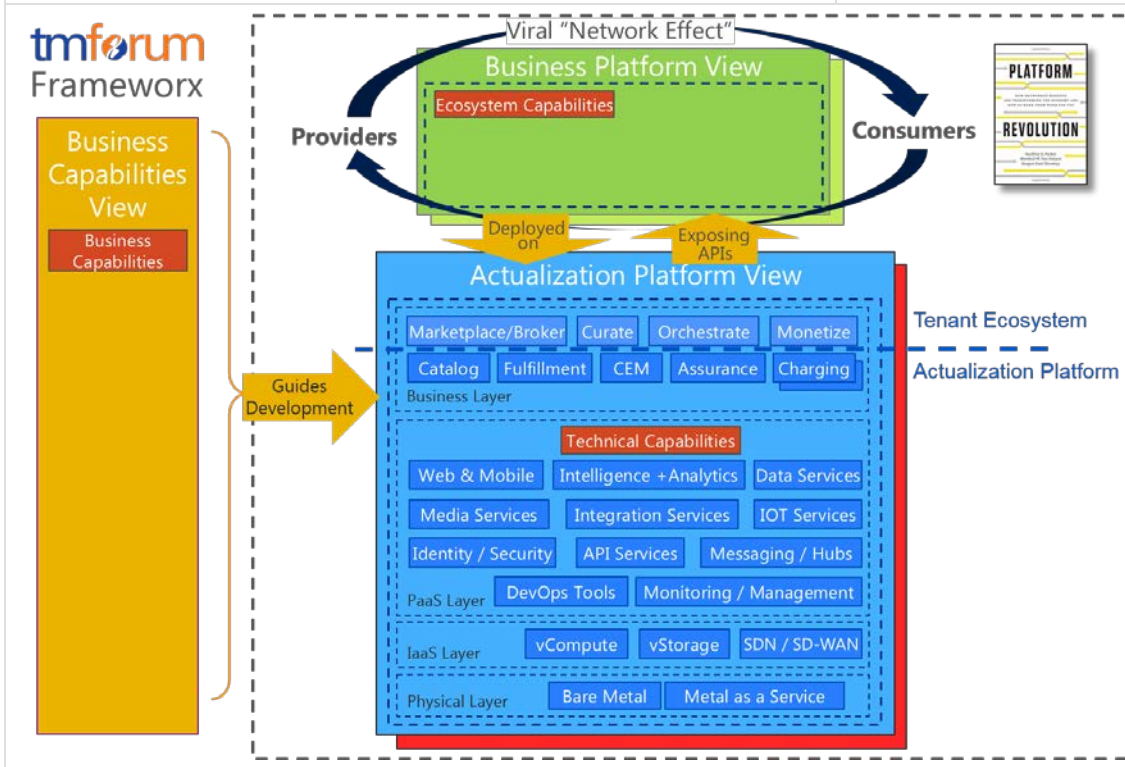
There is an implicit assumption that two kinds of services exist – *functional* services, most of which are real-time, and a set of supporting *management* services, many of which are not real-time, or at least not in the direct flow path of the service, but rather allow it to be discovered, configured, monitored etc. This distinction is of growing importance as the old paradigm of “network hardware” vs “OSS software” becomes increasingly obsolete. Most (but not all) of these platform capabilities, whether real-time or not, fall into the category of “*management*” services.

One of the collaboration models is an API wholesale model, where an API provider with an API platform, uses APIs that are provided by another “backend” API provider owning a platform exposing APIs. In this model, the former API provider, can offer high value-added APIs by mashing up APIs, including those from the “backend” API provider.

The diagram hereinafter lists the platform capabilities that the TM Forum believes are necessary and desirable to support thriving, robust digital ecosystems. In all cases the objective is summarized. For some, if an industry best practice was generally accepted, that best practice may be noted. In other cases, TM Forum either has an interface/service that meets the objective, or plans to specify one. These too will be noted. More information is available in the [DPRA whitepaper](#) and on the [TM Forum’s ODE website](#).

 [IG1157 DPRA Concepts & Principles](#)

 [Open Digital Ecosystem website](#)



2.4. Related TM Forum Initiatives R17.5

To what other TM Forum initiatives does the DSRA relate?

The TM Forum Digital Platform Reference Architecture (DPRA) describes the business architecture, a set of best practices and technical standards, metrics and open APIs that will be needed for the communications industry to collectively enable platform business, with reference to TM Forum assets that are useful in this context. The DSRA maps to the Actualization Platform view described in the DPRA.	 IG1157 DPRA Concepts & Principles
The TM Forum Internet of Everything Program's B2B2X Partnering initiative provides tools for forming and maintaining partnerships within a digital ecosystem.	 B2B2X
The TM Forum ZOOM project and the DSRA are related in that they represent specific uses of Frameworkx layering. DSRA is a more general use case applicable to a broad array of digital services. ZOOM (Zero-Touch Orchestration, Operations and Management) is focused on a specific set of digital services including the management aspects of virtualized network services (NFV) such as SDN Controllers or Evolved Packet Core (EPC) related services. The ZOOM project also defines the Hybrid Infrastructure Platform (HIP) in TR262 Hybrid Network Management Platform Blueprint. HIP is a specific implementation of the DPRA and DSRA.	 ZOOM  TR262 Hybrid Network Management Platform Blueprint
There is an ongoing effort within TM Forum to define Management APIs. For more information visit: Open Digital Ecosystem.	 APIs  Open Digital Ecosystem

2.5. Who Should Get Involved R17.5

Who should get involved?

The TM Forum is harnessing the best practices of its member companies and is working with other standards development organizations to deliver a reference blueprint for digital services, including APIs, information models, business processes, and extensive best practices for partnership and on-boarding. Active support and contributions by member company decision makers are critical.	
The specification of this ecosystem marks the first comprehensive effort to understand and meet the needs of all digital services stakeholders: consumers, developers, providers, operators, and regulators. The beneficiaries extend beyond the communications industry to all stakeholders in the digital economy including Automotive, Media, Healthcare, Government and the Internet of Things (IoT).	 IoT
The TM Forum welcomes representatives from all different industries to take part, actively contribute to, and benefit from this initiative.	
For more details on the information described herein or to become more actively involved in the initiative, visit us at the Open Digital Program website.	 Open Digital Program

3. DSRA Business Alignment Architecture R17.5

What are the characteristics of the business environment?

This section of DSRA Guide document is intended for any company or person engaged in the implementation of Digital Services. The primary target audience are Business and Implementation Architects who establish the fundamental approaches and infrastructure their companies will use to participate in the Digital Service Ecosystem. In this section, “Digital Service” is an umbrella term for any kind of virtualized service such as cloud computing, storage, applications, content or network. The key objectives of the DSRA Guide are to describe the: • Properties that the Digital Service components must have, as well as the business benefits that those properties will deliver. • Principles and rules for making Digital Services.	• Business drivers, requirements and models R17.5 • Interaction patterns and orchestration R17.5 • Main IT requirements and principles R17.5 • API identification and design R17.5
In this context, the Business Architecture: • Helps to ensure the implementation is purely driven by business requirements derived from the relevant business models (business aligned implementation) • Is demonstrating the business value and requirements of subsequent Technology Architecture work to key stakeholders, • Defines the business drivers and scope, reflects the business models with related collaborations and partnering requirements. It also shapes the relevant process decomposition • Presents the resulting implementation requirements (i.e. organization, processes and IT requirements) and principles that will be detailed in next sections	
So, it will describe from a business point of view the: • Identified business drivers • Business requirements to be considered when defining Digital Service Reference Architecture	

- Major business models to be supported, their common characteristics and resulting business interaction between involved parties
 - o Digital Business Models assume a high degree of operations automation and systems support for human activities. The duration to deliver value once contracted to do so by a consumer is assumed to be supported by not only a technically enabled process, but is also assumed to be instrumented in detail to enable the management of the fulfillment of the order, the billing and payment for the value, and the delivery of sufficient fine-grained detail to provide for error correction and continuous improvement of the process.
 - o High maturity of analytics capability is expected. Much of the value of a digital business is a rich stream of data that needs analytics to identify actionable trends. It is this stream of available information that informs business choices and delivers the data driven paradigm found in Digital Business models.
 - o A high degree of flexibility with regards to strategy with utilization of infrastructure and products is expected. While automation of these aspects is not yet mature, maturing Digital Businesses are turning to manage market risk by providing for the capability for rapid adoption, change and deprecation of these traditionally stable portions of the business model. In its mature form, this adaption of the business model is referred to as a Platform Business model.

In addition, it will describe from an IT point of view the:

- disruptive changes regarding business requirements
- functional scope covered by DSRA
- proposed implementation principles and architectural frameworks
- business requirements impact on processes and function to be handled by DSRA





Figure 1: The Digital ecosystem value fabric

The digital ecosystem is a value fabric, a mesh of interwoven, cooperating organizations and individuals or parties.

To illustrate the scope covered by the DSRA Business Architecture, Figure 1 presents the digital service ecosystem in the form of a value fabric.

The value fabric is a mesh of interwoven, cooperating organizations and individuals or parties. It is an ideal concept for showing interaction among collaborating organizations and individuals, collectively called parties, in the digital ecosystem.

- [Business drivers, requirements and models R17.5](#)
- [Interaction patterns and orchestration R17.5](#)
- [Main IT requirements and principles R17.5](#)
- [API identification and design R17.5](#)

3.1. Business drivers, requirements and models R17.5

What are the prevalent business drivers, requirements and models in digital services ecosystems?

This section addresses:

- [Business drivers R17.5](#)
- [Business models R17.5](#)
- [Business requirements R17.5](#)

3.1.1. Business drivers R17.5

What are the business drivers related to Digital Services Development?

The major business drivers and disruptions related to digital services development can be summarized as follows:

- [A growing complexity R17.5](#)
- [Needs for agility and simplicity R17.5](#)
- [The development of new business and revenue opportunities R17.5](#)
- [The need to control costs has consequences R17.5](#)

A growing complexity R17.5

What complexity is driving and disrupting business?

A growing complexity can be observed on many fronts.

This growing complexity results in the need for a complete service management in a complex ecosystem.

The **Market** is becoming more complex as growth of the digital world is driving increasing complexity for the providers who must deliver services that work reliably and deliver a great customer experience

Services are becoming more complex as the market is increasingly fragmenting into discrete services because service providers introduce next generation network services combined with cloud hosting

Players relationships are more complex as more complex value creation networks implies multi-lateral business interactions between players within the telecommunication industry and other adjacent industries



This growing complexity results in the need for a complete service management in a complex ecosystem:

- as the number and types of service and service provider are mushrooming

- each of them needing to work with the other requires having standards to work together in an open ecosystem,
- It is important to understand that the challenges of end-to-end service management exists regardless the set of business partnerships, and the complexity is multiplied exponentially as the number of partners in the service delivery chain increases.

Needs for agility and simplicity R17.5

What are the agility and simplicity needs?

There is a need for agility and simplicity.

Keeping the business architecture simple is the only way to gain in agility.

Agility means:

- The capability of companies to develop and deploy complex services extremely rapidly, by assembling resources from several players in an automated way
- The ability to quickly and cheaply adapt to changes regarding market, competition, business model and legislative requirements
- The ability to set up IT resources rapidly and to flex and scale quickly the amount of resources an enterprise needs in order to meet demand.

Keeping the business architecture simple is the only way to gain in agility and is also a business concern.



The development of new business and revenue opportunities R17.5

What is the impact on the development of new business and revenue opportunities?

The development of new business and revenue opportunities is impacted in relation with [business models](#), capability exposure and ability of 3rd parties to invent innovative services:



New <u>business models</u>: • Focus on wholesale, partners and enabling services such as: ◦ Consumer oriented software and applications ◦ Vertical industries (e.g. simplifying financial transactions, optimizing energy distribution, improving healthcare and education) • With increasing partnership possibilities to deliver innovative services • Integrating third party's building blocks in the end product/service delivered to the consumer (Open Innovation)	 <u>Business models</u>
Service, Data and networks assets are exposed: • Via API such as: ◦ Network API (access network selection, location information, etc.) ◦ Service API (e.g. RCS, TV, OCC ...) ◦ Information System API (third party billing, sponsored data, authentication, customer profile, traffic patterns...) • Allowing assets monetization as intermediary products, towards third parties that will integrate them into an end-product delivered to the consumer • Providing additional revenue for the company exposing its asset and saving time and investment cost for company using these assets	

Giving third parties freedom to invent innovative services.

The development of new business and revenue opportunities is impacted in relation with business models, capability exposure and ability of 3rd parties to invent innovative services.

The need to control costs has consequences R17.5

What are the consequences of the need to control costs?

The need to control costs has consequences that manifest themselves in many forms.



Need to simplify the way we do business.

The **need to simplify the way we do business**, to reduce costs and cycle time i.e. doing more with less, or remaining cost neutral, as the digital service ecosystem is becoming more and more complex

There is the need for minimizing customer and other players' interaction costs, as it is a key element of operational design and a major driver for solving the problems of managing across service boundaries.

This can be achieved by providing all the appropriate information immediately accessible and actionable either on a customer portal or, possibly, via a customer service agent.

3.1.2. Business models R17.5

What is the challenge in defining business models for Digital Services?

This section presents the most common business models used for digital services based on the [TM Forum Business Model Compendium](#) that leveraged [Alexander Osterwalder Business Model Canvas](#) to structure the relevant information about each business model and to find commonalities.



[TM Forum Business Model Compendium](#)

For [Alexander Osterwalder](#), a business model describes the rationale of how an organization creates, delivers and captures value.



[Alexander Osterwalder](#)

So, the [Business Model Canvas](#) is a strategic management template for developing new or documenting existing business models. It is a visual chart with elements describing a firm's value proposition, infrastructure, customers, and finances. It assists firms in aligning their activities by illustrating potential trade-offs.



[Business Model Canvas](#)

The starting point for business model innovation is a shared understanding of what a business model is. To reach this goal, we need a business model concept that everybody understands, that facilitates description and discussion, and to start from the same point and talk about the same thing.



[B2B2X web delivery](#)

The challenge is that the concept must be simple, relevant, and intuitively understandable, while not

oversimplifying the complexities of how enterprises function.

[Alexander Osterwalder](#) believes that a business model can best be described through nine basic building blocks that show the logic of how a company intends to make money.

The challenge is that the concept must be simple, relevant, and intuitively understandable, while not oversimplifying the complexities of how enterprises function.

These nine building blocks, foundation of the Business Model Canvas, are:

1. Customer Segment: CS;
2. Value Proposition: VP;
3. Channels: CH;
4. Customer Relationship: CR;
5. Revenue Stream: RS;
6. Key Resources: KR;
7. Key Activities: KA;
8. Key Partnerships: KP;
9. Cost Structure: CO.

Business Model Canvas

Key Partners	Key Activities	Value Propositions	Customer Relationships	Customer Segments
Who are our key partners? Who are our key suppliers? Which key resources are we acquiring from partners? Which key activities do partners perform?	What key activities do our Value Propositions require? Our Distribution Channels? Customer Relationships? Revenue Streams?	What value do we deliver to the customer? Which one of your customer's problems are we helping to solve? What bundles of products and services are we offering to each Customer Segment? Which customers' needs are we satisfying?	What type of relationship does each of our Customer Segment expect us to establish and maintain with them? Which ones have we established? how are they integrated with the rest of our business model? How costly are they?	For whom are we creating value? Who are our most important customers?
	Key Resources		Channels	
	What Key Resources do our Value Propositions require? Our Distribution Channels? Customer Relationships? Revenue Streams?		Through which Channels do our Customer Segments want to be reached? How are we reaching them now? How are our Channel integrated? Which ones work best? Which ones are most cost-efficient? How are we integrating them with customers routines?	
Cost Structure		Revenue Streams		
What re the most important costs inherent in our business model? Which Key Resources are most expensive? Which Key Activities are most expensive?		For what Value are our Customers really willing to pay? For what do they currently pay? How are they currently paying? How would they prefer to pay? How much does each Revenue Stream contribute to overall revenue?		

Figure 2: Alexander Osterwalder Business Model Canvas

Alexander Osterwalder is focusing on how the value proposition is provided to the customer segments and which key partners, resources and activities (concept-to-market) are required and their implication on the cost and the revenue side of the business model.

TM Forum enhanced Osterwalder's model to accommodate information on partner interaction (touchpoints) and drivers and barriers to establish the business model in the market.

Documents already published by TM Forum provide a wide range of typical business models such as:



[TM Forum Business Model Compendium](#)



[B2B2X web delivery](#)

TM Forum B2B2X web delivery R17.5

What business models are covered by the TM Forum B2B2X Partnering Step by Step Guide?

The [TR211 Online B2B2X Partnering Step by Step Guide](#) distinguishes following business models:

- "SaaS store"
- "Smart TV"



[TR211 Online B2B2X Partnering Step by Step Guide](#)

The "**SaaS store**" is the business model of an offer sold by several CSP where application and tools for SMB hosted in CSP Cloud and available in SaaS mode for CSP and Non CSP customers with free trial period and ATAWAD access.

The "**Smart TV**" is the business model of an offer where a content provider and a CSP partner to provide a premium product based on higher quality delivery of streaming video leveraging their respective capabilities through common API.

TM Forum Business Model Compendium R17.5

What business models are covered in the TM Forum Business Model Compendium?

The TM Forum Partnering Guidebook: Concepts and Examples TR211 distinguishes following business models: • "Internet proxy" • "Consumer brand" • "Digital assets marketplace" • "SmartGrid" • "Cloud Service Broker"	 TR218 B2B2X Partnering Guidebook Step by Step Guide  TR211 Online B2B2X Partnering Step by Step Guide
The "Internet Proxy" is a business model based on an offer where mobile operators propose limited free access to popular Internet sites bundled with their data packages offerings. Specifically, it is based on 0.facebook.com offer available through more than 53 mobile operators in 45 countries and territories	 0.facebook.com
The "Consumer brand" is the business model of a converged telecommunications and Internet service based on an operator built and owned platform for rich communication services offered to their own subscriber base; the platform allows 3rd party to contribute to the offering, 3rd party brand bundled inside but services are offered as part of the operator branded product. It is based on China Mobile's 139.com service enabling those of China Mobile's post-paid subscribers who sign up for it to access a range of voice, messaging and Internet services, using multiple devices attached to a single SIM.	 139.com
The "Digital assets marketplace" is a business model where operators sell access to Network and Content Service Enablers to 3rd Party Developers that will have revenue sharing over their applications usage by customers. It is based on Portugal Telecom/SAPO Service Delivery Broker Marketplace which is a concept-to-cash solution, supporting services management from Development to the Marketplace	
The "SmartGrid" presents the business of a next-gen utility service provider enabled by one or more CSPs with an increased use of communications and a greater use of IT systems, that can support real-time services, including an increased interaction and integration of formerly separated systems.	

The **"Cloud Service Broker"** presents a business model for Brokerage and management of all cloud service relevant aspects towards the upstream partners and customers and downstream partners. Cloud Services can be Software as a Service, Infrastructure as a Service, Platform as a Service.

3.1.3. Business requirements R17.5

What are the business requirements for Digital Services Development?

In conjunction with major business drivers and disruptions previously presented, the main business requirements can be summarized along the following themes:

- [Ability to develop new business and revenue opportunities R17.5](#)
- [Digital Service Lifecycle R17.5](#)
- [Digital Service Stakeholder interactions R17.5](#)
- [High quality of experience R17.5](#)
- [Miscellaneous Digital Service Characteristics R17.5](#)
- [Partnership capabilities increase R17.5](#)
- [Simplification and automation R17.5](#)

Ability to develop new business and revenue opportunities R17.5

How is the ability to develop new business and revenue opportunities enabled?

The ability to develop new business and revenue opportunities is enabled by:

- new business models
- allowing new capabilities



New [business models](#) must be supported including capabilities such as:

- Billing on behalf of (BoBo);
- Split billing;
- Business Process as a Service.



[Business models](#)

New capabilities must be allowed:

- Easy personalization of offers and promotions targeting the buyer by using the buyer's profile;

- Extreme flexibility for sellers to promote their products, allowing creation of dynamic product bundles, promotions, pricing, targeted offers, and even entirely new types of products;
- Ability to expose and monetize data and networks assets.

Digital Service Lifecycle R17.5

What are requirements from a Digital Service Lifecycle perspective?

The Digital Service Reference Architecture must be able to support the digital services lifecycle from developer to commercial actors & processes.

The digital service lifecycle is organized in 3 phases summarized in the diagram below:

1. On-boarding digital service provided by another player
2. Adding value to this digital service by adding complementary services
 - o Construction of operational E2E functional services
 - o Creation of management services to operate the functional service
3. Delivering it to final customer i.e. operating the E2E services



Figure 3: Digital service generic lifecycle

Digital Service Stakeholder interactions R17.5

What are requirements related to the interactions between the different Stakeholders?

Interactions allowing building the digital service can become more and more complex depending on the number of stakeholders involved in service construction and delivery.



This is illustrated by the diagram below where:

- Customer 1 buys service A from Digital Service Provider 1 (DSP1)
- DSP1 has composed service A with:
 - o a syndicated service Z from DSP3;
 - o a federated service Y from DSP2.
- DSP3 has composed syndicated service Z with:
 - o a purchased service X;
 - o a federated service W;
 - o a syndicated service V coming from DSP4.

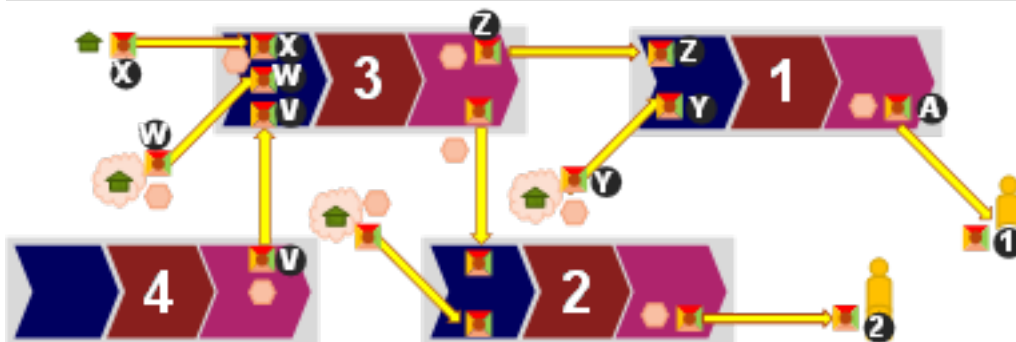


Figure 4: Complex digital Service interactions

High quality of experience R17.5

What is the quality of experience?

The quality of experience is manifested for the user, the developer and operation.



High quality user experience, whatever the service or content the user is consuming, for both:

- Seller: Provide support for the end-to-end product and offer lifecycle;
- Buyer: Provide rich and intuitive search, browse and navigation

High quality developer experience: The developer community must be provided with the tools and guidance needed to build easily the types of applications necessary to deliver great experiences	
High quality operation experience: All stakeholders in the service delivery and assurance processes need to be able to manage their services efficiently and effectively, notwithstanding that they rely on components hosted by different services providers in different clouds	

Miscellaneous Digital Service Characteristics R17.5

What are the general Digital Service Characteristics Requirements?

The Digital Service Reference Architecture must be able to support the digital services business scaling, security, service awareness and versioning.	
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Partnership capabilities increase R17.5

What are capabilities partnerships should embrace?

Partnership capabilities must increase in terms of: • End-to-end (i.e. complete) partnership management • Flexible partner relationship • Agile partner relationship	
End to end partner management: • After implicit or explicit commercial agreement, automated Partner registration and on boarding including developer journey for API integration • Federated catalogue management supporting aggregation of multiple 3rd Parties catalogues with a customized seamless experience. • Automatic calculation of settlement and invoice for partners • Automated “business process as services” (e.g. Billing on behalf of or trouble ticketing) subscription channel	

Flexible partner relationship:

- Provide a flexible set of business relationships options for the participants in the ecosystem. For instance, a partner might be able to choose a relationship:
 - o where the ecosystem markets, enable and charge for the usage of a partner's service and provide settlement periodically to the creator of the service;
 - o that enables and operates the service while managing himself charging and billing functions, to only leverage the ecosystem to market.

**Agile partner relationship:**

- Ecosystem participants can continuously improve the joint business model and are able to adapt to changes imposed by the market, competition, business model or legislative requirements. To enable this all participants must have the ability to:
 - o measure KPIs and provide immediate feedback to their partners
 - o adopt quickly and cheaply to changed requirements
 - o expose new features quickly if required

***Simplification and automation R17.5***

What are simplification and automation requirements?

Simplification and automation requirements are:	
• Automated processes whenever possible	
• Homogeneous solution independently of the partner	
• Easy integration between partners	 Partnership capabilities increase
• Fluidity of interactions and roles	 

“real time” expectation • High availability system allowing to do installations of new functionality while the service continues to run with no impact on users • Real-time creation of new products • Real-time change of existing products	
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3.2. Interaction patterns and orchestration R17.5

What are the interaction patterns and orchestration in digital services ecosystems?

Interaction patterns and orchestration for digital services ecosystems are described in this section in terms of:

- [Common Characteristics of Business Models R17.5](#)
- [Customer / Partner interactions R17.5](#)



3.2.1. Common Characteristics of Business Models R17.5

What are common characteristics of business models in digital services ecosystems?

The table below summarizes the common characteristics of the major part of the business models presented in the [Business models section](#).

Business Model	Name	Short Description	
	Market players/competition	Maturity level	Priority
	Customer Segment	Channel	Relationship
	Multi-sided markets	Online and automated on	• Self Service and

Customer / Market	Retail: Enterprises and Mass Market Partners: Digital Services Providers resellers	any web enabled device	automated Services (internet or mobile) • Co-selling with 3rd parties
Product Offering	Value Proposition		
	Benefits: • Self-service capabilities • One stop shopping • Cost reduction • Easy-to-use • Easy-to-access		
C2M	Activities	Resources	Partners
	Platform set up API and Portal set up SDK development Partners management • on boarding • contract negotiation Offer packaging Vendors training Marketing campaign	Network, hardware, device, software or content resources FTE for personal assistance: customer support and Sales forces Platform development/ operations API / Portal development / maintenance Partner management Data Analytics and BI	Communication Service Providers Device Providers Service Delivery Platforms Providers Resource Providers Software Providers Content Providers
	Cost Structure		Revenue Streams

Enterprise Management	Partnerships costs: revenue sharing Marketing campaigns and packages FTE for Customer support and Sales forces Infrastructure set up and operations, Platform set up and operations Portal development and maintenance	Customer services: Recurring fees / One shot fees Partner services: commission Customer loyalty and ARPU increase New partnership and revenue sources Advertisement
Barriers	Business related	Technical
Drivers	Strategic portfolio	Revenue Streams
Touchpoints/Use Cases	Upstream	Downstream
	Catalogue distribution Request for delivery and delivery status Service request and answer Settlement note and payment Technical problem and solution Deactivation request SLA data request	Order receipt Order delivery Service request and answer Invoice and Payment Bill dispute Trouble ticket submission and resolution Order cancellation
	Business Process Model (a.k.a. eTOM)	Information Model (a.k.a. SID)
Impact on other TM Forum teams	Other teams	

(Critical issues etc.)	
Partners	

The most important common characteristics highlighted are as follows:

- **Customer Segment** type is multi-sided market involving enterprise and mass market customer and partners such as digital services providers, etc.;
- **Channel** type is mostly online and automated;
- **Customer relationship** is mostly automated and self-service and can be handled by third parties;
- **Value proposition** focuses on simplicity (one stop shopping, self-service, ATAWAD) and ease of use;
- **Key activities** performed are mostly focused on software development (API, SDK, SPF), partner management and offer marketing;
- **Key Resources** are Network, hardware, device software and content one plus FTE for assistance, IT development and partner management;
- There is an extensive **Partner** Network;
- The **Cost structure** is value driven, variable, mostly IT and revenue sharing costs;
- **Revenue streams** come from customer service and service provided to partners.

3.2.2. Customer / Partner interactions R17.5

What are Customer / Partner interactions in digital services ecosystems?

New business models consist in multi-actor scenarios with dependencies on other party's processes and actions, as well as various interactions between the different actors.	 Business models
The diagram below presents the interactions between customer and partners during digital service lifecycle. It is a typical example based on a mix of real world offers (a mobile security offer and a medical image sharing offer) and it depicts interactions in a simple use case where only two partners are involved in digital service management.	

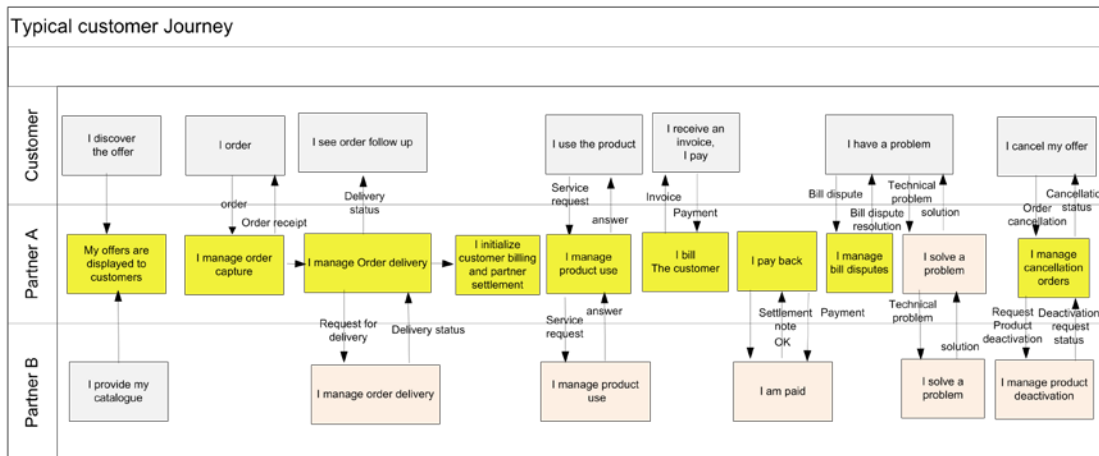


Figure 5: Typical customer / partner journey

In this use case, Partner A has the commercial relationship with final customer and partner B provides a product or service that partner A resells to final customer.

Main touch points between partners involve:

- Catalogue management;
- Digital service ordering (including cancellation);
- Digital service delivery;
- Digital service use;
- Billing and settlement;
- Trouble tickets management;
- SLA management (not displayed in this diagram).

3.3. Main IT requirements and principles R17.5

What are the IT requirements and principles?

Main IT requirements and principles are grouped according to following themes:

- [Disruptive changes for IT regarding business requirements R17.5](#)
- [IT Architecture requirements R17.5](#)
- [Proposed IT principles / architectural framework R17.5](#)
- [Impacted Business processes and positioning versus eTOM R17.5](#)
- [Functions to be handled by DSRA and positioning versus TAM R17.5](#)

3.3.1. Disruptive changes for IT regarding business requirements R17.5

What are the disruptive changes for IT regarding business requirements?

Regarding business requirements presented previously, the most disruptive changes brought about for IT are related to its ability to:

- Support development of new business and revenue opportunities relying on:
 - o New business models including features such as Billing on Behalf of, split billing, business processes as a service;
 - o New capabilities such as personalization of offers or flexibility for their promotion;
 - o Ability to expose and monetize data and network assets;
 - o Seamless access to DSP offers & products catalogues.
- Support the digital services lifecycle from offer creation to customer delivery and support with the following characteristics:
 - o Automated processes whenever possible;
 - o Flexible, homogeneous and generic solution;
 - o Easy integration between partners leading to the need of open interfaces;
 - o “real time” behavior.
- Manage partners industrially meaning:
 - o An automated and end to end (from partner registration to billing and settlement) partner management;
 - o A flexible partner relationship allowing partners to choose between several levels of services.
- Provide a great user, developer and operation experience, based on an end to end digital service management and monitoring.

3.3.2. IT Architecture requirements R17.5

What are the IT Architecture requirements?

The IT Architecture requirements are described in terms of:

- [The Functional scope to be covered by DSRA R17.5](#)
- [Types of APIs managed in the DSRA R17.5](#)



The Functional scope to be covered by DSRA R17.5

What is the functional scope to be covered by DSRA?

From a functional point of view, the Digital Service Reference Architecture must cover:

- Digital Service (as a product) lifecycle management from new product concept to possible withdrawal:
 - o Service construction;
 - o Service monitoring.
- Digital service fulfilment, assurance, billing and revenue management:
 - o Customer identification and authentication;
 - o Catalog management;
 - o Ordering and delivery management;
 - o Billing and invoicing;
 - o Customer financial management (payment, collections);
 - o Customer problem management;
 - o SLA management;
- Partners and interactions with partners management:
 - o Partner on-boarding including developer journey management;
 - o Contract management;
 - o Partner catalog management;
 - o Billing and settlements including assessment and approval;
 - o Partner support management;
 - o Physical goods delivery management (for example, if partner provide a connected object).

Some impacted processes and functions are not specific to digital services.

The diagram below illustrates DSRA functional scope.

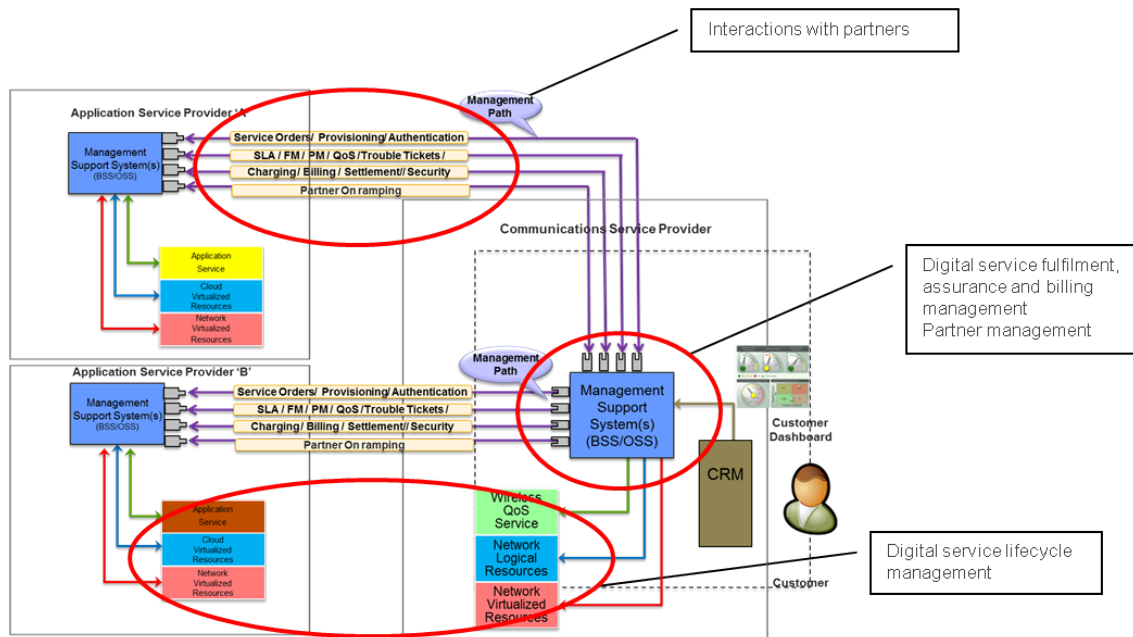


Figure 6: Digital services functional architecture scope

Types of APIs managed in the DSRA R17.5

What are the type of API types managed by the DSRA?

API managed in Digital Services Reference Architecture can be classified into 2 types:

1. Service management API also called management API;
2. Service usage API also called Functional API.



APIs managed in the Digital Services Reference Architecture can be classified into 2 types: management APIs and Functional APIs.

A **Functional API** reflects the service provided and used alone or bundled to deliver directly the service or to build an associated application. It exposes assets and encompasses a wide range of service / resources such as:

- Video;
- Communications;
- Cloud;
- Security;

• Data; • Network; • Etc.	
A Management API allows to perform all management operations before, during and after the use of a service such as: • Customer registration; • Customer account creation; • Service ordering; • Service delivery; • Partner settlement; • Service monitoring; • Trouble management; • Etc.	
The diagram below illustrates the positioning of functional and management API.	

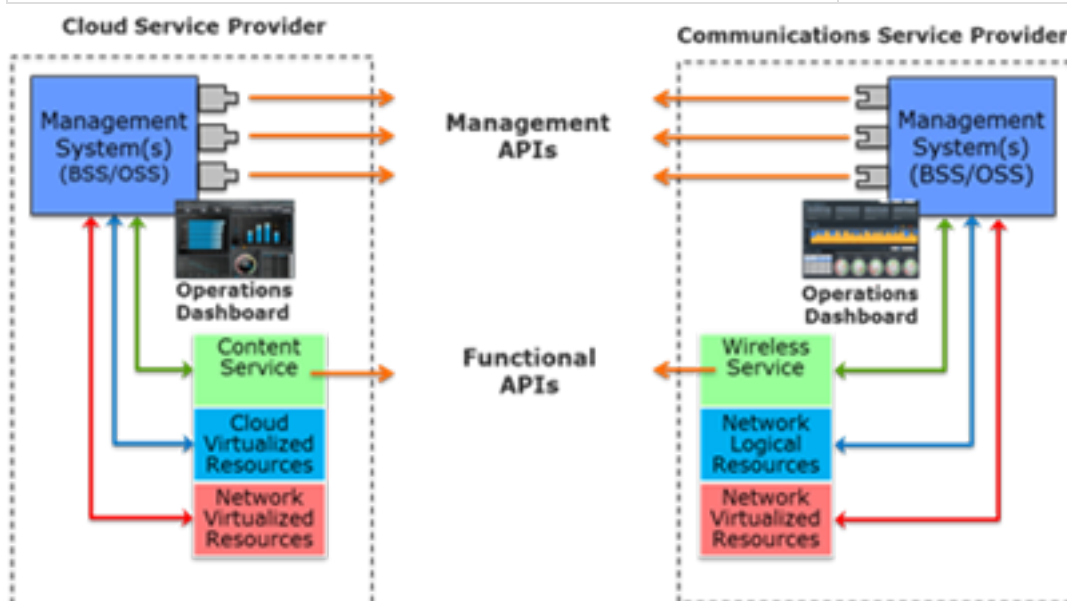


Figure 7: Functional API and management API

3.3.3. Proposed IT principles / architectural framework R17.5

What are the proposed IT principles and Architectural Framework?

The IT principles and Architectural framework for DSRA are:

- [Layered exchange architecture R17.5](#)
- [Flexible, simple and robust set of functions R17.5](#)
- [Open API exposure R17.5](#)
- [Use of a standard management API R17.5](#)
- [Federated catalog management R17.5](#)
- [Modular capabilities in business process management R17.5](#)
- [Flexible role and rights management for 3rd parties R17.5](#)
- [Analytics capabilities R17.5](#)
- [Runtime performance R17.5](#)
- [Online autonomy and availability for services and functions R17.5](#)
- [Business contribution importance of each IT principle R17.5](#)



This section ends with an overview of [the business contribution importance of each IT principle](#).

Layered exchange architecture R17.5

All API based exchanges between platforms are handled by a technical architecture composed of 3 layers:

- **Implementation layer:** network, infrastructure or IT platform delivering services such as telecommunications services, storage or processing services, cloud services, all types of data and video services;
- **Exposition layer:** brokerage, federated identity and profile management, transaction management, event notification management, service catalogue...;
- **Consumption layer:** channels, marketplace, store...

The business architecture is also layered and will differentiate roles like infrastructure enabler, service enabler, aggregator, service provider, etc., each business role needing the 3 technical layers as described above.

The diagram below illustrates this architecture.



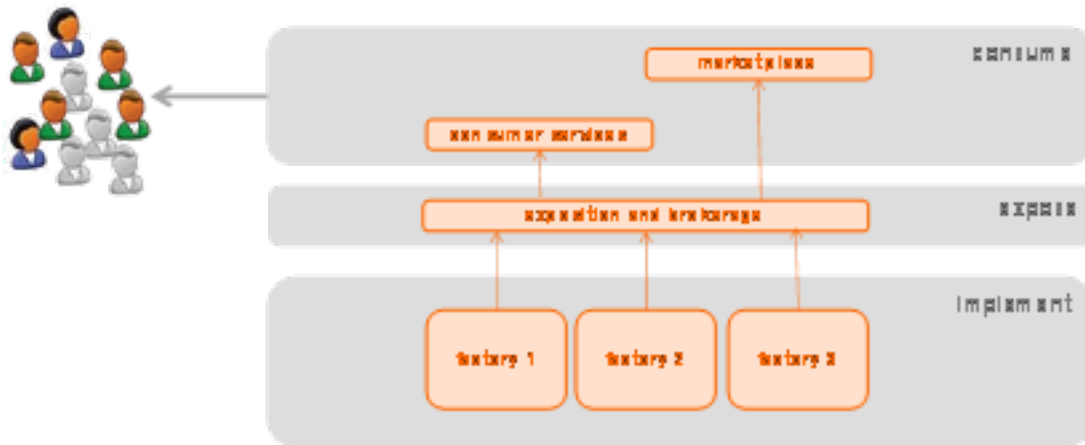


Figure 8: Layered Architecture (simplified)

Flexible, simple and robust set of functions R17.5

The DSRA must support the following BSS functions:

- Partner management (registration, Contracts management, invoicing, settlements, support management, etc.);
- Retail customer management (customer registration, billing account administration, invoicing, financial management support, etc.);
- Order management (capture, validation,) for partner and retail customer;
- Offer & Product catalog management;
- Installed offer and products management;
- Rating and bill calculation.

Main relation between partner and retail customer management lies in billing domain as the partner will be invoiced / settled based on retail customer consumption.

Having a flexible set of functions is mandatory to be able to create value networks based on "business model factories" managing [B2B2X](#) contractual relations. In such situation, the offer is not a classic product but an offer (such "as "Billing on Behalf of" for example) based on and operated by a collection of processes aka "business process as a service" and determining each player role in the value chain. So, flexible assignment of functions among players is essential to provide such capabilities and support different kinds of [business models](#) especially B2B2X ones.



[B2B2X](#)



[Business models](#)

Open API exposure R17.5

API exposure means: • Accessibility before run phase by developer; • Use during run phase. APIs must be extremely easy to use and access: • To access via a simple self-service web portal: to maximize reach and adoption among developers; • To use via the exposition layer: to facilitate communication and integration of software between a company and its business partners. Offering APIs in a self-service mode: • To increase dramatically agility with partners; • To reduce integration costs; • To spur innovation and open innovation. DSRA will allow exposing 3 kinds of API: • <u>Private APIs</u> used internally by each player (a company or its business partners) to facilitate the integration of different applications and systems used within its IT • <u>Restricted APIs</u> used to facilitate communication and integration of software between a company and some selected business partners • <u>Public APIs</u>, by contrast, allow companies to publicly expose information & functionalities of one or various systems & applications to third parties that do not necessarily have a business relationship with them	
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Use of a standard management API R17.5

To facilitate interoperability with partners and reduce integration cost, the DSRA must rely on a set of well documented standard management API such as the TM Forum ones.	
APIs currently specified by the TM Forum encompass: • Customer and Party management; • Product, service and resources Ordering; • Product, service and resources catalogue; • Product, service and resources inventory; • Trouble ticket management;	

• Usage management; • Billing account (CustomerBillSpec) management; • Invoice line (applied customer billing charge) management; • Settlement note advice Management.	
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Federated catalog management R17.5

The term “catalog” designates a database collecting all useful information concerning offers, the products, services, resources and entities of a company.

With development of partnerships, the catalog is extended to information about offers that customers can buy from a company’s partner and about products and services that a company can purchase from a partner to constitute an offer or as an element of his offer.

This leads to the need of federated catalog management.

A federated catalog:

- Supports aggregation and translation of multiple catalogs into a single federated catalog;
- Supports inclusion of partner/supplier/developer offers as product offers into a federated catalog;
- Support multiple ISVs (Independent Software Vendors) and their catalogs with a customized catalog experience;
- Is multi-tenant and hosts multiple catalogs in the most cost-effective manner providing isolation between them.
- Support associated business processes (ordering, fulfillment, assurance and charging for these offers etc.)

Modular capabilities in business process management R17.5

DSRA must allow building business processes based on a set of modular and reusable sub-processes and functions. Several processes must be able to share agnostic functions and sub-processes thanks to a modular architecture based on a normalized and shared services approach.

This principle allows proposing a flexible set of business relationship options also allowing functions and data mutualization. In addition, it improves Information Systems (IS) interoperability and facilitates service exposure towards partners. To do so, use of a BPM engine allows ability to:

- Describe sub processes that can be used or not depending on the player position in value chain;

- Standardize and automate the exchange process between players.

Flexible role and rights management for 3rd parties R17.5

For each party, DSRA manages associated roles and rights (usage, operation, and administration) stored in repositories depending on its position in the value chain.	
For example, the Assignment of a role to a partner (supplier, distributor, customer, etc.) opens to him rights to: • Access to partner portal; • Access to all APIs associated with the chosen offer; • Supply its product catalog. In addition: • the roles of a party must be easily and quickly modified depending on party lifecycle events; • DSRA allows consent management, i.e. a way to authorize parties give other parties access to data. Similarly, rights related to a role must be easily and quickly modified.	

Analytics capabilities R17.5

Analytics provide the ability to interrogate Event Logs in a common and standard way. It collects information from any source and exposes services related to these data exploitation.	
It allows improving • User experience by providing: ◦ Proactive notification of issues. • Developer experience by providing: ◦ Dashboard; ◦ Reporting / analysis capabilities • Operations experience by providing ◦ Additional security to manage and perform correctly Information Security processes; ◦ Fault tracking; ◦ Reporting / analysis possibilities enabling architecture scaling;	

o Cost Analysis.	
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Runtime performance R17.5

DSRA must allow to:

- Install new functionality while the service catalog continues to run without impact on users;
- Create new products or change existing products in real time;
- Support multi-tenancy;
- Provision / de-provision environments rapidly with no associated hardware provisioning;
- Reconfigure dynamically provider chains to work around failures as soon as they are detected.

Online autonomy and availability for services and functions R17.5

DSRA provides a web portal and automated interfaces (standard API, URL call, sending files).

The Partner management portal exposes self-service APIs.

Availability of web portal and automated interfaces allow customer and partners to be autonomous and interact quickly with service provider.

Web portal and automated interfaces allow managing all interactions between a company and its customers or partners such as:

- Customer/ partner management;
- Online ordering;
- Installed Offer and Product Management;
- Customer / partner support;
- E-billing

**Business contribution importance of each IT principle R17.5**

The following matrix shows the business contribution importance of each IT principle.

This matrix can be read two ways: by line, it shows the contribution of IT principle to business objectives; by column it gives the IT initiatives which should be achieved to provide expected benefits for a given business requirement.

Overall, it appears that a business driver cannot be satisfied with a single IT principle and that an IT principle provides benefits for several business drivers.

Business requirement	Layered architecture	Flexible, simple and robust set of functions	Open APIs exposure	Use of standard API	Federated catalog management	Modular capabilities in business process management	Flexible role and rights management for 3 rd parties	Analytics capabilities	Run time performance	Online autonomy and availability for services and functions
Support of new business models	✓	✓								
Ability to expose and monetize data and network assets			✓							
"real time" behavior				✓						✓
Flexible, homogeneous and generic solution		✓		✓	✓					
Automated process				✓		✓				
Easy integration between partners	✓			✓	✓					
Automated and end to end partner management				✓						✓
flexible partner relationship						✓	✓			
Great user,								✓	✓	✓

Business requirement	Layered architecture	Flexible, simple and robust set of functions	Open APIs exposure	Use of standard API	Federated catalog management	Modular capabilities in business process management	Flexible role and rights management for 3 rd parties	Analytics capabilities	Run time performance	Online autonomy and availability for services and functions
developer and operation experience										

3.3.4. Impacted Business processes and positioning versus eTOM R17.5

What are the impacted business processes?

The table below summarizes the eTOM business process being part of DSRA scope. For each of them, it provides: - The actors involved in the process: - The digital service provider is the one selling the service to the final customer; - Infrastructure / enabling service providers supply to digital service provider services or resources used to build the offer sold to final customer. - The IT principles described above impacting the process; - The high-level touchpoints between actors. It is partly based on a summary of technical specifications of B2B Interaction Business Processes contributed by NBNC0	   Business Process Framework
Impacted Assurance processes R17.5 Impacted Billing and Revenue Management processes R17.5 Impacted Fulfillment processes R17.5 Impacted Operation Support and Readiness processes R17.5 Impacted Transversal processes R17.5	

Impacted Assurance processes R17.5

What are the impacted business processes?

Category	Business Process	Actors involved	DSRA impact	Touchpoints
Assurance	Problem Handling	Digital service provider	Modular capabilities in business process management Online autonomy and availability for services and functions	→ Submit Trouble Ticket ← Notify Trouble Ticket information (e.g. amendment, status change, etc.)
	Customer QoS/SLA Management	Digital service provider Infrastructure / enabling service providers	Analytics capabilities Run time performance Use of standard API	→ Request SLA information ← Notify SLA information
	Service Problem Management	Digital service provider Infrastructure / enabling service providers	Run time performance Use of standard API	→ Submit Trouble Ticket ← Notify Trouble Ticket information (e.g. amendment, status change, etc.)
	Service Quality Management	Digital service provider Infrastructure / enabling service providers	Analytics capabilities Run time performance	→ Request Service Quality information! Notify Service Quality information
	Resource Trouble Management	Digital service provider Infrastructure / enabling service providers	Analytics capabilities Use of standard API	→ Submit Trouble Ticket ← Notify Trouble Ticket information (e.g. amendment, status change, etc.)

Category	Business Process	Actors involved	DSRA impact	Touchpoints
	Resource Performance Management	Digital service provider Infrastructure / enabling service providers	Analytics capabilities Run time performance	→ Request resource performance information ← Notify resource performance information
	S/P Problem Reporting & Management	Digital service provider Infrastructure / enabling service providers	Layered architecture Use of standard API Open APIs exposure	→ Submit Trouble Ticket ← Notify Trouble Ticket information (e.g. amendment, status change, etc.)
	S/P Performance Management	Digital service provider Infrastructure / enabling service providers	Analytics capabilities Run time performance	→ Request performance information ← Notify performance information

Impacted Billing and Revenue Management processes R17.5

What are the impacted business processes?

Category	Business Process	Actors involved	DSRA impact	Touchpoints
Billing and revenue management	Bill Invoice Management	Digital service provider	Flexible, simple and robust set of BSS functions Modular capabilities in business process management Online autonomy and availability for	N/A

Category	Business Process	Actors involved	DSRA impact	Touchpoints
			services and functions	
	Bill Payments & Receivables Management	Digital service provider	Flexible, simple and robust set of BSS functions Modular capabilities in business process management	N/A
	Bill Inquiry Handling	Digital service provider	Flexible, simple and robust set of BSS functions Modular capabilities in business process management Online autonomy and availability for services and functions	N/A
	Manage Billing Events	Digital service provider Infrastructure / enabling service providers	Flexible, simple and robust set of BSS functions Modular capabilities in business process management Use of standard API	← Notify Billing Event → Acknowledge Billing Event
	Charging	Digital service provider	Flexible, simple and robust set of BSS functions Modular capabilities in	N/A

Category	Business Process	Actors involved	DSRA impact	Touchpoints
			business process management	
	Manage Balances	Digital service provider	Flexible, simple and robust set of BSS functions Modular capabilities in business process management	N/A
	Service Guiding & Mediation	Digital service provider	Flexible, simple and robust set of BSS functions Modular capabilities in business process management	N/A
	Resource Mediation & Reporting	Digital service provider	Flexible, simple and robust set of BSS functions Modular capabilities in business process management	N/A
	S/P Settlements & Payments Management	Digital service provider Infrastructure / enabling service providers	Modular capabilities in business process management Online autonomy and availability for services and functions Use of standard API	→ Notify Partner Settlement ← Acknowledge Partner Settlement → Notify Partner payment ← Acknowledge Partner Payment

! Notify Billing Event

"Acknowledge Billing Event!Acknowledge Partner Payment

Impacted Fulfillment processes R17.5

What are the impacted business processes?

Category	Business Process	Actors involved	DSRA impact	Touchpoints
Fulfilment	Order Handling	Digital service provider Infrastructure / enabling service providers	Flexible, simple and robust set of BSS functions Modular capabilities in business process management Online autonomy and availability for services and functions Use of standard API Federated catalog management	→ Submit Product Order ← Notify Product Order information (e.g. status: accepted, rejected, completed, etc.)
	Service Configuration & Activation	Digital service provider Infrastructure / enabling service providers	Use of standard API Federated catalog management	→ Submit Service Order ← Notify Service Order information (e.g. status: accepted, rejected, completed, etc.)
	Resource Provisioning	Digital service provider Infrastructure / enabling service providers	Use of standard API Federated catalog management	→ Submit Resource Order ← Notify Resource Order information (e.g. status: accepted, rejected, completed, etc.)

Category	Business Process	Actors involved	DSRA impact	Touchpoints
	Resource Data Collection & Distribution	Digital service provider Infrastructure / enabling service providers	Analytics capabilities	→ Request Resources Data (Resource Inventory?) ← Notify Resource Data
	S/P Requisition Management	Digital service provider Infrastructure / enabling service providers	Modular capabilities in business process management Online autonomy and availability for services and functions Flexible role and rights management for 3rd parties Federated catalog management	→ Submit Partner Order ← Notify Partner Order
	S/P Interface Management	Digital service provider Infrastructure / enabling service providers	Layered architecture Use of standard API Open APIs exposure	All above mentioned fulfillment touchpoints

"Submit Product Order

!Notify Product Order information (e.g. status: accepted, rejected, completed, etc.)

Impacted Operation Support and Readiness processes R17.5

What are the impacted business processes?

Category	Business Process	Actors involved	DSRA impact	Touchpoints
Operation support and readiness	CRM - Support & Readiness	Digital service provider	Flexible, simple and robust set of BSS functions Modular capabilities in business process management Online autonomy and availability for services and functions	N/A?
	Service Management & Operations Support & Readiness	Digital service provider Infrastructure / enabling service providers	Use of standard API Analytics capabilities	→ Request Service Inventory ← Notify Service Inventory
	RM&O Support & Readiness	Digital service provider Infrastructure / enabling service providers	Use of standard API Analytics capabilities	→ Request Resource Inventory ← Notify Resource Inventory
	S/PRM Support & Readiness	Digital service provider Infrastructure / enabling service providers	Layered architecture Use of standard API Modular capabilities in business process management Flexible role and rights management for 3rd parties	→ Request Partner (party) Information ← Notify Partner (party) Information "On-Board Partner"

Impacted Transversal processes R17.5

What are the impacted business processes?

Category	Business Process	Actors involved	DSRA impact	Touchpoints
Transversal	Customer management	Digital service provider	Flexible, simple and robust set of BSS functions Modular capabilities in business process management	N/A
		Digital service provider Infrastructure / enabling service providers	Modular capabilities in business process management Online autonomy and availability for services and functions Use of standard API	→ Submit customer information ← Notify customer information

3.3.5. Functions to be handled by DSRA and positioning versus TAM R17.5

What are the impacted functions?

Based on relevant business processes, the table below summarizes the [TAM](#) Applications being part of DSRA scope. For each of them, it provides:

- The actors exposing or consuming the function;
- The standard API to be exposed.



[Application Framework](#)

- [Impacted Assurance Functions R17.5](#)
- [Impacted Billing Functions R17.5](#)
- [Impacted Fulfillment Functions R17.5](#)
- [Impacted Operation Support and Readiness Functions R17.5](#)
- [Impacted Supplier/Partner Functions R17.5](#)

Impacted Assurance Functions R17.5

What are the impacted functions?

Category	Application	Actors involved	Standard API exposed
Assurance	Customer SLA Management	Digital service provider Infrastructure / enabling service providers	SLA management API
	Customer Problem Management	Digital service provider Infrastructure / enabling service providers	Trouble ticketing API
	Service Problem Management	Digital service provider Infrastructure / enabling service providers	Trouble ticketing API?
	Service Performance Management	Digital service provider Infrastructure / enabling service providers	Performance management API?
	Service Quality Management	Digital service provider Infrastructure / enabling service providers	Service quality management API
	Resource Performance Management	Digital service provider Infrastructure / enabling service providers	Performance management API?
	Usage Management	Digital service provider Infrastructure / enabling service providers	Usage API

Impacted Billing Functions R17.5

What are the impacted functions?

Category	Application	Actors involved	Standard API exposed
Billing	Collection Management	Digital service provider	Not available

Category	Application	Actors involved	Standard API exposed
	Receivables Management	Digital service provider	Payment API (available with release FX 17.0)
	Billing Events Management	Digital service provider Infrastructure / enabling service providers	Customer Bill API (available with release FX 17.0)
	Billing Account Management	Digital service provider Infrastructure / enabling service providers	Account API (available with release FX 17.0)
	Bill Calculation	Digital service provider	Customer Bill API (available with release FX 17.0)
	Charge Calculation and Balance Management	Digital service provider	Not available
	Voucher management	Digital service provider	Not available

Impacted Fulfillment Functions R17.5

What are the impacted functions?

Category	Application	Actors involved	Standard API exposed
Fulfilment	Channel Sales Management	Digital service provider Infrastructure / enabling service providers	Not available
	Customer Self-Management	Digital service provider	Not available
	Customer order management	Digital service provider Infrastructure / enabling service providers	Product ordering API
	Service Order Management	Digital service provider	Service ordering API

Category	Application	Actors involved	Standard API exposed
		Infrastructure / enabling service providers	
	Resource Order Management	Digital service provider Infrastructure / enabling service providers	Resource Ordering API

Impacted Operation Support and Readiness Functions R17.5

What are the impacted functions?

Category	Application	Actors involved	Standard API exposed
Operation support and readiness	Catalog management	Digital service provider Infrastructure / enabling service providers	Catalog API (Product service and resource)
	Customer Information Management	Digital service provider Infrastructure / enabling service providers	Customer management API
	Service Inventory Management	Digital service provider Infrastructure / enabling service providers	Service inventory API
	Resource Inventory Management	Digital service provider Infrastructure / enabling service providers	Resource inventory API

Impacted Supplier/Partner Functions R17.5

What are the impacted functions?

Category	Application	Actors involved	Standard API exposed
Supplier / Partner	Partner management	Digital service provider Infrastructure / enabling service providers	Party management API Party Role API (available with release FX 17.0) OnBoarding API (partnership type)
	Wholesale billing	Digital service provider Infrastructure / enabling service providers	Not available

3.4. API identification and design R17.5

How is API identification and design done?

The present section focuses on API identification and design. It shows step by step how to identify management APIs, define their type and specify accurately their signatures.

The overall proposed approach consists of the following 3 phases:

- [Phase 0: Use cases identification R17.5](#)
- [Phase 1: Distribute processes R17.5](#)
- [Phase 2: Data Architecture R17.5](#)



3.4.1. Phase 0: Use cases identification R17.5

How are use cases identified?

In Phase 0, the Use cases are identified as follows:

- List the main uses cases to implement
- List the involved roles (customer/provider, operator/distributor...)
- Describe the E2E process



3.4.2. Phase 1: Distribute processes R17.5

How are processes distributed?

In Phase 1, the processes are distributed according to involved roles and systems:

- Assign tasks to involved roles
- Distribute tasks between systems
- Identify information flow exchanged between systems / roles
 - o Event (order validated, usage detected....)
 - o Sub process delegation (purchase, payment, ...)
 - o Function call
- Identify data shared by systems

Process distribution among systems may be realized according to 2 axes:

1. Vertical distribution:

- o Partner A hands over the lead to partner B.
- o Sub process 1 is completed and Partner A is not waiting for a response from Partner B
- o Event 1 is transmitted from A to B

2. Horizontal distribution:

- o Sub process 3 is delegated to partner C by Partner B. Partner B is waiting for a response from Partner C

- Service call from B to C



3.4.3. Phase 2: Data Architecture R17.5

How is API data distribution done?

Phase 2 deals with Data Distribution. About shared data, a choice must be made between:

- Authorize access to the data through APIs by function calls (get contract, get Bill, ...)



- Transmit a replication of shared data: data are transported by the flows, identified in the previous step



3 solutions are possible for data distribution:

1. Create repositories that can be accessed by every application
2. Allow a partner to handle other partner's data through function calls (CRUD)
3. Transmit a copy of the concerned data to other applications

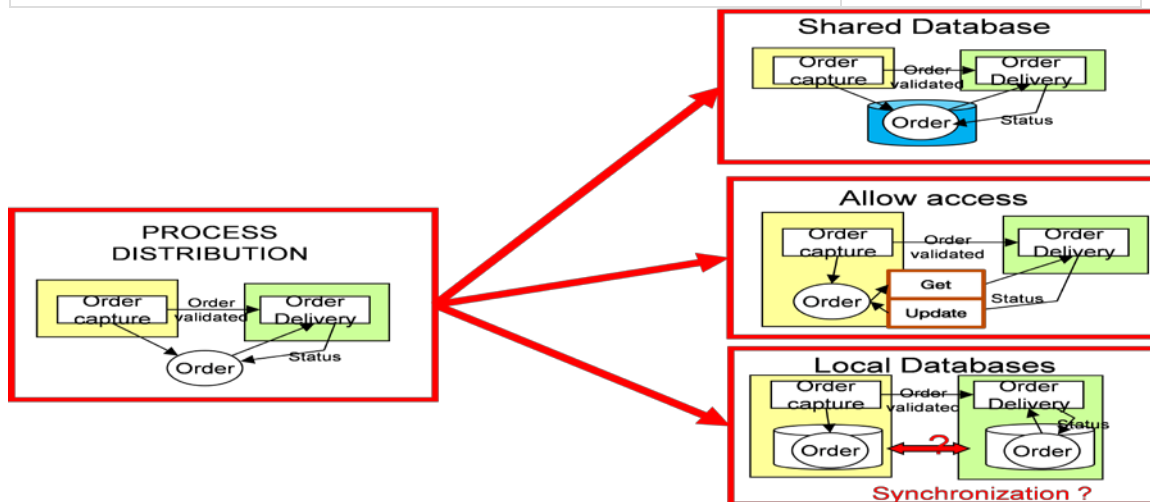


Figure 10: data distribution

4. DSRA Technology Architecture R17.5

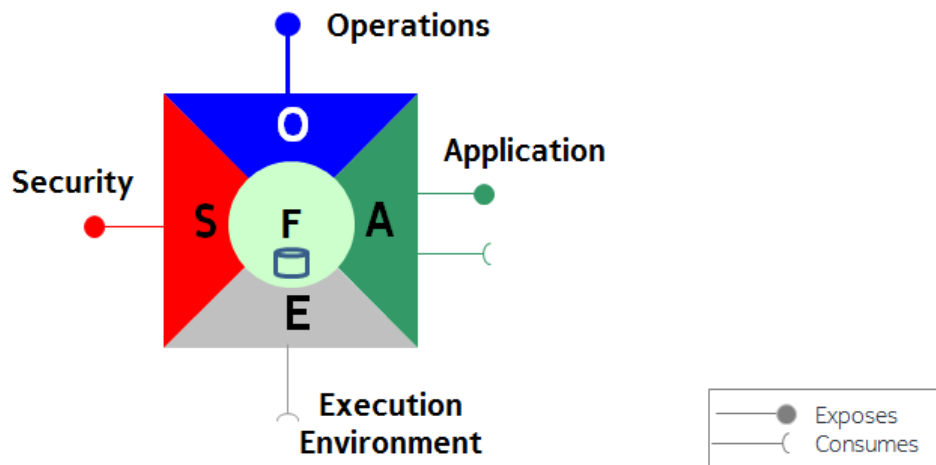
What is the technology supporting the DSRA?

- [Digital Service Component R17.5](#)
- [Digital Service Management API R17.5](#)
- [DSRA Platform Capabilities R17.5](#)
- [API's mapped on DSRA Platform Capabilities R17.5](#)
- [DSRA Standards Index R17.5](#)
- [Web Technology Utilization R17.5](#)



4.1. Digital Service Component R17.5

What is a digital service component?



The figure represents a digital service component with interfaces, as defined in [IG1118 OSS/BSS Futures - Architecture](#).



[IG1118 OSS/BSS Futures - Architecture](#)

The component definition is one of the crucial concepts in DSRA. In the figure you see "consuming" and "exposing" APIs.



The component is hosted in an execution environment, and consumes the environment functionality through the Execution API .	
There is also a Security API , and at the center there is the Functionality the component is providing, as well as the component Data store .	
The component functionality is exposed thru the Application API . The component can also consume functionality from other components when it is participating in a service composition.	
The Operations API exposes operations to configure, activate or suspend; to manage health and to monetize the digital service. This is referred to as the management interface.	
TM Forum is currently defining the Digital Services Management API for that purpose	
	 SM-API  TM Forum Ecosystem API Portal

4.2. Digital Service Management API R17.5

How do you manage composed Digital Services end-to-end across a multi-sided ecosystem?

The Digital Service Reference Architecture enables consistent end-to-end management of digital services exposed by and across different service providers domains and technologies. It delivers the ability to configure, activate, or suspend a digital service instance	 SM-API
---	--

execution and to both receive or be notified of any kind of metrics, health state and detailed information about eventual failures.

TM Forum is defining the Digital Services Management API, as an evolution of the [SM-API](#), with the following operations that are typically offered to manage a digital service:

- Activation of a digital service: Making the digital service available for a particular context (Deploying the digital service).
- Provisioning of a digital service: Configuring the settings of a digital service or a digital service instance.
- State monitoring of a digital service: Querying the history and status in terms of life cycle management (for a specific instance of the digital service) and listening for status updates.
- Usage monitoring of a digital service: Querying for usage metrics from the digital service instance or listening for usage metrics reports or alarm (e.g., if metrics conditions imply notifications).
- Health monitoring of a digital service: Querying for health metrics from the digital service instance or listening to alarm from the resource.
- Update of a digital service: Modification of the setting or life cycle management status of a digital service instance.
- De-activation of a digital service: making the digital service unavailable in a particular context.
- Configuring the communication endpoint address the digital service instance must use to deliver a report containing information about its health, execution state, eventual failures and metrics.
- Removing the communication endpoint address the digital service instance was using to deliver a report



TMF665 Digital Services Management API

containing information about its health, execution state, eventual failures and metrics.	
The Digital Services Management API that is defined as a selection from the existing APIs. See the TM Forum Ecosystem API Portal .	 TMF665 Digital Services Management API  TM Forum Ecosystem API Portal

4.3. DSRA Platform Capabilities R17.5

What is the set of Platform Capabilities that are necessary to support digital service ecosystems and should exist in each DSRA platform implementation?

The Technology Architecture describes the structure and interaction of the platform capabilities, and logical and physical technology components.

According to TOGAF , <u>the Technology Architecture describes the structure and interaction of the platform capabilities, and logical and physical technology components.</u>	 TOGAF 9.1
Based on business requirements and IT principles, this chapter focuses on defining a set of re-usable supporting services that multi-tenant (typically cloud-based) platforms can offer to support one or more digital ecosystems. The digital platforms should can provide these by hosting 1st party or 3rd party supporting services or by providing federated access to the necessary services.	 The DSRA Approach

- [Business Layer Technical Capabilities R17.5](#)
- [PaaS Layer Technical Capabilities R17.5](#)
- [IaaS Layer Technical Capabilities R17.5](#)

4.3.1. Business Layer Technical Capabilities R17.5

What are the Business Layer Technical Capabilities?

The Business Layer contains those services that a digital ecosystem creates and uses to operate their business, their digital ecosystem. A developer would consume IaaS and PaaS services offered by an Actualization Platform and create the necessary business services following the guidelines published by the TM Forum Framework. For example, all BSS/OSS that any given CSP or DSP would need to run their business would fall into this category. After these systems are developed, they are deployed onto an Actualization Platform for runtime operations to form the Business Layer of the Actualization Platform. Increasingly today, the Actualization Platform is a cloud platform hosted in one or more datacenters.

Following Business Layer Technical Capabilities are considered:

- [Catalog Lifecycle Management & Federation Platform Capability R17.5](#)
- [Fulfillment Platform Capability R17.5](#)
- [Customer Experience Management Platform Capability R17.5](#)
- [Assurance Platform Capability R17.5](#)
- [Billing & Revenue Management Platform Capability R17.5](#)

Catalog Lifecycle Management & Federation Platform Capability R17.5

What is catalog lifecycle management and federation?

For Catalog Federation following aspects are considered:

- [Catalog context R17.5](#)
- [Catalog user story R17.5](#)
- [Catalog Lifecycle Management R17.5](#)
- [Catalog Federation R17.5](#)
- [Catalog Architectures R17.5](#)
- [Catalog Federation Direction R17.5](#)
- [Catalog Federation Business Processes R17.5](#)
- [Catalog management and API exposure R17.5](#)

Catalog context R17.5

What is the context of Catalog Lifecycle Management & Federation?

There are many personas involved in a Digital Ecosystem, partners, suppliers, application developers, enterprise owners, government, and operators.

There will be federated catalogs within a large ecosystem and there will be the need to utilize catalog data to provide the data necessary for storefront/portal operations.

As a partner, I want to seamlessly join the ecosystem, sign agreements, receive credentials and then upload product offers that contain my Digital Service - API(s). Each product offer may have one or more types of price: SID Product Offer Price - Usage, one time and recurring fees.

The partner can onboard themselves thru a portal but will likely send the product offer over the catalog management API. Each product offer is versioned in the catalog and maintains the version provided by the partner.

To get access to the open environment of a digital ecosystem, a candidate partner needs to on-board the ecosystem. The On-boarding function allows different parties such as operators, social networks, service contributors, product creators, developers and others, to apply for partnership and if approved, to create accounts with the appropriate permissions for the role they want to play in the ecosystem.

Once the partner has been on-boarded, he can define products that need to be on-boarded. This also allows to establish commercial terms and conditions, sign agreements, define payment and settlement methods. Once a partner's product has been on-boarded, it will be published in the Digital ecosystem catalog.

The Digital ecosystem catalog stores product offer APIs within the Resource entitled Software Resource.

The on-boarding process is described in the [B2B2X Partnering Guide](#).

On-boarding takes place in Marketplace that is described in the [DPRA Concepts & Principles](#).



[TR211 Online B2B2X Partnering Step by Step Guide](#)



[DPRA Concepts & Principles](#)

Catalog user story R17.5

What is a typical user story for Catalog Lifecycle Management & Federation?

User story

"GoodAds" pays developers for targeted ad insertions. It is a component service to which a profile and ad request is sent, and based on that profile, the most appropriate ad is returned.

"Movies4U" sells premium streaming movies at a significant discount, if users opt-in to its targeted ads. Movies4U wants the product to periodically "call" GoodAds to deliver ads that it inserts.

Movies4U, being a partner with a “Supplier role” in the Digital Ecosystem means that the Movies4U Product Manager can log into the partner section of the portal to onboard their product offers. The Product Manager can upload a handful of offers thru the portal or can submit bulk product offers via the TM Forum Product Catalog Management API. The portal will lead the product manager thru the type of information requested in each section or the Product Catalog Management API will dictate the format. Once the data is received the data is stored in the Catalog as a product offer in design lifecycle state.	  Product Catalog Management API
A product offer requires service management capabilities to be delivered: order fulfillment, assurance, policies, billing/charging. These Service management components need to be associated to the product offer. In our user story the Movies4U product manager wants the ecosystem to handle all service management and exposure of the movies4U product offers. The ecosystem expects the product offers to contain the executable API for the offer.	
User story “GoodAds” pays developers for targeted ad insertions. It is a component service to which a profile and ad request is sent, and based on that profile, the most appropriate ad is returned. “Movies4U” sells premium streaming movies at a significant discount, if users opt-in to its targeted ads. Movies4U wants the product to periodically “calls” GoodAds to deliver ads that it inserts. Since GoodAds and the preferred LTE carrier are partners in a digital ecosystem, Movies4U decides to apply for partnership in that same ecosystem. The Movies4U Partnership Manager registers on the portal of the digital ecosystem by completing an account profile including the Supplier, Developer and Reseller roles, and general as well as billing information. Upon completion of the account profile, the Movies4U Partnership Manager submits the account profile for partnership approval. During this process he can check the approval status through the portal. When the partnership candidature is approved, Movies4U receives an acceptance email which includes a reference to the partnership agreement, which includes the terms and conditions. The Partnership Manager reviews, signs and submits the agreement. Movies4U is now listed as digital ecosystem partner.	

Movies4U, being a partner with Supplier role now, the Movies4U Product Developer can log into the partner section of the portal to define offers. Through that section integration documentation can be consulted, offers can be submitted for approval and published after approval.

The Service Design subsection of the portal, allows to define an End-to-End Composite Product offer. The Movies4U Product Developer defines a name, a description and a representative icon for its streaming movie offer. After this is set, the Product Developer browses the catalog for his preferred LTE mobile transport and the GoodAds ad service. And finally, he defines the movie streaming offer as a composition of these 3 digital services. This includes the management policy for the Composite Product that determines whether only the management interface of the Composite Product, or the management interface of the Composite Product as well as some of the Service Components needs to be consumed to manage the movie streaming offer. He defines the configurable attributes of the product offer, including a means to opt-in and out of targeted ads and the GoodAds ad service profile information. He also defines the pricing and subscription model(s), and Service Level Objectives (SLOs).

After completion of the composition, the Product Developer submits the offer for approval. When the offer is approved, the Product Developer submits the offer for publication, which makes it appear in the digital ecosystem catalog.

Catalog Lifecycle Management R17.5

What is Catalog Lifecycle Management?

An ecosystem must allow partners to on-board product offers.

And allow partners to discover available service management and service enablement components from the Ecosystem catalog to enable fully formed product offers. Offers that can be fulfilled, assured, charged for and to manage policies.

Each offer should provide the Resource definition for the API.

Each product offer is managed via lifecycle management.

Once in an active state the product offer is made available

Catalog Lifecycle Management & Federation Service automates everything possible on the support services and infrastructure layers to avoid a new integration project for every Partner, is crucial to economically sustain an

ecosystem and enable the trial of new products and services without great upfront effort.

The Catalog Lifecycle Management & Federation Service automates everything possible on the support services and infrastructure layers to avoid a new integration project for every Partner, is crucial to economically sustain an ecosystem and enable the trial of new products and services without great upfront effort.

The Catalog Lifecycle Management & Federation Service:

- Allows partners to define Service Components and End-to-end Composite Products by means of following information categories:
 - o General information (e.g. name, description, a representative icon) of the offer
 - o Configurable attributes
 - o Pricing and subscription model(s)
 - o Service Level Objectives (SLOs)
- Allows partners to submit a product offer for approval
- Evaluates product offer approval requests submitted by a partner
- Notifies a partner of the evaluation results of his product offer approval request
- Integrates with the "Catalog Lifecycle Management & Federation" Support Service to define candidate product offerings and publish approved product offerings into the digital ecosystem catalog

The "on-boarding" function should:

- Allow for candidate partners to register as partner of the digital ecosystem by completing an account profile with, but not limited to following attribute sets:
 - o Credentials
 - o The role the party wants to play such as, but not limited to following roles: Supplier, Developer, Reseller and Enterprise
 - o General information such as name, contact email and phone
 - o Billing Information such as denomination, fiscal number, postal address and debit account number
- Allow candidate partners to submit a partnership account approval request which includes a completed account profile
- Allow partners to re-submit a partnership account approval request if the account profile has been changed
- Allow (candidate) partners to check approval status
- Evaluate partnership account approval requests submitted by a candidate partner
- Notify a (candidate) partner of the evaluation result of his partnership account approval request
- Send a partnership agreement including terms and conditions, to the candidate partner upon partnership account approval request

- Allow candidate partners to review and sign electronically a partnership agreement which includes terms and conditions
- Allow candidate partners to submit a signed partnership agreement
- Register candidate partners as digital ecosystem partners upon receipt of a signed partnership agreement
- Grant access permissions to the partner based on its agreed upon role:
 - o As a Supplier, the partner will be able to on-board through the portal, Composite Products and Service Components
 - o As a Developer, the partner will be able to register an APP through the portal or to purchase Composite Products and Component Services
 - o As a Reseller, the partner will be able to create a bundled product offering
 - o As an Enterprise, the partner will be able to submit partner names to the digital ecosystem
- Make available proper documentation of the APIs, the partners need to develop and make available on their side:
 - o This is to avoid an ad-hoc integration project for each new Partner
 - o Rather the Provider can use a standard approach to provision the partner
 - o For example, every time a partner offer is sold, to check upon the Health state of the partner products and services, so that SLAs can be verified, and to be able to receive incident notifications and gather metrics values
- Allow partners to consult documentation that is required to integrate with the digital ecosystem, including API documentation
- Allow partners to define product offers
- Allow partners to discover available End-to-End Composite Products and Component Services from the digital ecosystem catalog
- Integrate with the “Catalog Lifecycle Management & Federation” Platform Capability to discover Component Services and End-to-End Composite Products from the digital ecosystem catalog
- Allow partners to define Service Components and End-to-end Composite Products by means of following information categories:
 - o General information (e.g. name, description, a representative icon) of the offer
 - o Configurable attributes
 - o Pricing and subscription model(s)
 - o Service Level Objectives (SLOs)
- Allow partners to submit a product offer for approval
- Evaluate product offer approval requests submitted by a partner
- Notify a partner of the evaluation results of his product offer approval request
- Integrate with the “Catalog Lifecycle Management & Federation” Support Service to define candidate product offerings and publish approved product offerings into the digital ecosystem catalog

Automate everything possible on the support services and infrastructure layers to avoid a new integration project for every Partner, is crucial to economically sustain an ecosystem and enable the trial of new products and services without great upfront effort

Catalog Federation R17.5

What is Catalog Federation?

Catalog Federation is the transparent combination of multi-catalog arrangements, whether located within a single enterprise or extended across external supplier-provided catalogs, using standardized inter-catalog interface.

- Federated Catalogs refer to a logical and physical partition of a catalog database along some classification criteria.
- Criteria can be geographical, supplier/partner or content related (product/services/resources).
 - o Even though each of the partitions in a federated catalog will be quite autonomous, for certain use cases a federated catalog may still present itself as a single (logical) catalog.

Catalog Federation is the transparent combination of multi-catalog arrangements, whether located within a single enterprise or extended across external supplier-provided catalogs, using standardized inter-catalog interface.

Catalog Architectures R17.5

What are catalog architectural principles?

- Goal: provide a common view on Catalog data
- Centralized PLM; enables Business to be more involved (step-away from the pure Project Management)
- Product modeling performed by Product Management (with IT owning the technical components)
- Central Catalog, 2 approaches:
 1. Master Catalog (MDM approach)
 - o Catalog is the owner of all relevant data
 - o Slave systems are fed by Central Catalog in a controlled manner
 2. Federated Catalog
 - o Catalog data distributed across systems
 - o Master system for each piece of data

Catalog Federation Direction R17.5

What is catalog federation direction?

Single Direction Client Catalog

Discover catalog information from a Source Catalog, use portions of the Source company's fulfillment process.



Catalog Supports Single Point of Truth configurations

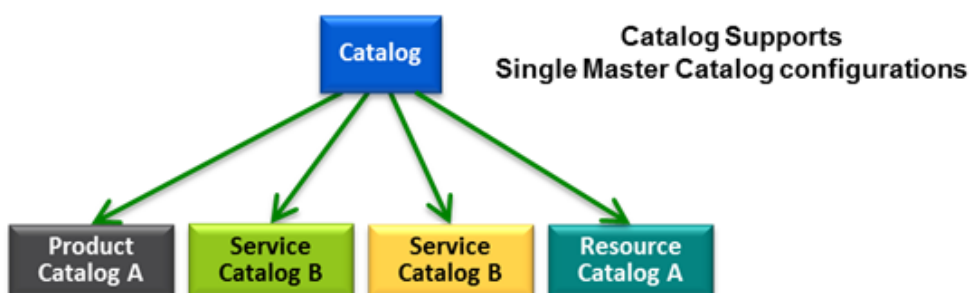
Single Direction Source Catalog

Share catalog information with a Client Catalog, allow the Client Catalog to access catalog information and use the Source Catalog's fulfillment processes.

Bi-directional Client/Source

Both companies can discover and share information by forming Client and Source relationships with each other.

Each company shares and discovers catalog information and each allows the other to utilize fulfillment processes.



Federated Catalog architecture

- Offer/Product/Service/Resource definition spread over one or more Catalogs
- 1 master system for each piece of data
- Key requirement; interface between the Catalogs as simple as possible
- Solution; 1:1 mapping between the "handshake objects"

Architectural decisions

- Mapping on CFS level or Product Spec level
- Which system holds the definition of the mapped object and which holds the reference only (master/slave controlled redundancy is not the optimal approach)

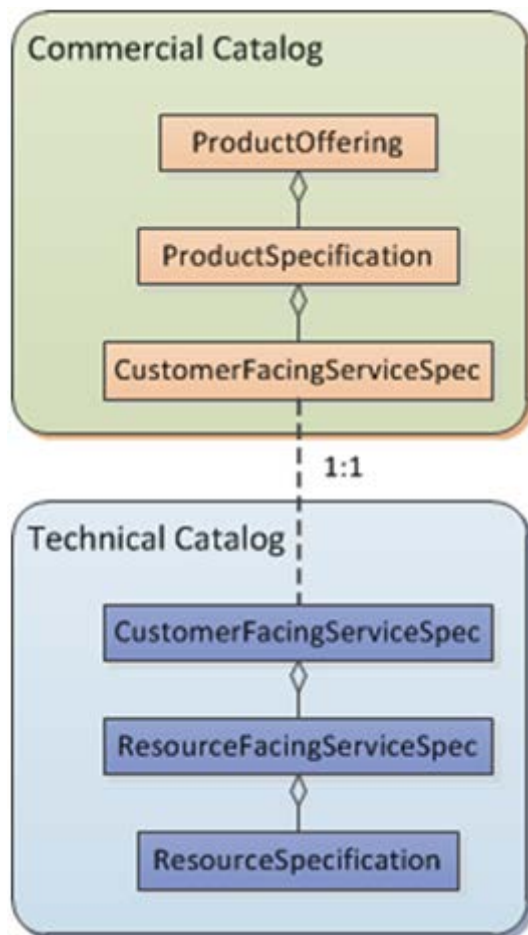


Figure: Commercial to Technical Catalog

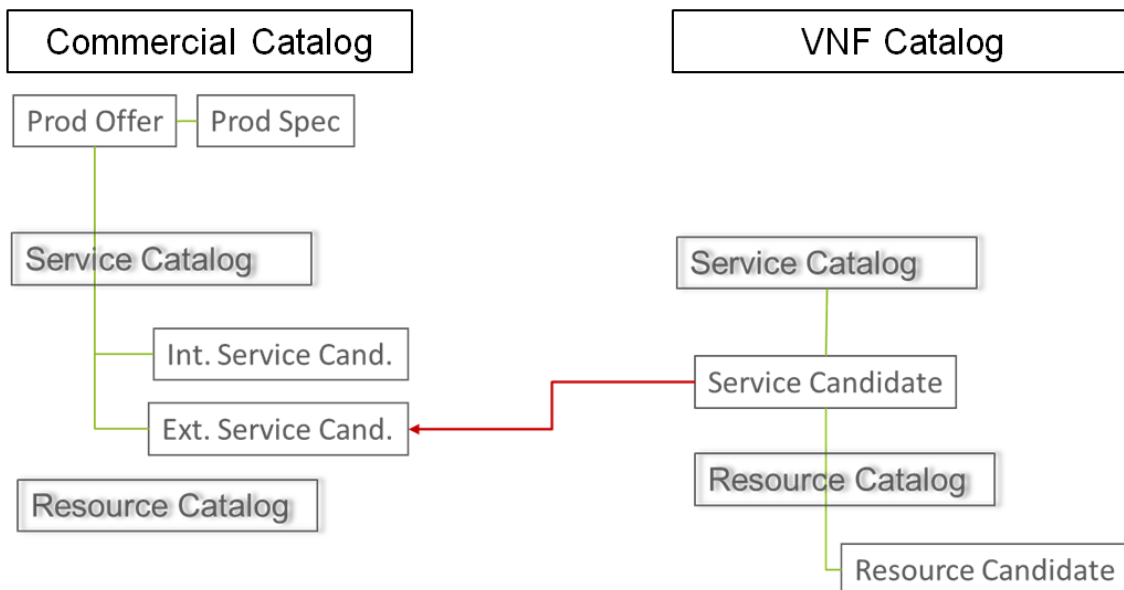


Figure: Commercial to VNF Catalog

Federation increases requirements

Catalog federation means that Catalog data (Product, Service, Resource, Pricing) is distributed across multiple systems. There is 1 master system for each piece of data,

The information being managed causes increased pressure where:

- Catalog is acting as both master and slave
- Catalog interface needs to fully support CRUD operations
- Catalog configurable workflow capabilities are very significant in this approach (as the data maintenance goes across systems for most of the processes)
- PLM processes are more complex, so PLM features are very important

Catalog Federation Business Processes R17.5

What are Catalog federation business processes?

Business Process Federation

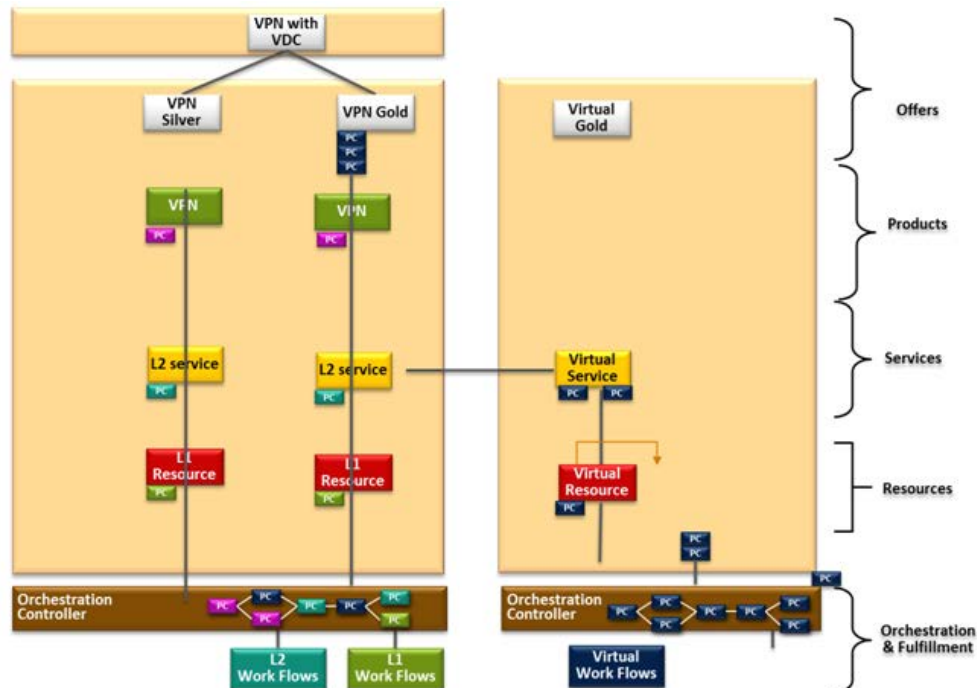
The transparent combination of multiple autonomous systems with seamless integration to create an ostensibly single process

- Idea to Offer
 - o Create product offers
 - o Associate fulfillment flows, assurance models, billing/charging, policies
- Order to Service
 - o Orchestrate multi party fulfillment
- Service to Cash

- o Charge for the service
- o Bill for the service
- o Execute policies on the service

Federated Business Processes can be a grouping of process flows, some executed in parallel and some executed in sequence, both with the objective to achieve some overall business goal (processing a customer order, rolling out network infrastructure, etc.).

Example Diagram:



PC is process component

Catalog management and API exposure R17.5

What APIs are involved in Catalog Lifecycle Management and Federation?

In Framework 17.5, the API reference implementation is available for:

- [Product Catalog](#)



[TMF620
Product Catalog
Mgmt API REST
spec.](#)

• Service Catalog	 TMF633 Service Catalog Mgmt API REST spec.
• Resource Catalog	 TMF634 Resource Catalog Mgmt API REST spec.
• Entity Catalog	 TMF662 Entity Catalog Mgmt API REST spec.
The APIs provide a means for adding partners' product offers into a Catalog. It allows to: • Feed digital ecosystem partners systems with the technical description of the available product offerings • Publish, expose and query for Component Services and Composite Product offerings in a standard and consistent way • Compose End-to-End Composition Product Offerings	
The following APIs are also available: • Party Role API • On-boarding API (partnership type) • Party Management API • Agreement API allows management of partner agreements (create, update, notification).	

Fulfillment Platform Capability R17.5

How do you fulfill Digital Component Service(s) and End-to-End Composite Products across the ecosystem?

Context

A partner of a digital ecosystem may browse the digital ecosystem catalog and discover a Component Service or End-to-End Composite Product that suits the product offering he wants to make available to users. He may


[Catalog Lifecycle Management &](#)

decide to deploy the discovered Component Service or End-to-End Composite Product. When instances of that deployed digital service are created, user and service configuration data needs to be send to the individual Component Service(s), as well as to the End-to-End Composite Product. This configuration could involve but is not limited to end user related information (authorization credentials, account establishment) or authorizing legitimate server addresses.	Federation Platform Capability
User story Movies4U is a partner of a digital ecosystem and has its movie streaming product offering published in the digital ecosystem catalog. The Movies4U Deployment Manager has deployed the product offering which exposes the offering on the Movies4U web portal. Jennifer has been looking for interesting movie streaming offers, discovers the Movies4U offering through a friend, and decides to try it out. She registers for an account on the Movies4U web portal, including payment method(s), the geographical area she resides in. She also opts for the targeted ads and completes a survey with areas of interest. This triggers the instantiation of the movie streaming Product Offering and subsequent configuration of the instance based on the information provided by Jennifer when registering. This includes but is not limited to the targeted ads opt-in/out attribute, the GoodAds ad service profile information and appropriate discount information. Upon completion, Jennifer receives a confirmation and is directed to the movie directory section of the Movies4U portal. Jennifer will receive – until she opt-out – targeted ads based on her location and areas of interest. Appropriate discounts will be applied.	
Desired features – The “Fulfillment Platform Capability” should: • Allow for setting configurable properties of a Composite Product instance within the constraints defined in the respective catalog item	
• Allow for setting configurable properties of a Component Service and Composite Product instances within the constraints defined in the respective catalog items	

Understand and execute the Composite Product management policy as defined in the catalog item	
<u>Consume the Management Interface of the Composite Product and/or Component Service instances as defined in the management policy, to set configurable properties</u>	  Types of APIs managed in the DSRA

The Fulfillment Platform Capability consumes the Management Interface of the Composite Product and/or Component Service instances as defined in the management policy, to set configurable properties.

Related TM Forum APIs and work – Following TM Forum API applies to this Platform Capability:	  TM Forum Ecosystem API Portal
- Digital Service Management API - Activation & Config API - Product Catalog Management API - Product Ordering API - Product Inventory API - Service Qualification API - Quote API - Address API - Appointment API - Service Ordering API - Service Catalog API - Service Test Management API - Change Management API - Service Inventory API - NFV Entity Provisioning API - Resource Ordering API - Resource Catalog API - Resource Inventory API - Resource Function Configuration & Activation API	

Frameworkx includes following definitions that relate to this "Fulfillment Platform Capability":	
• Service Order Management • Resource Order Management	
Gaps in existing work • None identified at this point.	

Customer Experience Management Platform Capability R17.5

How to manage Customer Experience across the ecosystem?

Description

The Customer Experience Management Platform Capability provides the necessary functionality for continuous monitoring of operational factors that qualify the customer experience, and take appropriate actions to provide a unique customer experience.

APIs exposed

The following APIs are available:

- SLA Management API
- Service Quality Management API

Assurance Platform Capability R17.5

How to know the health of the composite Product and of any underlying Component Services across the ecosystem?

Context

The DSRA platform hosts Services that are provided by multiple service providers, likely from across different industries. The intention is that a developer (who may, or may not, also be a service provider) can assemble several of these into a Product. In this instance, the developer must be able to know the health of any of the under-lying Component Services on which his product depends, and know the health of his product(s) – in effect the composite view.



[Definition of Terms](#)

A corollary of this is that a developer must be able to “trace” the health of his product to the underlying Component Services, and therefore drill down to understand the root cause of any problems. Similarly, it is likely that a service provider will want to know what E2E products his Component Service is a component of, since any availability or performance issues will impact all E2E products.

Even if the service provider does not know what products his Component Service is being used in, it is likely that he will be providing an SLA for his Component Service to the Digital Service Provider and will thus need to maintain a record of his customer and the values of the committed performance measures.

User Story

“GoodAds” pays developers for targeted ad insertions. It is a service to which a profile and ad request is sent, and based on that profile, the most appropriate ad is returned. “Movies4U” builds a product that sells premium streaming movies at a significant discount if users opt-in to its targeted ads.

The movie product periodically “calls” GoodAds to deliver ads that it inserts. The entire stream, with the movie and in-line targeted ads, is delivered over another service provider’s LTE network. If GoodAds is unavailable or has poor performance, Movies4U is affected, either in terms of revenues (no ad and payment) or user experience (delays mid movie).

Movies4U will need to discern whether unavailability or poor performance is a problem with GoodAds or the LTE network over which it is being delivered.



Desired Features

The intention of this section is to define what functionally is needed, while allowing significant latitude for various DSRA vendors to provide their own innovative solutions; “the what, not the how”). The DSRA Assurance Platform Capability should:

- Allow Component Service providers to provide a “health” feed for their service – whether the platform provider’s own running locally, or a remote feed from the 3rd party SP’s data center.
- Allow developers to view and/or link to those feeds
- Allow developers to aggregate individual feeds into a composite E2E health view of any product they develop on the DSRA platform.

• Allow Component Service providers to know what Products utilize their Component Services • Enforce proper permissions such that only legitimate service owners or those who consume/bought them have access to the Component Service.	
Some of the traceability data is delivered from a different service – the catalog management service, which should be the repository of the mapping between composite products and underlying components.	
Related TM Forum APIs and work Following TM Forum APIs exist that apply to this capability, and should provide further guidance. 1. The Digital Service Management API provides a RESTful structure and schema to expose simple health status of software-application based services such as are expected to dominate the DSRA. This applies to both the individual components and to composite E2E products. 2. The SLA Management API provides a RESTful structure and a suggested schema to expose the SLA compliance of services. This applies to both the individual components and to composite E2E products. 3. The Performance Management API provides standardized mechanism for performance management such as creation, partial or full update and retrieval of the resources involved in performance management (Measurement Production Job, Measurement Collection Job, and Ad hoc Collection). It allows also notification of events related to performance. 4. The Product Catalog Management API 5. The Product Inventory Management API 6. The Appointment API 7. The Address API 8. The Service Problem API 9. The Service Test Management API 10. The Change Management API	  TM Forum Ecosystem API Portal  Digital Service Management API  TMF623 LA Management API  TMF628 Performance Management API REST Specification

11. The Service Inventory API 12. The Service Quality Management API 13. The Resource Inventory API The most critical need is that each Service Component exposes health information such that a developer can create an algorithm/function (either on its own, off-platform or using a capability on platform) to understand the impact of each Component Service on its E2E product, and to view the E2E health of the Product as a function of those sub-Component Services.	
Gaps in existing work These two interfaces (Digital Service Management API and SLA API), together with catalog management that maintains the structure and traceability of Component Services into Products, provide the basics needed for developers to assure an E2E composite Product offer. However, there are areas that require more attention – either to extend the work or to ensure that they apply in a multi-tenant DSRA environment: • There could/should be a service that facilitates the assembly of component health indicators into an E2E Product health indicator • There could/should be a method to automatically authorize developers and service buyers for access to the information • There could/should be a method for 3rd party service providers to link their platforms' health API/indicator directly to the DSRA platform to drive either Digital Service Management API or SLA-API (reference, map/encapsulate, expose)	
	 TM Forum Ecosystem API Portal

Billing & Revenue Management Platform Capability R17.5

What is the "Billing & Revenue Management" Platform Capability?

The Digital Services Platform hosts Component Services that are provided by multiple service providers, likely from across different industries. For examples of component services, continue reading below. The intention is that a developer (who may, or may not, also be a service provider) can assemble several of these into a Product. To facilitate flexible commercial arrangements, and to implement both end-user and wholesale charging, several capabilities are desirable:



- [3GPP](#) defines an online charging system as performing:
 - a. Usage data collection (by records or real-time protocol)
 - b. Rating and various levels of discounting
 - c. Account balance management, and
 - d. Transfer of required data to next stage of billing and / or accounting systems.
- The component service owner should be able to define a price and business model for the use of its service. This might be a simple wholesale charge structure, or it could constrain both end-user and wholesale models. This is common, for example, in digital rights for content.
- A component service owner should be able to define the charging/business model parametrically, so that a developer can choose within valid options or ranges. An example is various levels of quantity commitment and associated discounts.
- A developer should be able, in many cases, to choose a wholesale model (price) and define a retail price independently. In many cases the end user price of a component is zero (it is an invisible part of the E2E product offer) and the end user sees only an overall price for the product. Money flows from the end user to the developer to the component service providers.
- The component owner or product owner should be able to define the unit of sale, rate, discounts, and method of usage measurement. Methods of usage measurement might range from simple (\$X for each use of the API) to more complex (\$Y for each video delivered without a network or server associated SLA violation, plus \$Z revenue from targeted advertising embedded in the stream). Note, in this example the \$Z revenue would be part of the charging/business model



service associated with a targeted advertising component service. • These attributes are configured via the catalog – they are commercial attributes of products at the component or E2E product offer level • This service must interact with some charging/rating function (possibly a real-time charging system, possibly an on-board function) and its capabilities will be limited to the functionality of that system.	
In addition, Invoicing functions enable streamlining and simplifying the invoicing and payment activities in the ecosystem allowing all service providers to rely on a common service, avoiding the burden of creating a new one and providing a consistent customer experience	
User Story 1 “GoodAds” pays developers for targeted ad insertions. It is a component service to which a profile and ad request is sent, and based on that profile, the most appropriate ad is returned. “Movies4U” builds a product - which is also a composite service, e.g.: made up of more than 1 component - that sells premium streaming movies at a significant discount if users opt-in to its targeted ads. The movie product periodically “calls” GoodAds to deliver ads that it inserts. The ads are delivered over another service provider’s LTE network. There are 3 top-line component services in this product: 1. the Movies4U streaming movie service; 2. the LTE network transport bundled as a wholesale service from the carrier to Movies4U, and 3. the targeted ad service, offered by GoodAds. The money flows as follows: 1. A consumer pays Movies4U \$X per movie; 2. GoodAds pays Movies4U \$0. Y0 per placed ad, which subsidizes the end-user price; and 3. Movies4U pays the LTE carrier \$Z per movie for the transport necessary for one streaming move. All 3 have SLAs with conditions. All 3 have options (quantity discounts, etc.). Movies4U uses the charging service to identify consumer usage of its movies, and rate (price) the movie to the consumer. GoodAds uses the charging service to identify wholesale usage of its ads (by many firms, including	

Movies4U), and to rate these ads based on the plan selected by, for example, Movies4U.

The network operator uses the charging service to identify usage of its network for a wholesale plan (purchased in this case by Movie4U), and to rate this usage. In all cases the results of rating are passed to the settlement process.

In all cases the seller of a service provides a set of pricing package options in the catalog.

The buyer of the service selects from these packages.

These are in fact catalog management functions and ordering functions. Users are provisioned (for access to movies and for access to the network) by the order fulfillment process, which is in turn driven by catalog data.

User story 2

- As a connected objects provider
- I need to be able to lend my objects without deploying a complex billing system
- So that I can reach an extended customer base
- To do this I need to benefit from an invoicing function allowing billing on behalf of myself



Desired Features

The intention of this section is to define what functionally is needed, while allowing significant latitude for various DSRA vendors to provide their own innovative solutions; ("the what, not the how"). This DSRA platform capability should:

- Perform the following functions (Usage collection, Rating, Account balance management and Transfer of required data)
- Allow Component Service providers to define a business model under which they offer their service, with parameters if desired and supported
- Allow product developers to view and configure parameters within bounds (if supported) of the services that they consume (wholesale relationship) within their E2E service/product
- Define the execution steps for triggers (if applicable), usage data collection, rating and account updating
- Allow product developers to define a business model (same as "a") for an E2E product offer
- Allow component service providers and E2E composite product developers to view the status of accounts (usage data and rated /discounted accumulators)
- Enforce proper permissions such that only legitimate service owners or those who consume/bought them have access to the Charging & Business model Definition Service.

- Billing & Settlement Account Administration including account management (creation, retrieval, update, deletion) and major information management such as contact information, Tax information, Payment method, Invoicing language & currency, etc.
- Billing Administration including billing Parameter set up (Billing frequency, groups, cycles, etc.)
- Customer & Partner Invoicing including summary & breakdown of Charges display, invoicing languages, currencies & rounding rules management, invoice content presentation & specific invoice formats management
- Customer & partner payment management including ability to set up several payment methods (check, credit card, online, etc.), customer balance management, payment allocation rejection & recycling management and interface with banking sector
- Creation of accounting events to be sent to the general ledger
- Partner Settlement Management including production of a settlement note for each partner / product and Production of a statement summarizing all settlement and invoices for a given partner
- Bill Calculation including non-recurring & recurring charges (NRC) rating management, Aggregation of rated billing items and discount calculation, invoice number generation, tax calculation, tax exemptions & invoicing currency management, etc.

Most of the exposed parameters – rate or rate ranges, discounts, quantity levels, etc. are configured indirectly via the catalog. These are essential components of a Product Offer in the catalog.

Related TM Forum APIs and work

Two [TM Forum APIs](#) exist that apply to this capability, and should provide further guidance.



 [TM Forum Ecosystem API Portal](#)

The [Billing Management API](#), described in [TMF636](#) . Needs succinct summary. Needs significant discussion because it tends to mix the functional portion and the management portions into one. For example, much of a billing service's configuration occurs either through the catalog (configuration of the business model) or through the fulfillment service (adding a user to the service so that charging methods apply). We also need to ensure that we separate charging from invoicing.



[TMF636 Billing Management API Rest Specification](#)

The [Usage Management API](#) described in [TMF635](#), has overlap with the usage collection and exchange portions of charging. Probably best to summarize and do "deltas" after we complete our definition.



[TMF635 Usage Management](#)

	API Rest Specification
The Customer management API to manage customer balance	 TMF629 Customer Management API
The Customer Bill Management API	 Customer Bill Management API
The Account API	 TMF666 Account API
The Payment Methods API to manage operation on payment such as acquisition, rejection, refund, etc.	TMF670 Payment Methods API
The most critical need is that each Service Component exposes a business model and relevant parameters such that money flows correctly either between the component service provider and an end user, or between a component service provider and a developer. In addition, it is critical to have such a service so that the developer can define a business model between the composite product offer and the end user. Note that one party may play multiple roles, but they are distinct from the perspective of the charging service.	
From an eTOM perspective, the invoicing function supports the following level 2 processes: • Bill Invoice Management & Support Bill Invoice Management • Bill Payments & Receivables Management and support Bill Payments & Receivables Management • Bill Inquiry Handling & support bill inquiry handling • S/P Settlements & Payments Management From a TAM perspective, the invoicing function covers the following applications / functionality: • Account Receivables • Billing Inquiry, Dispute & Adjustment Management • Billing Account Management • Bill Calculation	 Business Process Framework  Application Framework

Gaps in existing work

For the time being, TM Forum doesn't propose to define an invoicing and payment API.

4.3.2. PaaS Layer Technical Capabilities R17.5

What are the PaaS Layer Technical Capabilities?

The Actualization Platform may contain a set of Platform as a Service (PaaS) capabilities. In general platform services are consumed without explicitly managing the underlying Compute, Storage and Networking resources. In mature cloud platforms, particularly hyper-scale cloud platforms, the management of the underlying Infrastructure services supporting each PaaS capability are handled automatically by the cloud platform itself.

The specific set of PaaS capabilities is optional. They may consist of first party services provided by the platform supplier or third-party capabilities exposed via the platform via a PaaS Catalog. The PaaS examples below are therefore a representative sample and is not meant to be prescriptive.

Following PaaS Layer Technical Capabilities are considered:

- [Intelligence & Analytics Platform Capability R17.5](#)
- [Data Platform Capability R17.5](#)
- [Media Platform Capability R17.5](#)
- [Integration Platform Capability R17.5](#)
- [IOT Platform Capability R17.5](#)
- [Messaging & Hub Platform Capability R17.5](#)
- [Identity Management Platform Capability R17.5](#)
- [API Management Platform Capability R17.5](#)
- [Cyber Security Platform Capability R17.5](#)
- [Developer Platform Capability R17.5](#)
- [Monitoring & Management Platform Capability R17.5](#)
- [Web & Mobile Platform Capability R17.5](#)

Intelligence & Analytics Platform Capability R17.5

What are benefits of analytics from a business and operations perspective?

The value of data as an organizational asset is increasing. Beyond customers' personal information, subscriptions, usage, etc., there is also relevant customer data available from external sources, which can enrich



our view of every customer's world. Because all this data has a value, data may quickly become just as important (or more) as any other company asset.

Today, organizations need to be able to perform real time updates to deliver a foundational set of attributes on every customer, gathering data from all channels, and all segments. Aggregating data from classic Enterprise Data Warehouses those elements will include (but are not restricted to) gender, age, location, job/occupation, hobbies, preferences, brand reputation, lifestyle, products and services usage, purchases, QoS perception, lifetime value, income, acquisition, retention, etc.

A large part of so called [“Big Data or Black Hole”](#) promise is not new, and many analytics methods have been available for a long time to perform statistical analysis on business data. However, we have a new set of conditions, eventually pushing us to think differently:

- Availability of low-cost hardware infrastructure and software, that can be used for parallelized computation and data distribution;
- Large volumes of relevant unstructured data from a rising number of innovative digital experiences, producing a vast amount of data about most aspects of what our customers do;
- Urgent need to maximize network and IT operations efficiency, to reduce costs and remain competitive.



[Big data or black hole](#)

After the creation of a [data lake](#) or other similar large scale unstructured data repository, there will be a need to enable a great self-service experience to business users, including those that lack expert analytics skills but are domain experts. It also must be possible for them to test new data correlations. For this goal, data exploration and visualization options become most relevant, to put in evidence many new aspects of information, that otherwise would be unnoticed.



[Data lake](#)

Enabling experimentation with data is also considered an important factor of innovation: *“By collecting and analyzing big data, a large share of which is provided by Internet users (consumers), Internet companies are able to automate their processes and to experiment with, and foster, new products and business models at much a faster rate than the rest of the industry. The [advanced use of data and analytics](#) enables Internet firms to scale their businesses at much lower costs than other ICT firms”.*



[OECD Data Driven Innovation Interim Synthesis](#)

The capacity to meter and process this enormous amount of data in real-time and non-real-time from many different digital sources is essential for both the business (tailoring the service to the needs, marketing new services) and operations (quality of service and security).

The capacity to meter and process this enormous amount of data in real-time and non-real-time from many different digital sources is essential for both the business and operations.

Given the rapid evolution of the market, the TM Forum is pointing out the importance of the availability of an analytics service to the platform, and data structure for all ecosystem partners/services. This service can enable data to be gathered from across all services and products operating on the digital platform, and draws inferences and reports from it.

The DSRA recognizes that somewhere in the ecosystem, there must exist a set of data and storage services. These data and storage services are separate from the Intelligence & Analytics platform capability described here. Examples are SQL-data, DocumentData-a-a-Service, Unstructured Data, etc. Privacy and Net Neutrality may need to be considered when combining these two kinds of services.

This implies that:

- 1) each service (component) creates usage records and saves it in a common format, and
- 2) the analytics service itself has the ability – without compromising privacy – to analyze various subsets with various goals – technical, marketing, customer satisfaction, etc.

Analytics might also be from a user's point of view, a product or service's point of view, or some collection (e.g.; a set of users, etc.)

Note that other services may be inputs to this service – e.g.: assurance and traceability for QoE data to be analyzed.

Analytics services for digital services

Functional analytics

To help sustain digital ecosystems, digital ecosystem platform can offer a range of business / data analytics services. These Business / Data Analytics services would be available to participants in the digital ecosystem including developers when building applications and users for ad hoc analysis.

Building block capabilities may be required across the following areas:

- Collection and aggregation
- Validation and transformation
- Exploration and visualization
- Streaming

- Event hub

Some of the analytics functions that could be offered within a suite of DSRA analytics services are described below.

Data Catalog

Data Catalog service provides a system of registration and system of discovery for enterprise data sources. Provides capabilities that enable any user – from analysts to data scientists to developers – to register, discover, understand, and consume data sources.

Event Hubs

A highly scalable data ingress service designed to ingest millions of events per second so that one can process and analyze the massive amounts of data produced by your connected devices and applications. Event Hubs acts as the "front door" for an event pipeline, and once data is collected into an event hub, it can be transformed and stored using any real-time analytics provider or batching/storage adapters. Event Hubs decouples the production of a stream of events from the consumption of those events, so that event consumers can access the events on their own schedule.

Event Hubs is an event processing service that provides event and telemetry ingress to the cloud at massive scale, with low latency and high reliability. This service is especially useful for:

- Application instrumentation
- User experience or workflow processing
- Internet of Things (IoT) scenarios

Other key Event Hubs capabilities are behavior tracking in mobile apps, traffic information from web farms, in-game event capture in console games, or telemetry data collected from industrial machines or connected vehicles.

Data Factory

A cloud-based data integration service that orchestrates and automates the movement and transformation of data. Just like a manufacturing factory that runs equipment to take raw materials and transform them into finished goods, Data Factory orchestrates existing services that collect raw data and transform it into ready-to-use information.

Data Factory works across on-premises and cloud data sources and SaaS to ingest, prepare, transform, analyze, and publish your data. Going beyond just a tabular monitoring view using rich visualizations of Data Factory to quickly display the lineage and dependencies between data pipelines. Monitor all data flow pipelines from a single unified view to easily pinpoint issues and setup monitoring alerts.



Stream Analytics

A managed real-time event processing engine that helps to unlock deep insights from data. Stream Analytics facilitates setting up real-time analytic computations on data streaming from devices, sensors, web sites, social media, applications, infrastructure systems, and more.

Machine Learning

A service using computers to run predictive models that learn from existing data to forecast future behaviors, outcomes, and trends.

These forecasts or predictions from machine learning can make apps and devices smarter. For instance, when a person shops online, machine learning helps recommend other products the customer might like based on past actions and purchases. When a credit card is swiped, machine learning compares the transaction to a database of transactions and helps banks do fraud detection.

Machine Learning can not only provide tools to model predictive analytics, but could also provide a fully-managed service to deploy predictive models as ready-to-consume web services. Quickly create, test, operationalize, and manage predictive models without having to buy any hardware or manually manage virtual machines.

Big Data / Hadoop

Hadoop ecosystem of components, which includes Storm and HBase clusters, as well as other technologies under the Hadoop umbrella. Big Data refers to data being collected in ever-escalating volumes, at increasingly high velocities, and for a widening variety of unstructured formats and variable semantic contexts.

Big data describes any large body of digital information, from the text in a Twitter feed, to the sensor information from industrial equipment, to information about customer browsing and purchases on an online catalog. Big data can be historical (meaning stored data) or real-time (meaning streamed directly from the source).

For big data to provide actionable intelligence or insight, not only must the right questions be asked and data be relevant to the issues be collected, the data must be accessible, cleaned, analyzed, and then presented in a useful way.

Hyper-Scale cloud computing is often required to discover the relationships between anonymized data sets and create the reference model. The reference model is then run against actual data to see the specific results.

The next version of the DSRA Guide, will elaborate on the following capabilities:

- Collection and aggregation
- Validation and transformation
- Exploration and visualization
- Streaming

Data Platform Capability R17.5

How to optimize handling of different types of data across the ecosystem?

Description

Collection of PaaS capabilities optimized around handling different types of data possibly including:

- SQL database - manage relational SQL Database as a service,
- SQL data warehouse - Elastic data warehouse as a service with enterprise class features,
- Document database - Manage NoSQL document database as a Service,
- Redis Cache - High throughput, low latency data access to support fast and scalable applications,
- Data Factory - Orchestrate and manage data transformations and movement
- API Apps - Easily build and consume Cloud APIs.

Media Platform Capability R17.5

How to optimize handling of different types of media across the ecosystem?

Description

A logical grouping of PaaS capabilities focused on enhanced accessibility, distribution, and scalability of media content. This could include platform capabilities optimized for:

1. ingesting digital content from multiple live feeds,
2. storing and indexing,
3. editing, reformatting, Advert inserting, meta data synching for dual screen content delivery
4. organizing for live streaming or on-demand delivery via multiple channels to any device.
5. Cost-effective mechanisms to stream content to millions of users in audiences both local and worldwide all while securely protecting that content with industry standard digital rights management.

Additional platform capabilities often used in conjunction with Media Services include VPN, Replication, and CDN services.

Integration Platform Capability R17.5

How to optimize enterprise integration across the ecosystem?

Description

Tools and capabilities to facilitate Enterprise Integration across the ecosystem / cloud.
Categories of capabilities that fall under this category could be:

- Logic Apps - To automate the access and use of data across clouds without writing code.
- Service Bus - To connect across private and public cloud environments
- API Management - To publish APIs to developers, partners and employees securely and at scale.
- Data Catalog - To get more value from data assets by spending less time looking for data, registering data assets, discovering and unlocking data assets, capture tribal knowledge, bridge gaps between IT, Network and Business, control access, Integrate into existing tools and processes with APIs.
- Data Factory - To orchestrate and manage data transformation and movement.

IOT Platform Capability R17.5

How to optimize handling of various IOT characteristics across the ecosystem?

Description

A set of capabilities optimized to handle specific characteristics of IOT data and applications across the ecosystem:

- Development tools to accelerate IOT device connectivity,
- Hubs to connect, monitor and control billions of IOT assets,
- Event hubs to receive telemetry from millions of devices,
- Stream Analytics for real-time data stream processing from millions of devices,
- Machine Learning predictive analytics tools to enable scenarios such as predictive maintenance,
- Notification hubs for a scalable push notification engine for quickly sending millions of messages.

Messaging & Hub Platform Capability R17.5

How to manage various types of messaging across the ecosystem?

Description

The Messaging Platform function provides message queuing with two key concepts: queues and topics.

A queue implements a one-way communication whereby the sender posts a message in the queue and a receiver collects the message at his discretion. The queue is a

durable concept, as a message placed on a queue is saved and remains for a long time in the queue, in case the receive is no able to collect.

A topic is similar to a queue, but a message sent to a topic can have a copy of that message forwarded to multiple queues. Rules associated with the topic, determine which messages from a topic will go to each queue involved.

A Pub/Sub implementation provides many-to-many, asynchronous messaging that decouples senders and receivers and allows for secure, highly available and scalable communication between independently written applications.

The Hub function handles subscription requests and delivers notifications to subscribers in a pub/sub setup.

The Hub function validates subscription requests.

When a specific topic URL has been updated, the Hub platform capability will forward notifications to the end-points that subscribed to that topic.



[WebSub](#)



[Microsoft Azure Notification Hubs](#)

Identity Management Platform Capability R17.5

How to authenticate and authorize users across services, provider and technology domains?

The DSRA requires that participating parties utilize either SAML 2.0, OpenID Connect or both as a means for authenticating and authorizing users.

The DSRA requires that participating parties utilize either SAML 2.0, OpenID Connect or both as a means for authenticating and authorizing users.

Context

Security Assertion Markup Language 2.0 (SAML 2.0) is a version of the SAML OASIS standard for exchanging authentication and authorization data between security domains. SAML 2.0 is an XML-based protocol that uses security tokens containing assertions to pass information about a principal (usually an end user) between an identity provider and a web service. SAML 2.0 enables web-based authentication and authorization scenarios including single sign-on (SSO).

SAML enjoys the dominant position in terms of industry acceptance and production federated identity deployments. SAML is deployed in tens of thousands of M2M Single Sign-On (SSO) connections, and thousands of large enterprises, government agencies and service providers have selected it as their standard protocol for communicating identities across the Internet. More details about SAML can be found at https://www.oasis-open.org/committees/tc_home.php?wg_abbrev=security



[SAML](#)

For those DSRA participants electing to support SAML for federated identity management, it is recommended that the participants use SAML 2.0 to authenticate to an OAuth2 (<http://oauth.net/2/>) provider, and then access a given service using the OAuth2 token that is provided by the OAuth2 provider. An example of how to use the OAuth2 SAML bearer profile can be found at: <http://tools.ietf.org/html/draft-ietf-oauth-saml2-bearer-17>


[OAuth2](#)

[OAuth2 SAML](#)

[OpenID Connect](#) is an open standard that describes how users can be authenticated in a decentralized manner, eliminating the need for services to provide their own ad hoc systems and allowing users to consolidate their digital identities. OpenID Connect is the latest version of the OpenID Foundation standards and though the standard only began finalization in February 2014, broad adoption of OpenID Connect, this latest version of OpenID Foundation standards, has already occurred.


[OpenID Connect](#)

[OpenID Foundation](#)

OpenID Connect is an open federated identity standard targeted towards the consumer and mobile world, allowing individuals Single Sign-On (SSO) to "relying party" sites from an OpenID Connect provider such as their email provider, CSP or social network. Large OpenID Connect providers such as Google and Microsoft allow all their users to authenticate using OpenID Connect. OpenID Connect is one of few federated identity standards that enable SSO without the need for a pre-existing relationship between the identity provider and the relying party, a feature that greatly fosters scalability. The OpenID Foundation web site provides a wealth of details about the standard: <http://openid.net/connect/>.

[OpenID Connect is one of few federated identity standards that enable SSO without the need for a pre-existing relationship between the identity provider and the relying party, a feature that greatly fosters scalability.](#)

OpenID Connect is an identity layer that operates on top of the OAuth2 protocol. As with the discussion above regarding using SAML 2.0, the result of successfully authenticating a user's identity with OpenID Connect is an OAuth2 token from the OAuth2 provider that can then be used to assert the user's identity across all the federated services.

[Through federated identity management, each service can interact with the user and the other services in a manner that simultaneously delivers a seamless user identity experience across the services while at the same time providing the appropriate level of protection to each service.](#)

The DSRA recognizes that all the participating parties operate in an ever increasingly connected fabric of digital services. Across all these services a user has an identity across each of the services and no one service can assert to be the sole provider of the user's authentication and authorization. Through federated identity management, each service can interact with the user and the other services in a manner that simultaneously delivers a seamless user identity experience across the services while at the same time providing the appropriate level of protection to each service.

"OpenID Connect" should not be confused with "OpenID". OpenID 2.0 is an earlier version of the OpenID Foundation standards. While OpenID Connect performs many of the same tasks as OpenID 2.0, it does so in a way that is API-friendly. OpenID Connect can be extended to include more robust mechanisms for signing and encryption. And OpenID Connect has OAuth 2.0 capability built into the protocol itself, whereas OpenID 2.0 required an extension for integration of OAuth 1.0a.



[What is OpenID Connect?](#)

Related to the Identity Management Platform capabilities, there are the Profile Management and Privacy Management services described in:

- [Privacy Management Function 17.5](#)
- [Profile Management Function R17.5](#)

User Story

Continuing the Movies4U use case, Movies4U is a partner of a digital ecosystem and has its movie streaming product included in a broad set of Provider offerings. Pre-condition of the Provider and Movies4U federating their identity services. Suzie is a customer of the Provider and signs up for the Movies4U service. Suzie authenticates to the Provider and selects the registration link for Movies4U. Suzie is asked if it is acceptable for the Provider to share details with Movies4U. Suzie confirm the sharing of the data and the Provider provides identity information to Movies4U along with an assertion that Suzie is really Suzie (no passwords are shared). Movies4U only asks for any remaining details needed to establish the service, identity was pre-filled and is not editable by Suzie. Suzie completes the registration process with Movies4U and begins to receive the service.

When Suzie attempts to access Movies4U, Suzie is prompted for the identity and password for the Provider. Movies4U passes this identity request to the Provider who asserts that Suzie is in fact who she claims to be. The Provider assertion is provided to Movies4U and Movies4U accepts the assertion as authentication. Suzie can now access the services of Movies4U. If Suzie has a valid cookie established from an

earlier login to the Provider's portal, Suzie is not prompted and the authentication happens entirely behind the scenes.

Additionally, a 3rd party intermediary can also be inserted into this chain. The Provider, Movies4U and now Facebook can all federate their identity services. When Suzie logs into the Provider, Suzie can provide her Facebook credentials and select a button that is labeled "Login with Facebook". Suzie is prompted by Facebook if it is OK for the Provider to have access and perform this action. Suzie confirms the Provider's access and Facebook responds to the Provider with an assertion that Suzie is Suzie. Now Suzie selects the Movies4U service like was performed above. Suzie wishes to subscribe to the service from the Provider's portal. Suzie selects the service and again is asked by Facebook if she agrees that Movies4U can have access and perform this action. Suzie confirms and the same process occurs with Movies4U gaining an assertion that Suzie is who she reports to be. Movies4U completes the registration process.

When Suzie access a movie from Movies4U in this scenario, if Suzie has a valid Facebook cookie then the authentication process happens entirely without Suzie's knowledge and Suzie immediately is presented with the Movies4U landing page. The advantage of a 3rd party intermediary such as Facebook is that Suzie is much more likely to have a valid cookie / session established with Facebook and as a result the overall experience is cleaner and simpler. The disadvantage is that the Provider does not have an opportunity to capture an impression with Suzie by having Suzie first login to the Provider's portal; the Provider can be dis-intermediated.

Privacy Management Function 17.5

How to manage personal identifiable information across providers?

Managing user-related data life cycle across several providers is a challenge, particularly in modern services, which require an increasing volume of data (more or less "personal"), and an increasing number of parties providing parts of the service. It becomes very complex for the end-user to understand which provider handles which part of the service. An example of such a complexity is online advertising, comprising the display web site, ad brokers, ad networks, ad servers, etc.

An example of such a complexity is online advertising, comprising the display web site, ad brokers, ad networks, ad servers, etc.

The principles that should be shared by DSRA participants are as follows:

- The DSRA Participant must inform the end-user about any personally identifiable information implied in its service. The provider has the responsibility to identify Personal Information handled by its service, the data flow, the interaction with internal or external parties, the data lifecycle from the creation/collection of data up to its deletion.

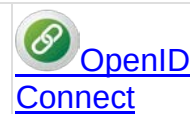
- The DSRA Participant must give accurate details about how the PII will be managed and used. This must be done using TM Forum TR243 Privacy Management specification, particularly the SID model including Privacy Profile Types and Privacy Profile. Depending the laws and regulations enforced within a geographical location, it will likely need to specify: - Who can access the PII: internal team: from marketing, commercial, technical, help desk, etc., external parties: supplier, provider, etc. - What the purpose is for the usage of this PII, related to the party - Where the data is kept and processed - How the user (data Subject) can access it, make modifications to it. - How long it can be used or kept for the usage/party.	 TR243 Privacy Management  Information Framework (SID)
This must be done using the TM Forum TR243 Privacy Management specification, particularly the SID model including Privacy Profile types and Privacy Profile.	
The DSRA Participant must provide the end-user the ability to 'opt-in' or 'optout' for usages of PII. Consent screens can be used to collect user choices, and the Dashboard to display current user settings and propose new 'opt-ins' or 'optouts'.	
When the PII comes from another DSRA Participant, in accordance with the Profile Management rules described above, the recipient must ensure the privacy choices of the end-user are correctly applied and enforced.	 Profile Management Function
The DSRA Participant must implement privacy APIs such as the TM Forum Privacy Management API, so that synchronization can be done between the DSRA Participants, and propagation of user choice regarding a PII can be done. Batch mechanisms can also be used for synchronization purposes. It enables the creation of an over-the-top privacy management dashboard, that can simplify end-user configuration.	 TMF644 Privacy Management API REST Specification
To ensure the "right to be forgotten", the API can be used to delete access rights to information, and the	

information itself. In the case of the DSRA Participant having to store locally PII coming from another DSRA participant (e.g.: for technical reasons), synchronization of the Privacy Profile can be an efficient way of enforcing the right to be forgotten across a chain of DSRA participants relying on the same user identity and profile information.

Profile Management Function R17.5

How to retrieve profile and graphing data across providers?

At the time of the DSRA publication, unlike Federated Identity with OpenID Connect, there is not a widely adopted standard that provides a single mechanism to retrieve profile and graphing data across providers. The Profile Management function is not required for DSRA participation.

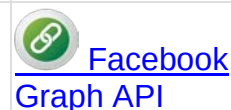


Context

There is not a widely adopted standard that provides a single mechanism to retrieve profile and graphing data across providers. The Profile Management function is not required for DSRA participation.

Viewed as a spectrum of capabilities, as more and more elements are included then the overall user experience and value all the parties involved increases. A significant number of very successful partnerships and eco-systems today are not federating profiles. So great success can be had if Profile Management is excluded, however as much as federated identity has become a given over the past 5 to 7 years, federated profiles are going to quickly become expected. While identity management and profile management are related and sometimes implemented together, this does not mean that they are the same nor should they be combined.

Facebook's Graph API is implemented by itself separate from identity and is a good example of a federated profile management system. In the end, ecosystems can exist without identity management and profile management, having these two capabilities in place results in a richer overall user experience.



Ecosystems can exist without identity management and profile management, having these two capabilities in place results in a richer overall user experience.

Under [IETF](#) leadership [SCIM](#) is a published standard for federating profiles with version 2.0 under development and version 1.1 released in 2012. SCIM 2.0 holds strong promise to address many of the needs of a DSRA participant around Federated Profile Management; DSRA participants are strongly encouraged to monitor and participate with the work IETF is doing with SCIM. With the publication of the DSRA the question of SCIM being able to address all profile information across an extremely diverse set of cloud and service providers is yet to be answered. DSRA participants should expect SCIM v2.0 to be widely adopted for general or basic profile information about users, with proprietary APIs from the various providers continuing to be used for more elaborate or domain specific data the provider makes available regarding the user (e.g. Twitter's graph API for follower count, last tweet, etc.)



[IETF](#)



[SCIM 2.0](#)

DSRA participants agree to three areas Federated Profile Management. First, the DSRA participant agrees that if the participant gathers profile or graphing data about the user, the DSRA participant shall provide an API to allow other providers to consume the DSRA participant profile data. If the DSRA participant does not already have a widely adopted basic profile API, then supporting SCIM v2.0 is suggested. If the DSRA participant has an existing widely supported basic profile API, one of the possible implementations of the API can be [REST](#) over HTTP with [JSON](#) queries.



[REST](#)



[JSON](#)

The DSRA participant agrees that if the participant gathers profile or graphing data about the user, the DSRA participant shall provide an API to allow other providers to consume the DSRA participant profile data.

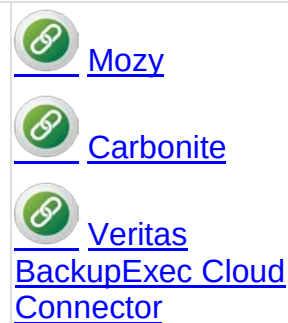
The second, the DSRA participant agrees that DSRA participants do not persist a 3rd party's profile or graphing data. Short-term caching a 3rd party's profile and graphing data for performance or network optimization is acceptable, during the browser session and for some short period after, however persisting 3rd party profile data is prohibited under the DSRA. DSRA participants agree to retrieve the references provided by the profile and graph API, effectively retrieving elements as they are needed for the user experience at run-time from the source provider of the data.

The DSRA participant agrees that DSRA participants do not persist a 3rd party's profile or graphing data.

Finally, it is recommended that each DSRA participant, where possible, goes beyond the basic profile and federates the rich profile using the provider's proprietary profile or graphing API if available. Until there is wide adoption of a single standard API, like SCIM, that include the rich volume of data found in the various proprietary APIs, the richness of the experience is at its best when elements of the user's profile from various cloud providers are included and blended into the session that the DSRA participant is creating. DSRA participants agree at a minimum to retrieve basic profile information from all providers to whom the DSRA participant federates identity. When possible and practical the DSRA participant agrees to go beyond the basic profile data and include the richest possible profile and graphing data possible from the provider.

Each DSRA participant, where possible, goes beyond the basic profile and federates the rich profile using the provider's proprietary profile or graphing API if available.

There are many examples where a federated identity between the DSRA participant and the cloud provider is available and practical, however a federated profile is not needed or available an online backup service (e.g. [Mozy](#), [Carbonite](#), [Veritas BackupExec Cloud Connector](#)) for example. A DSRA participant can deliver a strong user experience when brokering or including an online backup service by providing a federated login to the online backup service. However, retrieving a user's online backup profile may not be possible; neither with proprietary nor standard profile federation APIs like SCIM. In addition, the online backup provider may have little need to provide an API for profile federation, and the usefulness of federating profile data about online backup could be questioned. So, while federated ID is required under DSRA, both providing and consuming a profile is strongly suggested but not required by the DSRA.



User Story

Continuing the use case example of Movies4U, Suzie is now accessing the Movies4U as a service from her Provider. Suzie is prompted by the Provider's web site if she would like to see what movies at TV shows her Facebook and Twitter contacts are talking about. Agreeing to the request, the Provider contacts Facebook on behalf of Suzie and a Facebook redirect page is presented asking Suzie if she gives permission for the Provider to access this information. The Provider advises Suzie to check back later for results. Suzie agrees and the Provider uses the Facebook graph API to perform a search across Suzie's Facebook content for recent posts about movies. The Provider performs the same action

against Twitter, looking for movie titles or specific keywords.

Returning to the Provider page, Suzie sees a listing of the movies and TV shows that her contacts have been talking about on various social feeds. A posting from Suzie's friend Stan shows that Stan just saw the Clint Eastwood movie The Good, The Bad and The Ugly last week. Next to this Facebook posting that is rehosted on the Provider's site are two buttons labeled "Watch now on Movies4U" and "Add to Later on Movies4U".

Scrolling down the feeds, Suzie sees a Twitter post from her co-worker Jill that a new preview about Season 9 of Game of Thrones is on Pay Per View / On-Demand. Next to this twitter posting is a button labeled "Watch now on TV". Suzie, presses that button on the web page and sees that her TV changes from the Real Madrid FC recap that she was watch to the Game of Thrones Season 9 preview on PPV.

API Management Platform Capability R17.5

How to manage APIs across the ecosystem?

Context

A "Policy based routing" function of the API management capability, allows late binding of client applications to services accessible through APIs. The API endpoints can change over time and can be reached thanks to the policy based routing service. This indirection allows different policies such as load balancing, fail over, hand over, context based API calls (e.g. finding an API implemented by different service providers according to the mobile number).

The policy based routing function abstracts a client application from a set of different resources which could be represented by different endpoints, all exposing the same API type.



User story

A mobile client application may need to access different user profiles provided by the customer communication service providers. The mobile application needs to work across the different service providers in the same region. It will call a generic profile API exposed via the policy based routing service which then will use the mobile number to find out the right communication service provider and the corresponding profile API. If a new communication service provider joins the ecosystem, it



will be fully transparent for the client application. No change will be needed on the client side.	
API Management helps organizations publish APIs to external, partner and internal developers to unlock the potential of their data and services. Businesses are looking to extend their operations as a digital platform, creating new channels, finding new customers and driving deeper engagement with existing ones. API Management Services aims to provide the core competencies for a successful API program through developer engagement, business insights, analytics, security and protection. This platform capability would be used by the Actualization Platform to manage all the APIs used throughout the platform. These APIs would then be published via a Catalog Service in the Business Layer.	

Description

The policy based routing service abstracts a client application from a set of different resources which could be represented by different endpoints, all exposing the same API type.

The policy based routing service abstracts a client application from a set of different resources which could be represented by different endpoints, all exposing the same API type. The decision to route to one endpoint or the other is taken by a gateway supporting the "policy based routing service" according to a policy. A policy is a way to define the expected routing behavior without impacting the code of the client application. Typical routing policies are:

- load balancing: a load balancing policy will route the different API calls to different endpoints to parallelize processing and optimize API throughput
- fail over: a fail over policy will define how a server node is elected in case of a node failure and how the APIs calls can be rerouted to the right server node
- profile based policy: a profile policy will retrieve some attributes of the user profile (e.g. the mobile number of the device supporting the client application) and will route the calls to the right API (e.g.: in the case the right API of the right communication service provider)

The policy based routing service also provides a way to upload and configure new policies and attach policies from a policy catalog to a specific API and use case.

The policy based routing service is in fact a specific case of a more general "API management" pattern which defines how management policies can be attached to APIs and not only for routing. We should probably define "API management" as a transverse foundation of the DSRA and distinguish it from "digital service" management services such as identity, profile, configuration

and activation, invoicing etc. which can be described using TM Forum APIs, management policies and the API management pattern.

"API management" is now a very important trend. It is not addressed in the frameworks. TM Forum should probably define it, articulate it with service management. Many of the "digital service management" problems will use the "API management pattern" and can be explained using a policy based API management architecture. DSRA is probably the best place to do that. We should involve API management vendors to join TM Forum and help us provide an industrial specification.

The policy based routing service is in fact a specific case of a more general "API management" pattern which defines how management policies can be attached to APIs and not only for routing.

Desired features

API Management deals with the Inventory and Lifecycle Management of APIs used throughout the platform including version control and policies governing usage and charging.

In case of an API wholesale collaboration model, there is the requirement that the "backend" API provider should be hidden from end users and application developers.

If we take the example of a model whereby there is an end user, a content provider, an operator - as a wholesaler - and a resource service provider - the wholesale customer. In this case, it may be necessary that the end user and the content provider are not able to identify the resource service provider.

Cyber Security Platform Capability R17.5

How to manage cyber security across the ecosystem?

Description

A cloud platform should provide security and privacy as a built-in priority. The platform itself should be built using software development processes that embed security requirements into the entire software lifecycle, from planning through deployment. An example is the [Microsoft Security Development Lifecycle \(SDL\)](#)

 [Microsoft Security Development Lifecycle \(SDL\)](#)

Platform operational activities should follow the similar security priorities. A documented example is the [Microsoft Operational Security Assurance process \(OSA\)](#).

 [Microsoft Operational Security Assurance process \(OSA\)](#)

Security should be able to first protect from and then detect threats. Platforms are increasingly able to provide very

robust security protection perimeters but they can never be totally impenetrable. Platforms need to supplement protection with efficient and fast reacting detection mechanisms. Passive and active countermeasures can then mitigate and defeat threats.	
Consumer Privacy is another aspect to the cybersecurity regulatory challenge. Over the past year the TM Forum has focused on privacy as driven primarily by EU legislation (general data protection legislation GDPR) which focusses on giving citizens control over their data and places requirements on organizations collecting data to handle it appropriately (protection, access, etc.).	
Some specific cyber-security capabilities that should be present on mature, cybersecurity enabled platforms include:	
Managing and controlling identity and user access to environments, data, and applications by federating user identities to and enabling multi-factor authentication for more secure sign-in. Biometric capabilities on devices such as fingerprints or artificial intelligence enhanced facial recognition enable stronger identity and role-based security.	
Encrypting communications and operation processes. For data in transit use of industry-standard transport protocols between user devices and datacenters and within datacenters themselves. For data at rest, a wide range of encryption capabilities up to AES-256, giving the flexibility to choose the solution that best meets each need.	
Securing networks. Infrastructure necessary to isolate tenants and to securely connect virtual machines to one another both with within one datacenter (such as with CLOS VL2) and between multiple, networked datacenters as in hybrid cloud use cases. Capability to block unauthorized traffic to and within datacenters, using a variety of technologies. Software Defined Virtual Networking to extend on-premises networks to the cloud through site-to-site VPN.	
Managing threats. To protect against online threats, Antimalware for cloud services and virtual machines. Robust intrusion detection, denial-of-service (DDoS) attack prevention, regular penetration testing, and data analytics and machine learning tools to help mitigate threats to the platform.	
Protecting the privacy of Customers. Time-tested approaches to privacy and data protection including	

maintaining organizations' ownership of and control over the collection, use, and distribution of their information.	
Owning your own data. Customers own customer data—that is, all data, including text, sound, video, or image files and software. Owners should be able to access their customer data at any time and for any reason without assistance. Customer data or derive information from it should not be used for advertising or external data mining.	
Trust in the Rule of Law for responses to government and law enforcement requests to access data. When a government wants customer data—including for national security purposes—it must follow the applicable legal processes, in the applicable jurisdiction when serving a court order for content or a subpoena for account information.	

Developer Platform Capability R17.5

How to facilitate development across the ecosystem?

Description

A set of capabilities and tools for teams to share code, track work, ship software:

- DevOps tools and capabilities for implementing applications and services on a cloud platform whether leveraging PaaS, IaaS or even Bare Metal. Tools in this category could include Visual Studio or Eclipse, container hosting environments with industry standard API endpoints for your chosen orchestrator such as Microsoft Service Fabric, DC/OS, Docker Swarm, or Kubernetes,
- DevTest environments,
- Application Insight tools for detection, triage, and diagnosis of issues in web apps and services.
- Tools to deploy mobile apps, collect feedback, crash reports and monitor usage from a developer point of view.

Monitoring & Management Platform Capability R17.5

How to manage and monitor tenant and platform resources across the ecosystem?

Description

PaaS capabilities and tools to facilitate Build, Management and Monitoring or all tenant and platform resources from one console. Capabilities a platform could offer under this category are:

1. Management Portal to facilitate Build, Management and Monitoring of all tenant and platform resources from one console,
2. Resource Manager to simplify the specification of resource requirements for applications, services and components,
3. Application Analytics tooling to detect, triage, and diagnose issues with apps and services,
4. Log Analytics to collect, search, and visualize machine data from private and public clouds,
5. Automation to simplify cloud resource management with process automation,
6. Backup for all applications, servers, VMs and Containers,
7. Site Recovery to orchestrate protection and recovery of private clouds,
8. Scheduler to run jobs on simple or complex recurring schedules.

Web & Mobile Platform Capability R17.5

How to facilitate the development and deployment of digital services across the ecosystem?

Description

Tools and capabilities to facilitate and automate the development and deployment of Applications Services including:

- Web Apps - Quickly create and deploy mission critical Web Apps at Scale.
- Mobile Apps - Build and host the backend of any mobile application.
- Logic Apps - Automate the access and use of data across clouds without writing code.
- API Apps - Easily build and consume Cloud APIs.

4.3.3. IaaS Layer Technical Capabilities R17.5

What are the IaaS Technical Platform Capabilities?

IaaS Technical Platform Capabilities are for those situations where the solution architect and developer are going to have to not only design the software architecture but take an explicit role defining the Compute, Storage and Networking resources to run their applications. Unlike PaaS where the developer can simply specify the services to be consumed and performance targets desired and let the platform allocate and deallocate resources automatically, in IaaS the developer assumes responsibility for the entire design and its operational configuration.

IaaS Platform Capabilities include:

- [Compute Platform Capability R17.5](#)
- [Networking Platform Capability R17.5](#)
- [Storage Platform Capability R17.5](#)

Compute Platform Capability R17.5

How to provision virtual machines across the ecosystem?

Description

Platform capabilities designed to provision Virtual Machines including various CPU and GPU configurations. Capabilities in this category could include:

1. Linux VMs
2. Windows VMs
3. Virtual Machine Scale Sets
4. Web Apps
5. Container Services
6. Functions that process events with server less code
7. Batch to run large-scale parallel and batch compute jobs

Networking Platform Capability R17.5

How to provision network services across the ecosystem?

Description

Set of IaaS capabilities to facilitate the rapid provisioning of network resources. Examples in this category could include:

1. Virtual Networks
2. Load Balancers to deliver high availability and network performance to applications / Network Functions.
3. Application Gateways
4. VPN Gateways
5. CDN
6. Traffic Manager to route incoming traffic for high performance and availability
7. MPLS Connectivity Services
8. SD-WAN Connectivity Services.

Storage Platform Capability R17.5

How to provision storage resources across the ecosystem?

Description

Set of platform services designed to facilitate the rapid provisioning of various types of storage including Blob Storage, Queue Storage, File Storage, and Premium low latency, high throughput storage, Big Data Analytics repositories, storage backup and site recovery / orchestration.

4.4. API's mapped on DSRA Platform Capabilities R17.5

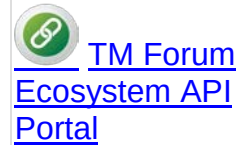
How are TM Forum APIs mapping to the DSRA Platform Capabilities?

APIs	Billing & Revenue Mgmt	Assurance	Customer Experience Management	Fulfillment	Catalog Lifecycle Mgmt & Federation	Intelligence & Analytics	Identity / Security	Integration Services	API Management	Monitoring / Management	DevOps Tools	Messaging / Hubs	IOT Services
Digital Service Mgmt API	✓	✓		✓	✓	✓		✓	✓	✓	✓	✓	✓
Activation & Config API				✓									
SLA Mgmt API		✓	✓							✓			
Customer Bill Management API	✓												
Account API	✓												
Customer Mgmt API	✓			✓									
Payment Methods API	✓												
Product Catalog Mgmt API		✓		✓	✓						✓		
Party Role API					✓						✓		
On-boarding API (partnership type)					✓						✓		
Party Mgmt API					✓						✓		
Agreement API					✓						✓		

APIs	Billing & Revenue Mgmt	Assurance	Customer Experience Management	Fulfillment	Catalog Lifecycle Mgmt & Federation	Intelligence & Analytics	Identity / Security	Integration Services	API Management	Monitoring / Management	DevOps Tools	Messaging / Hubs	IOT Services
Usage Mgmt API	✓												
Performance Mgmt API		✓											
Product Inventory Mgmt API		✓		✓									
Product Ordering API				✓									
Identity API / Open ID Connect					✓		✓				✓		
Privacy Management API					✓		✓				✓		
Service Qualification API				✓									
Quote API				✓									
Address API		✓		✓									
Appointment API		✓		✓									
Service Problem Management API (SPM)		✓											
Service Ordering API				✓									
Service Catalog API				✓	✓						✓		
Service Test Management API		✓		✓							✓		
Change Management API		✓		✓							✓		
Service Inventory API		✓		✓									
Service Quality Management API		✓	✓										
NFV Entity Provisioning API				✓									
Resource Ordering API				✓							✓		
Resource Catalog API				✓	✓						✓		
Prepay Balance Management API	✓												
Entity Catalog API				✓							✓		
Resource Inventory		✓		✓									
User Roles & Permissions API											✓		

APIs	Billing & Revenue Mgmt	Assurance	Customer Experience Management	Fulfilment	Catalog Lifecycle Mgmt & Federation	Intelligence & Analytics	Identity / Security	Integration Services	API Management	Monitoring / Management	DevOps Tools	Messaging / Hubs	IOT Services
Resource Function Configuration & Activation API (FX 17.0)				✓							✓		

See [Open API Table \(archive\)](#) and [TM Forum Ecosystem API Portal](#)



4.5. DSRA Standards Index R17.5

What standards are relevant for the DSRA?

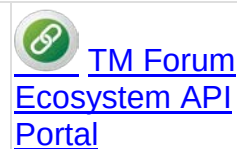
Standard	Relevance	Reference	Comments
SAML 2.0	Security Assertion Markup Language 2.0 (SAML 2.0) enables web-based authentication and authorization scenarios including single sign-on (SSO).	https://www.oasis-open.org/committees/tc_home.php?wg_ab_brev=security	
OAuth2	For those DSRA participants electing to support SAML for federated identity management, it is recommended that the participants use SAML 2.0 to authenticate to an OAuth2 provider, and then access a	http://oauth.net/2/	

Standard	Relevance	Reference	Comments
	given service using the OAuth2 token that is provided by the OAuth2 provider.		
OpenID Connect	OpenID Connect is one of few federated identity standards that enable SSO without the need for a pre-existing relationship between the identity provider and the relying party, a feature that greatly fosters scalability.	http://openid.net/connect	
SPML	Service Provisioning Markup Language (SPML) is the open standard for the integration and interoperation of service provisioning requests.	https://www.oasis-open.org/standards#spmlv2.0	
SCIM	The SCIM standard was created to simplify user management in the cloud by defining a schema for representing users and groups and a REST API for all the necessary CRUD operations. It standardizes the interactions with a directory server	http://www.simplecloud.info/	

Standard	Relevance	Reference	Comments
	enabling user provisioning.		
SOAP	DSRA can use SOAP to integrate with different Partner existing information system and then transform it into REST when deliver it to ender user or partner.	http://www.w3.org/TR/soap/	
JSON	DSRA can use JSON for resource model and exchange format in DSRA API, it is easy for manipulating and presentation using java, PHP? HTML/JavaScript ? JSON will reduce the information exchange overhead.	http://json.org/	
WS-*	WS-* is popular in enterprise computing environment, but not fit for light weight internet style and resource constrained environment.	http://www.w3.org/2002/ws/	
REST	REST is recommended as API style in DSRA	http://en.wikipedia.org/wiki/Representational_state_transfer	
SM-API	The Simple Management API (SM-API) enables the creation of a single management	https://www.tmforum.org/DeveloperPackSimple/14254/home.html	

Standard	Relevance	Reference	Comments
	interface for any digital service, independent of platforms, networks or other resource specific technologies. It provides a simple set of operations to allow services to interact with management systems anywhere in the value chain in a consistent way.		

See also the [TM Forum Ecosystem API Portal](#).



4.6. Web Technology Utilization R17.5

What protocols and standards are relevant for the DSRA?

SOAP DSRA can use SOAP to integrate with different Partner existing information system and then transform it into REST when deliver it to ender user or partner.	 SOAP
JSON DSRA can use JSON for resource model and exchange format in DSRA API, it is easy for manipulating and presentation using java, PHP? HTML/JavaScript? JSON will reduce the information exchange overhead.	 JSON
WS-* WS-* is popular in enterprise computing environment, but not fit for light weight internet style and resource constrained environment.	 WS-*

REST

[REST](#) is recommended as API style in DSRA

Advantage for using REST

1. REST API can reduce transmission overhead which fit for the mobile network/app, thus saving the battery life
2. Using standard HTTP operator like PUT, POST, GET, DELETE etc. make development easier and get familiar with DS API.
3. Easy transformation. Using API Broker, or API Orchestration, SOPA, WS-* interface can be easily transformed to REST API (using XSLT or scripting).



[Representational
state transfer
\(REST\)](#)

5. Business Scenarios R17.5

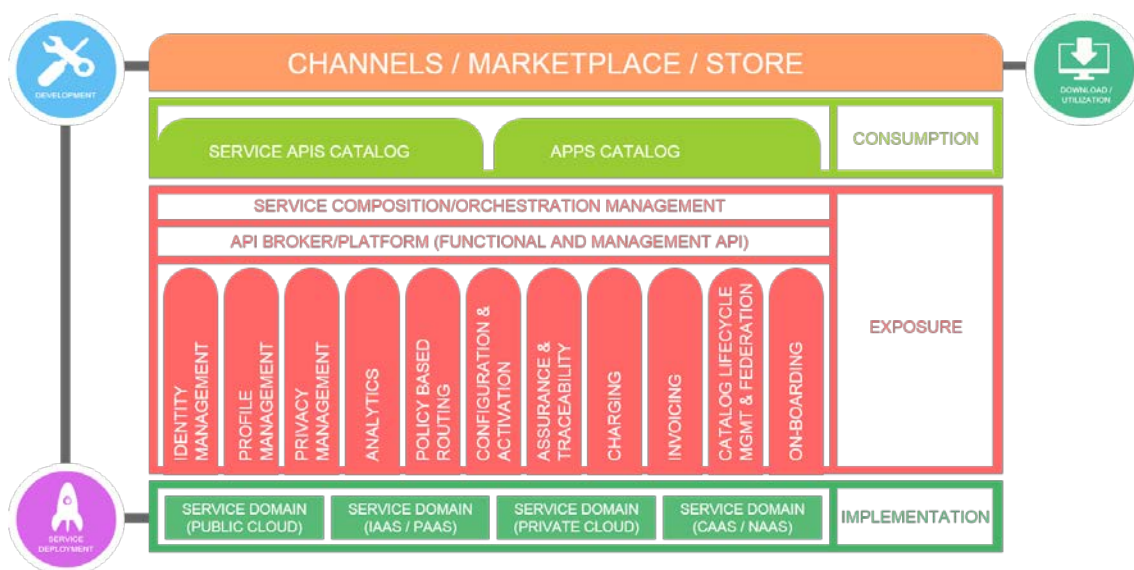
What are business scenarios that implement DSRA concepts?

In this chapter we will illustrate the DSRA concepts, by means of a set of business scenarios. These scenarios were developed in the context of the [B2B2X workstream](#) of the Internet of Everything program.



[B2B2X](#)

These scenarios were developed against the version 16.5 of the DSRA, where 11 Support Services were identified as shown in the EXPOSURE layer of the diagram below.



With the definition of the [Digital Platform Reference Architecture](#), Support Services have either been reclassified as [Platform Capabilities](#), or functions within a Platform Capability. Also, in [DSRA R17.0](#), the number of Platform Capabilities were extended.

This chapter however is still referring to the [DSRA 16.5](#) Support Services. As a reading guide, the following table provides the mapping:



[IG1157 DPRA Concepts & Principles](#)



[DSRA Platform Capabilities](#)

v16.5 Support Service	Change	v17.5 Platform Capability
Identity Management	No	Identity Management

v16.5 Support Service	Change	v17.5 Platform Capability
Profile Management	Function of Identity Management	Identity Management
Privacy Management	Function of Identity Management	Identity Management
Analytics	Renamed	Intelligence & Analytics
Policy Based Routing	Function of API Management	API Management
Configuration & Activation	Renamed	Fulfillment
Assurance & Traceability	Renamed	Assurance
Charging	Combined with Invoicing	Billing & Revenue Management
Invoicing	Combined with Charging	Billing & Revenue Management
Catalog Lifecycle Management & Federation	No	Catalog Lifecycle Management & Federation
On-boarding	Considered to be more of a process than a specific capability. During the on-boarding process data is stored in the catalog. Hence refer to the Catalog capability. Note that on-boarding takes place in the market place and in the actualization platform.	Catalog Lifecycle Management & Federation

Following business scenarios are described:

- [Business Scenarios overview R17.5](#)
- [Digital Health Business Scenario R17.5](#)
- [On-demand manufacturing Business Scenario R17.5](#)
- [Stadium App Business Scenario R17.5](#)
- [API Wholesale Business Scenario R17.5](#)

5.1. Business Scenarios overview R17.5

How do the business scenarios apply the DSRA concepts?

In this chapter we will illustrate the DSRA concepts, by means of a set of business scenarios. These scenarios were developed in the context of the [B2B2X workstream](#) of the Internet over Everything program.



[B2B2X](#)

Scenario	IDENTITY MANAGEMENT	PROFILE MANAGEMENT	PRIVACY MANAGEMENT	ANALYTICS	POLICY BASED ROUTING	CONFIGURATION & ACTIVATION	ASSURANCE & TRACEABILITY	CHARGING	INVOICING	CATALOG/LIFECYCLE MGMT & FEDERATION	ON-BOARDING
Digital Health	✓			✓		✓		✓		✓	✓
Smart Cities				✓			✓			✓	✓
On-demand manufacturing	✓			✓				✓	✓	✓	✓
Stadium App	✓			✓		✓	✓	✓	✓	✓	✓
API Wholesale	✓			✓		✓				✓	✓

5.2. Digital Health Business Scenario R17.5

What is the Digital Health business scenario?

In this chapter we will illustrate how the Digital Health catalyst implements the DSRA concepts by means of the following:



- [Description - Digital Health Business Scenario R17.5](#)
- [Use Cases - Digital Health Business Scenario R17.5](#)
- [DSRA Support Services overview - Digital Health Business Scenario R17.5](#)

5.2.1. Description - Digital Health Business Scenario R17.5

What is the digital health business scenario?

This chapter describes the Digital Health business scenario by means of the following:	
Problem Statements	 Problem Statements
Value Proposition	 Value Proposition
Key Participants	 Key Participants

5.2.2. Use Cases - Digital Health Business Scenario R17.5

What are the digital health business scenario use cases?

Use Cases	  Use Cases
---------------------------	--

5.2.3. DSRA Support Services overview - Digital Health Business Scenario R17.5

Which DSRA Support Services does the digital health business scenario use?

Support Service Name	In scope (Y/N)	Comment
Identity Mgmt	Y	To manage Partner and customer identity
Profile Mgmt	N	Out of scope?
Analytics	Y	To provide information about customer and partner behavior
Policy based routing	N	out of scope
Configuration & Activation	N	out of scope

Support Service Name	In scope (Y/N)	Comment
Assurance & Traceability	N	out of scope
Charging	Y	To calculate commission fees
Invoicing	Y	To invoice final customer To invoice / payback designer and payback Design software provider location service provider and Payment service provider
Catalog Lifecycle Mgmt & Federation	Y	To manage customizable goods in catalogue
Onboarding	Y	To onboard partners

DSRA Support Services use - Digital Health Business Scenario R17.5

How do the digital health business scenario use cases make use of the DSRA support services?

Collection	Use Case	IDENTITY MANAGEMENT	PROFILE MANAGEMENT	ANALYTICS	POLICY BASED ROUTING	CONFIGURATION & ACTIVATION	ASSURANCE & TRACEABILITY	CHARGING	INVOICING	CATALOG LIFECYCLE MGMT & FEDERATION	ON-BOARDING
Idea to Offering	On-board Partner									✓	✓
	On-Board Partner Service									✓	✓
	Set Up Product Area										
	Develop App										
Order to Service	Download & Register App										
	Order Equipment & Service					✓				✓	

Collection	Use Case	IDENTITY MANAGEMENT	PROFILE MANAGEMENT	ANALYTICS	POLICY BASED ROUTING	CONFIGURATION & ACTIVATION	ASSURANCE & TRACEABILITY	CHARGING	INVOICING	CATALOG LIFECYCLE MGMT & FEDERATION	ON-BOARDING
	Register Equipment										
Service Execution	Collect Device Measurements						✓				
	View Individual Report										
	Notify Event						✓				
	Notify Patient to Medicate						✓				
	Analyze Public Health Information			✓							
Usage to Payment	Collect and Rate Usage							✓			
	Generate Invoice								✓		
	Accept Payment								✓		
	Settle with Partners								✓		

5.3. On-demand manufacturing Business Scenario R17.5

How does the on-demand manufacturing business scenario implement DSRA concepts?

In this chapter we will illustrate how the On-demand Manufacturing business scenario implements the DSRA concepts.



- [Description - On-demand manufacturing Business Scenario R17.5](#)
- [Use Cases - On-demand manufacturing Business Scenario R17.5](#)
- [DSRA Support Services overview - On-demand manufacturing Business Scenario R17.5](#)
- [APIs - On-demand manufacturing Business Scenario R17.5](#)
- [Example sequence flows - On-demand manufacturing Business Scenario R17.5](#)

5.3.1 Description - On-demand manufacturing Business Scenario R17.5

What is the On-demand manufacturing business scenario?

This chapter describes the On-demand manufacturing business scenario by means of the following:

- [Context - On-demand manufacturing Business Scenario R17.5](#)
- [Drivers - On-demand manufacturing Business Scenario R17.5](#)
- [Key Participants - On-demand manufacturing Business Scenario R17.5](#)
- [Problem Statement - On-demand manufacturing Business Scenario R17.5](#)
- [Value Proposition - On-demand manufacturing Business Scenario R17.5](#)

Context - On-demand manufacturing Business Scenario R17.5

What is the On-demand manufacturing business scenario?

The on-demand market place enables designers to create 3D designs with software provided by a Design Software Provider.	
The designer can publish his designs in the marketplace catalog and when a customer orders goods based on such a design, a 3D printer location close by the customer is selected by the Location service provider.	
The goods are manufactured by the manufacturer and shipped by the on-demand Manufacturing marketplace owner.	
Payment is arranged by the payment service provider that also takes care of settlements with all partners involved.	

Drivers - On-demand manufacturing Business Scenario R17.5

What are the business drivers for the On-demand manufacturing marketplace?

Technological changes: Development of 3D printers or additive manufacturing equipment and decrease of their cost

Customers are looking for customized good quickly and at affordable price

Small manufacturers are looking for new area of growth with Low investments

Development of "buying local" mindset



Technological changes: Development of 3D printers or additive manufacturing equipment and decrease of their cost

Customers are looking for customized good quickly and at affordable price

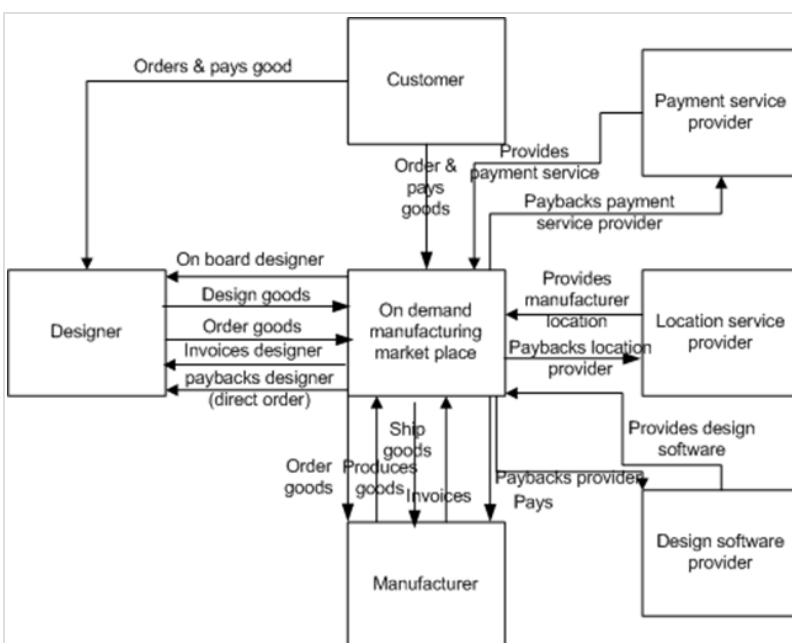
Small manufacturers are looking for new area of growth with Low investments

Development of "buying local" mindset



Key Participants - On-demand manufacturing Business Scenario R17.5

Who are the key participants in the On-demand manufacturing business scenario?



In some cases, an actor can play several roles i.e. the market place service provider can also be design software provider and / or manufacturing location service provider

Problem Statement - On-demand manufacturing Business Scenario R17.5

What challenges are the ecosystem participants facing?

As a final customer, • I need to purchase a customized /unique good at affordable price • So that I can have something meeting my exact needs • To do this I need to find someone able to manufacture it	
As a designer • I need to sell new products without financial risks • So that reach new customers and improve my revenue • To do this I need to find easy to use design tools and small series manufacturer	
As a Design software manufacturer • I need to deploy my tools widely • So that I can reach more designers and get new revenues • To do this, I need to make them available widely	
As a manufacturing market place provider • I need to connect final customer or designer, tools providers and manufacturer • So that I can get revenues • To do this I need to develop an IT system allowing goods design, manufacturer location, goods ordering, shipping and payment	
As a manufacturer	

• I need to be able to produce small series on demand • So that I can get additional revenue without financial risk • To do this I need to use manufacturing tools such as 3D printers and to reach new customers	
As a manufacturer location service provider • I need to monetize my data • So that I can get new revenue • To do this I need to make my location information available • such as 3D printers and to reach new customers	
As a payment service provider • I need to extend my customer footprint • So that I can increase my revenue • To do this I need to be able to propose payment service to any kind of Digital Service	

Value Proposition - On-demand manufacturing Business Scenario R17.5

What is the value proposition for the on-demand manufacturing eco-system participants?

For the final customer • To be able to purchase a customized or unique good at affordable price	
For the designer • To be able to sell customized good at reasonable price with reduced lead time because of: - o no upfront costs - o low inventory - o no financial risk	

For the design software provider To allow easy design of customized good by providing software allowing creation, manipulation and export of 3D models	
For the manufacturing marketplace provider - To allow customized goods production by: - connecting designers and manufacturer - providing an efficient, effective and cost saving supply chain service	
For the manufacturer - To be able to produce customized good at reasonable price by: - avoiding investments required by mass production - keeping inventory low - being sure to be paid	
For the payment service provider To provide simple payment means and get new revenue	
For the payment service provider To provide simple payment means and get new revenue	

5.3.2. Use Cases - On-demand manufacturing Business Scenario R17.5

What are the on-demand business scenario use cases?

In this chapter, the following use cases are described:

- [Manufacturer subscribes to marketplace - On-demand manufacturing R17.5](#)
- [Design software provider subscribes to marketplace - On-demand manufacturing R17.5](#)
- [Design software provider on-boards design software - On-demand manufacturing R17.5](#)

- [Designer registers on marketplace - On-demand manufacturing R17.5](#)
- [Designer subscribes to marketplace - On-demand manufacturing R17.5](#)
- [Designer designs goods - On-demand manufacturing R17.5](#)
- [Designer makes the designed goods orderable - On-demand manufacturing R17.5](#)
- [Customer registers on the website of the designer - On-demand manufacturing R17.5](#)
- [Customer orders goods from the designer - On-demand manufacturing R17.5](#)

Manufacturer subscribes to marketplace - On-demand manufacturing R17.5

How does a manufacturer subscribe to the marketplace?

The Manufacturer submits a request to be on-boarded as a partner.

Use Case Id		 Example sequence flows - On-demand manufacturing Business Scenario
Use Case Name	Manufacturer subscribes to the marketplace	
Summary	<u>The Manufacturer submits a request to be on-boarded as a partner.</u>	
Actor(s)	Manufacturer Manufacturing marketplace provider	
Lifecycle Phases		
Begins When	The manufacturer logs in on the Manufacturing marketplace partner portal and submits a Partner Application request.	
Description	To be completed in the next iteration. For now, see Example sequence flows - On-demand manufacturing Business Scenario .	
Ends When	The manufacturer is on-boarded as a partner of the Manufacturing Marketplace.	

Design software provider subscribes to marketplace - On-demand manufacturing R17.5

How does a design software provider subscribe to the marketplace?

The Design software provider submits a request to be on-boarded as a partner.

Use Case Id		 Example sequence flows - On-demand manufacturing Business Scenario
Use Case Name	Design software provider subscribes to the marketplace	
Summary	<u>The Design software provider submits a request to be on-boarded as a partner.</u>	
Actor(s)	Design software provider Manufacturing marketplace provider	
Lifecycle Phases		
Begins When	The design software provider logs in on the Manufacturing marketplace partner portal and submits a Partner Application request.	
Description	To be completed in the next iteration. For now, see Example sequence flows - On-demand manufacturing Business Scenario .	
Ends When	The design software provider is on-boarded as a partner of the Manufacturing Marketplace.	

Design software provider on-boards design software - On-demand manufacturing R17.5

How does a design software provider on-board design software?

The design software provider submits a request to on-board his design software as a product offering on the Manufacturing Marketplace.

Use Case Id		 Design software provider on-boards design software - flow
Use Case Name	Design software provider on-boards design software	
Summary	The Design software provider on-boards his design software on the Marketplace.	
Actor(s)	Design software provider Manufacturing marketplace provider	
Lifecycle Phases		

Begins When	<u>The design software provider submits a request to on-board his design software as a product offering on the Manufacturing Marketplace.</u>	
Description	On-boarding of design software. To be completed in the next iteration. For now, see Design software provider on-boards design software - flow .	
Ends When	The design software is on-boarded as a product offering on the manufacturing marketplace.	

Designer registers on marketplace - On-demand manufacturing R17.5

How does a designer register on the marketplace?

The Designer logs in on the Partner Portal and authenticates himself with existing credentials from an Identity Provider, the Marketplace integrates with.

Use Case Id		 Example sequence flows - On-demand manufacturing Business Scenario
Use Case Name	Designer registers to the marketplace	
Summary	<u>The Designer logs in on the Partner Portal and authenticates himself with existing credentials from an Identity Provider, the Marketplace integrates with.</u>	
Actor(s)	Designer Manufacturing marketplace provider	
Lifecycle Phases		
Begins When	Designer logs in on the Partner Portal	
Description	To be completed in the next iteration. For now, see Example sequence flows - On-demand manufacturing Business Scenario .	
Ends When	Designer has an approved partner account credentials for the market place.	

Designer subscribes to marketplace - On-demand manufacturing R17.5

How does a designer subscribe to the marketplace?

The designer submits a request to be on-boarded as a partner.

Use Case Id		 Designer subscribes to marketplace - flow
Use Case Name	Designer subscribes to the marketplace	
Summary	<u>The designer submits a request to be on-boarded as a partner.</u>	
Actor(s)	Designer Manufacturing marketplace provider	
Lifecycle Phases		
Begins When	The designer logs in on the Manufacturing Marketplace partner portal and submits a Partner Application request	
Description	To be completed in the next iteration. For now, see Designer subscribes to marketplace - flow .	
Ends When	The designer is on-boarded as a partner of the Manufacturing Marketplace.	

Designer designs goods - On-demand manufacturing R17.5

How does a designer design goods?

The Designer designs goods and on-boards these as product offerings on the Marketplace.

Use Case Id		 Designer designs goods - flow
Use Case Name	Designer designs to the marketplace	
Summary	<u>The Designer designs goods and on-boards these as product offerings on the Marketplace.</u>	
Actor(s)	Designer Manufacturing marketplace provider	

Lifecycle Phases		
Begins When	Designer starts a design session using the design software from the Design software provider.	
Description	To be completed in the next iteration. For now, see Designer designs goods - flow .	
Ends When	The designed goods are on-boarded as a product offering in the Manufacturing Marketplace catalog.	

Designer makes the designed goods orderable - On-demand manufacturing R17.5

How does the designer make the design goods orderable?

The Designer complements his designed goods with additional service components to make them orderable.

Use Case Id		 Designer makes designed goods orderable - flow
Use Case Name	Designer makes the designed goods orderable	
Summary	<u>The Designer complements his designed goods with additional service components to make them orderable.</u>	
Actor(s)	Designer Manufacturing marketplace provider	
Lifecycle Phases		
Begins When	Designer starts a product composition session.	
Description	The designer creates a composed product offering consisting of his designed goods, manufacturing location, payment and shipment services. To be completed in the next iteration. For now, see Designer makes designed goods orderable - flow .	

Ends When	The orderable goods are on-boarded as a product offering in the Manufacturing Marketplace catalogue.	
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Customer registers on the website of the designer - On-demand manufacturing R17.5

How does a customer register on the website of the designer?

The Customer logs in on the Designers website and authenticates himself with existing credentials from an Identity Provider, the Marketplace integrates with.

Use Case Id		 Customer registers on the website of the designer - flow
Use Case Name	Customer registers on the website of the designer	
Summary	<u>The Customer logs in on the Designers website and authenticates himself with existing credentials from an Identity Provider, the Marketplace integrates with.</u>	
Actor(s)	Customer Designer	
Lifecycle Phases		
Begins When	The customer clicks on the login button on the website of the designer	
Description	To be completed in the next iteration. For now, see Customer registers on the website of the designer - flow .	
Ends When	The customer is registered on the website of the designer with credentials that will give them access to the customer portal of the Manufacturing Marketplace.	

Customer orders goods from the designer - On-demand manufacturing R17.5

How does a customer order goods from the designer?

The customer orders goods to the designer. After payment, the goods are manufactured and shipped to the address as requested by the customer.

Use Case Id		 Customer orders goods from the designer - flow
Use Case Name	Customer orders goods from the designer	
Summary	<u>The customer orders goods to the designer. After payment, the goods are manufactured and shipped to the address as requested by the customer.</u>	
Actor(s)	Customer Designer Manufacturing location provider Digital Service Provider Manufacturer Manufacturing marketplace provider Payment service provider Design Service Provider	
Lifecycle Phases		
Begins When	Customer places an order.	
Description	To be completed in the next iteration. For now, see Customer orders goods from the designer - flow .	
Ends When	The goods are shipped to the customer.	

5.3.3. DSRA Support Services overview - On-demand manufacturing Business Scenario R17.5

Which DSRA Support Services does the digital health business scenario use?

Support Service Name	In scope (Y/N)	Comment
Identity Mgmt	Y	To manage Partner and customer identity
Profile Mgmt	N	Out of scope?
Analytics	Y	To provide information about customer and partner behavior

Support Service Name	In scope (Y/N)	Comment
Policy based routing	N	out of scope
Configuration & Activation	N	out of scope
Assurance & Traceability	N	out of scope
Charging	Y	To calculate commission fees
Invoicing	Y	To invoice final customer To invoice / payback designer and payback Design software provider location service provider and Payment service provider
Catalog Lifecycle Mgmt & Federation	Y	To manage customizable goods in catalogue
Onboarding	Y	To onboard partners

- [DSRA Support Services use - On-demand manufacturing Business Scenario R17.5](#)

DSRA Support Services use - On-demand manufacturing Business Scenario R17.5

How do the On-demand manufacturing business scenario use cases make use of the DSRA support services?

Scenario	IDENTITY MANAGEMENT	PROFILE MANAGEMENT	PRIVACY MANAGEMENT	ANALYTICS	POLICY BASED ROUTING	CONFIGURATION & ACTIVATION	ASSURANCE & TRACEABILITY	CHARGING	INVOICING	CATALOG LIFECYCLE MGMT & FEDERATION	ON-BOARDING
Design software provider subscribes to marketplace	✓									✓	
Manufacturer subscribes to marketplace	✓									✓	✓

Scenario	IDENTITY MANAGEMENT	PROFILE MANAGEMENT	PRIVACY MANAGEMENT	ANALYTICS	POLICY BASED ROUTING	CONFIGURATION & ACTIVATION	ASSURANCE & TRACEABILITY	CHARGING	INVOICING	CATALOG LIFECYCLE MGMT & FEDERATION	ON-BOARDING
Designer subscribes to marketplace	✓									✓	✓
Customer orders goods from designer	✓			✓				✓	✓	✓	

5.3.4. APIs - On-demand manufacturing Business Scenario R17.5

What are the relevant APIs for the On-demand manufacturing business scenario?

Use case	Actors	API	Touchpoints
Design software provider subscribes to marketplace	Design software provider	Party Role On-boarding (partnership type)	
Manufacturer subscribes to marketplace	Manufacturer	Party Role On-boarding (partnership type)	
Designer subscribes to marketplace and design goods	Designer	Party Role On-boarding (partnership type) Agreement Product catalog	
Customer orders goods from designer	Customer	Customer Product Ordering	
Customer order goods to market place	Customer	Customer Product Ordering	
Marketplace request manufacturer location	Marketplace	Location	

Use case	Actors	API	Touchpoints
Marketplace sends order to manufacturer	Marketplace	Product ordering	
Manufacturer produces the good and informs marketplace	Manufacturer	Product ordering	
Marketplace ship good and deliver it to customer or designer	Marketplace	Shipping management (future API)	
Market place paybacks partners	Marketplace	Customer Bill	

5.3.5. Example sequence flows - On-demand manufacturing Business Scenario R17.5

What are example sequence flows for the on-demand manufacturing business scenario?

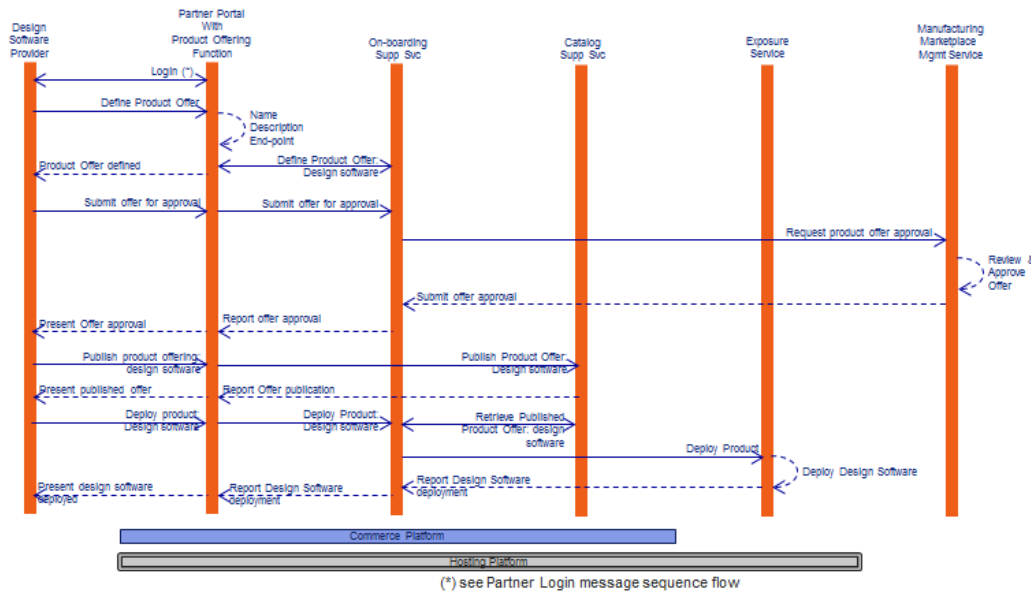
- [Design software provider on-boards design software - flow - On-demand manufacturing R17.5](#)
- [Designer subscribes to marketplace - flow - On-demand manufacturing R17.5](#)
- [Designer designs goods - flow - On-demand manufacturing R17.5](#)
- [Designer makes his goods orderable - flow - On-demand manufacturing R17.5](#)
- [Customer registers on the website of the designer - flow - On-demand manufacturing R17.5](#)
- [Customer orders goods from the designer - flow - On-demand manufacturing R17.5](#)

Design software provider on-boards design software - flow - On-demand manufacturing R17.5

How does a design software provider on-board design software?

The design software provider on-boards his design software tool as a product offering on the Manufacturing marketplace.

The design software provider on-boards his design software tool as a product offering on the Manufacturing marketplace.



Designer subscribes to marketplace - flow - On-demand manufacturing R17.5

How does a designer subscribe to the marketplace?

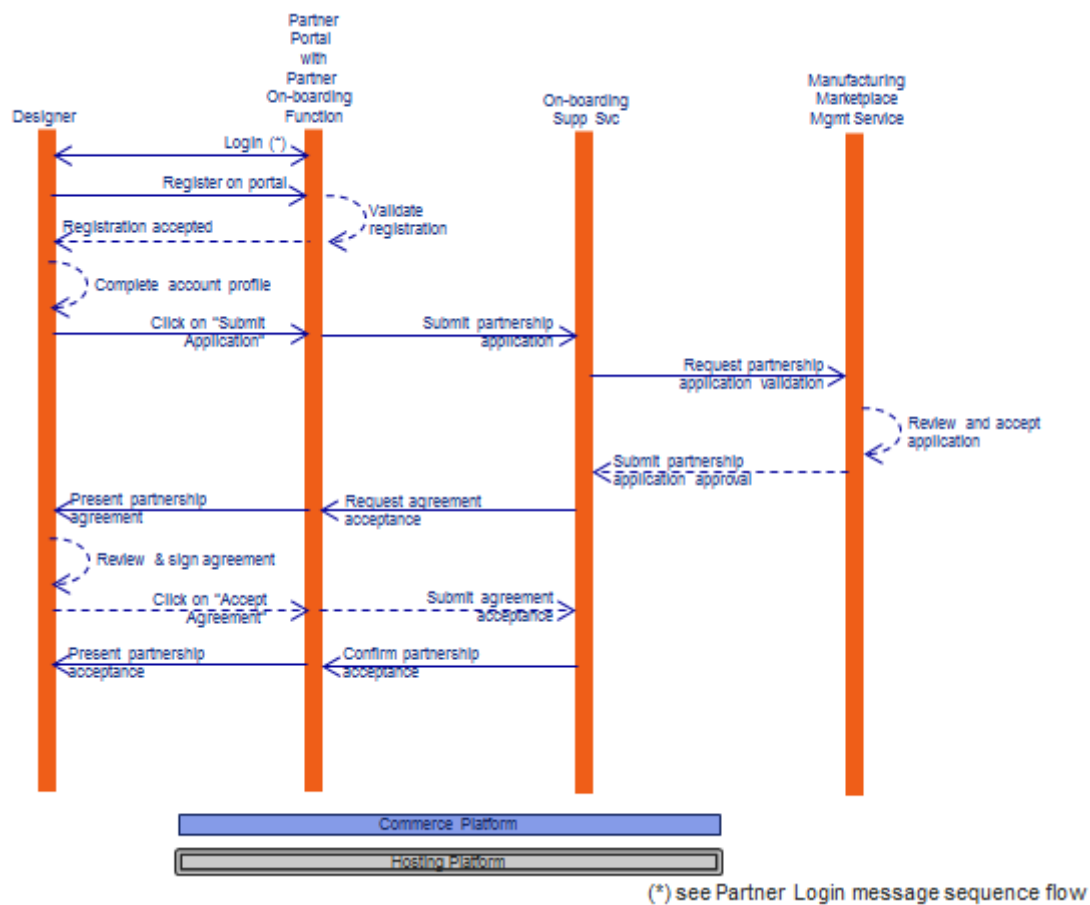
The designer registers as a partner and logs in on the Partner Portal of the Manufacturing marketplace.

The designer registers as a partner and logs in on the Partner Portal of the Manufacturing marketplace.

In the scenario that follows, the Designer selects one of several Identity Providers.



And the designer applies for on-boarding as a partner of the marketplace.

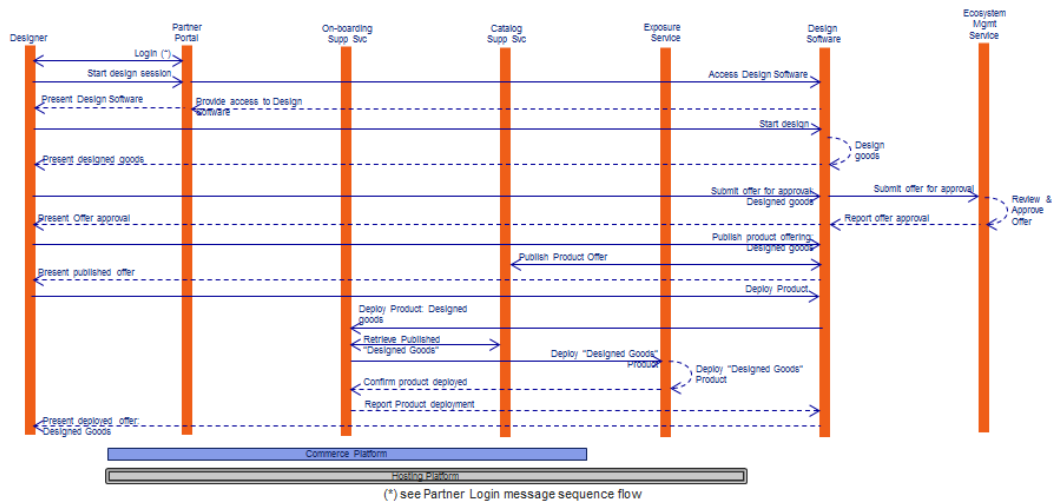


Designer designs goods - flow - On-demand manufacturing R17.5

How does a designer design goods?

The designer designs goods and on-boards these as a product offering in the Manufacturing Marketplace catalogue.

The designer designs goods and on-boards these as a product offering in the Manufacturing Marketplace catalogue.

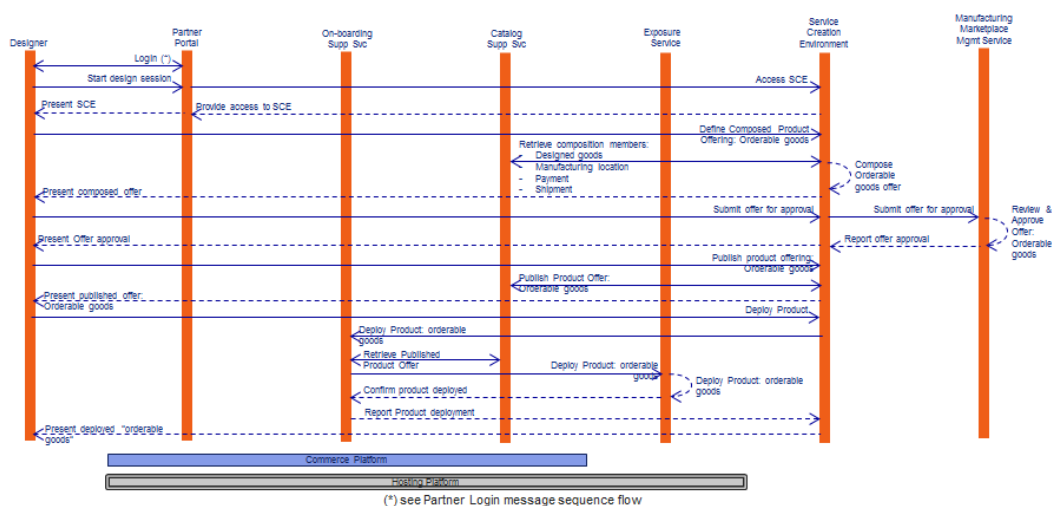


Designer makes his goods orderable - flow - On-demand manufacturing R17.5

How does a designer make goods orderable?

The designer defines the "orderable goods" composed product offering by combining his "designed goods", a manufacturing location, payment and shipment product composition members, and publishes this in the Manufacturing Marketplace catalogue.

The designer defines the "orderable goods" composed product offering by combining his "designed goods", a manufacturing location, payment and shipment product composition members, and publishes this in the Manufacturing Marketplace catalogue.



Customer registers on the website of the designer - flow - On-demand manufacturing R17.5

How does a customer register on the website of the designer?

The customer registers on the website of the designer and gets credentials that will grant him access to the customer portal of the Manufacturing Marketplace.

The customer registers on the website of the designer and gets credentials that will grant him access to the customer portal of the Manufacturing Marketplace.

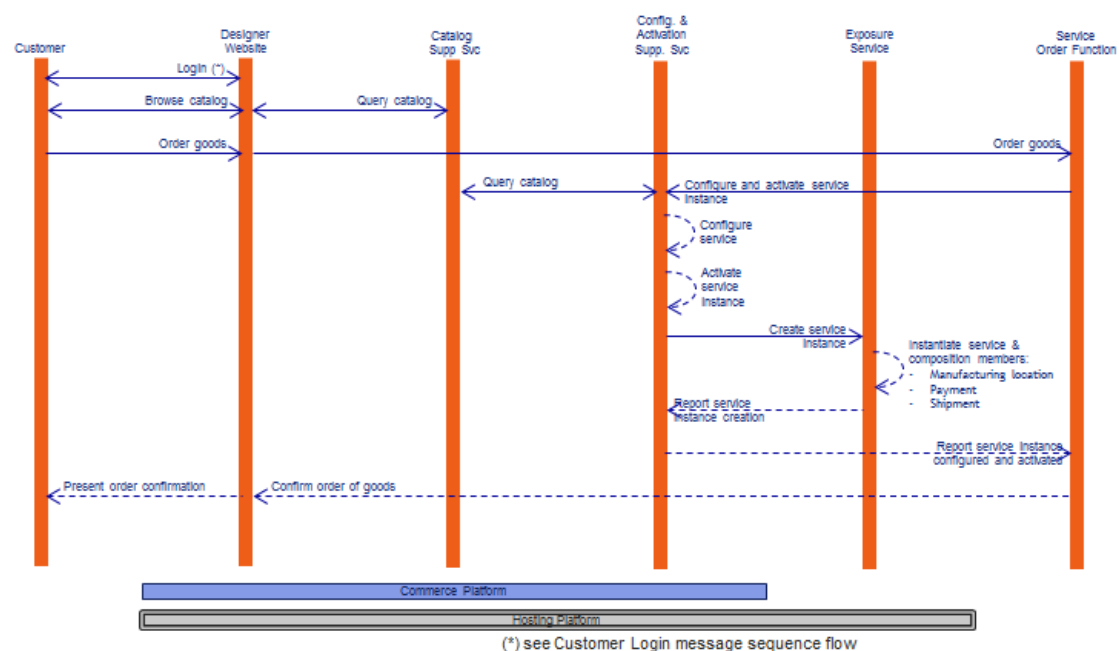


Customer orders goods from the designer - flow - On-demand manufacturing R17.5

How does a customer order goods from the designer?

The customer browses the catalog of orderable goods and places an order.

The customer browses the catalog of orderable goods and places an order.



(*) see Customer Login message sequence flow

5.4. Stadium App Business Scenario R17.5

How does the stadium app business scenario implement DSRA concepts?

This chapter illustrates how the Stadium App business scenario implements the DSRA concepts.



- [Description - Stadium App Business Scenario R17.5](#)
- [Use Cases - Stadium App Business Scenario R17.5](#)
- [DSRA Support Services overview - Stadium App Business Scenario R17.5](#)
- [APIs - Stadium App Business Scenario R17.5](#)
- [Example sequence flows - Stadium App Business Scenario R17.5](#)

5.4.1. Description - Stadium App Business Scenario R17.5

What is the stadium app business scenario?

- [Context - Stadium App Business Scenario R17.5](#)
- [Drivers - Stadium App Business Scenario R17.5](#)
- [Key Participants - Stadium App Business Scenario R17.5](#)
- [Problem Statement - Stadium App Business Scenario R17.5](#)
- [Value Proposition - Stadium App Business Scenario R17.5](#)

Context - Stadium App Business Scenario R17.5

What is the context of the Stadium app business scenario?

A sports stadium owner has a mobile app which allows visitors to get information about schedules, facilities in the stadium, order drinks and food. He wants to complement this with streaming to allow Customers to follow matches in other stadiums, improving their experience in the stadium.



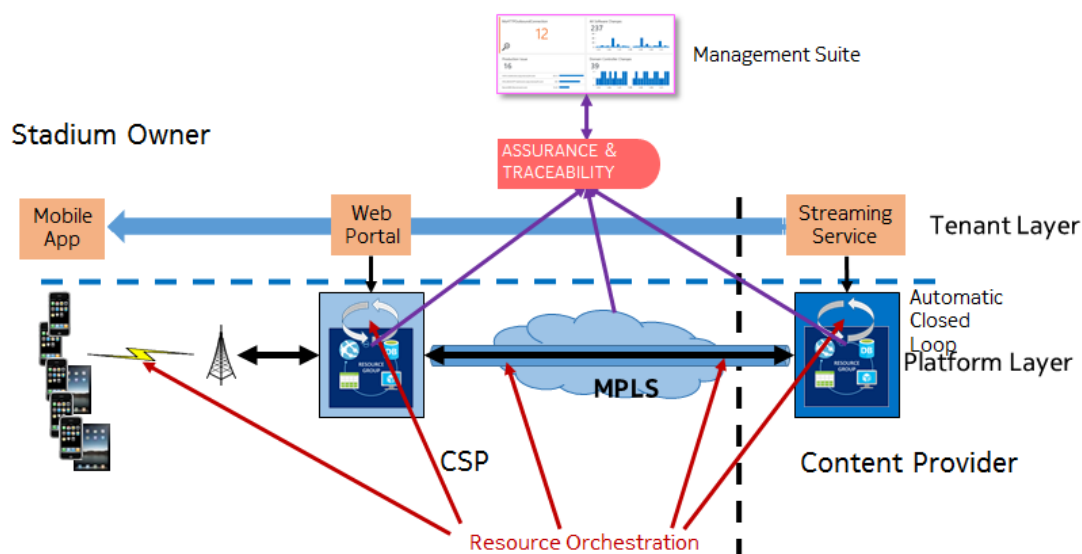
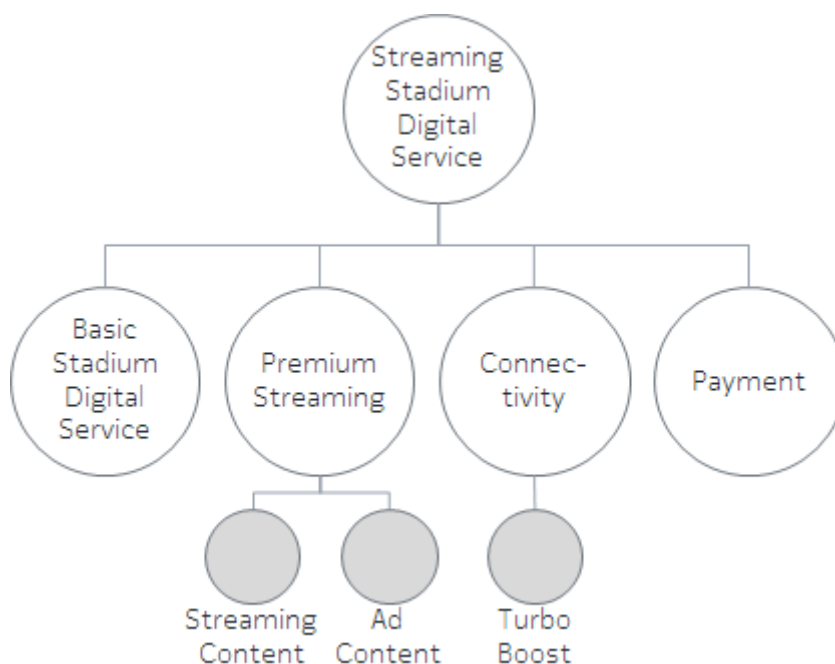
The Communications service provider offers connectivity and a web portal to allow partners to on-board and end users to register for services. Customers can also opt for

a Turbo Boost function for an additional charge. The Communications service provider also acts as the Sports entertainment ecosystem Manager.

Payments and settlements are taken care of by the Payment Provider.

The Content provider offers basic streaming content as well as premium whereby if a user opts in to get advertisements he gets the service for a lower price. The ad service is provided by the Advertisement provider. The content service is hosted in a cloud environment run by the Cloud Provider.

The following diagram illustrates the Digital Service composition.



The CSP and the content provider have established an SLA. As more and more users will make use of the stadium app, within the platform layer, resource orchestration will take place to enable more resources to take the extra load.	
The Management Suite of the CSP will consume the API of the Assurance & Traceability Support Service to retrieve information about the health of the composed service.	 Assurance & Traceability Support Service
The CSP will be able to monitor whether the SLA is respected.	

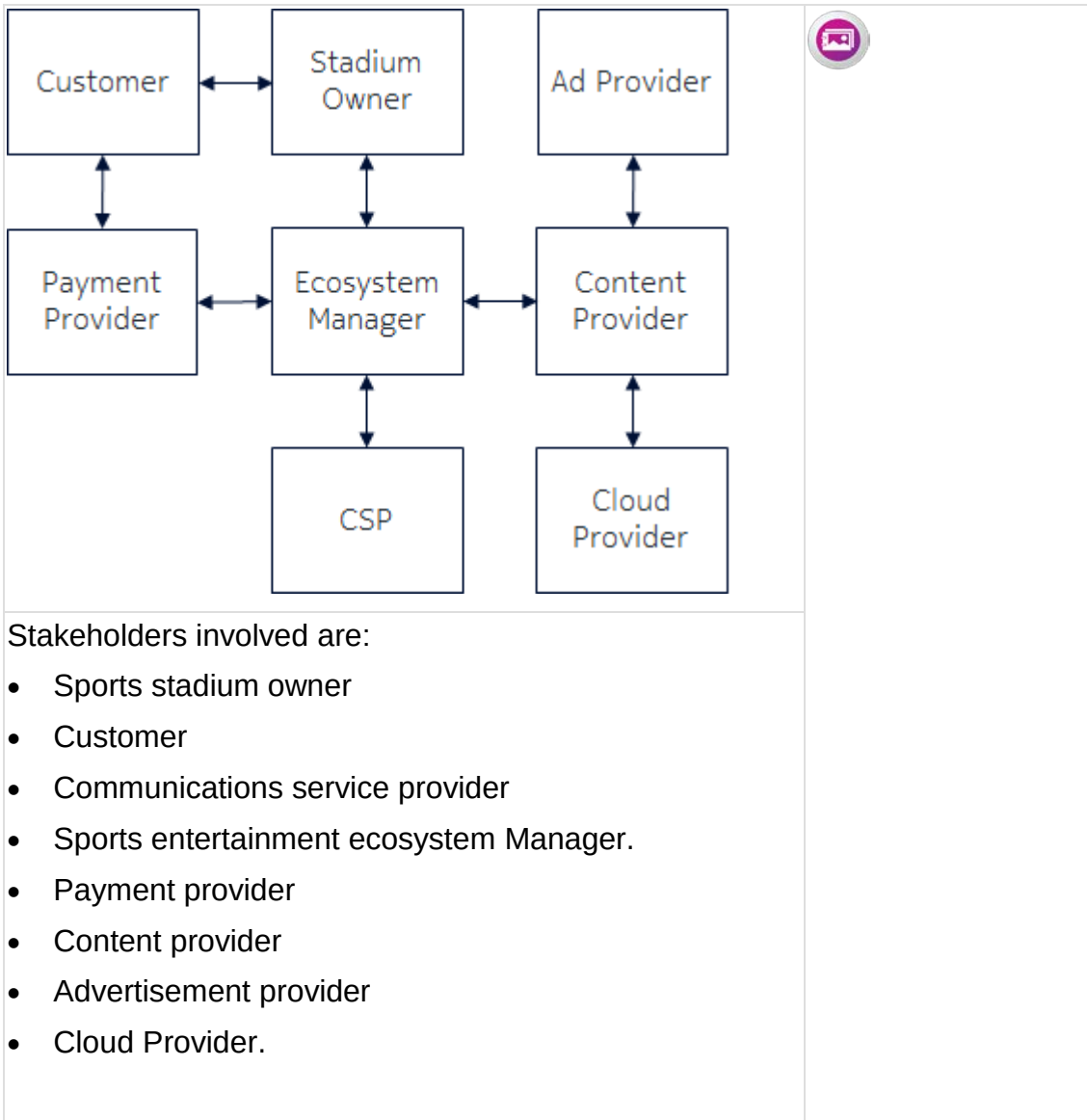
Drivers - Stadium App Business Scenario R17.5

What are the drivers for the Sports Entertainment ecosystem?

Technological changes: • Market penetration of smartphones • Availability of mobile internet (currently 4G with 5G coming)	
Customers are looking for access to streaming content anywhere, anytime.	
Sports entertainment providers are looking for new and innovative ways to reach customers.	
Content service providers are looking for alternative digital channels to reach customers beyond the broadcast model.	
Stadium owners are looking for ways to please their customers by offering them convenience services as well as giving them access to sports news, including live coverage of events.	
Communication service providers are looking for ways to create value on top of the connectivity services.	

Key Participants - Stadium App Business Scenario R17.5

Who are the key participants of the stadium app business scenario?



Problem Statement - Stadium App Business Scenario R17.5

What challenges are the ecosystem participants facing?

As a **Sports entertainment ecosystem manager**

- I need to
 - connect stadium owners, content owners, connectivity service providers and end customers
- So that I can
 - generate revenues
- To do this I need to

- o develop an ecosystem environment enabling offering and consuming streaming content - o implement a catalog where partners and customers can find the available product offerings	
As a Stadium Owner • I need to - o offer my customers a great experience when visiting my stadium • So that I can - o welcome them also in the future - o they also promote my services to friends • To do this I need to - o develop an app which provides them comfort while in the stadium, including watching games from other sports and locations - o collaborate with the appropriate partners	
As a Communications Service Provider • I need to - o create a communications service portfolio • So that I can - o attract content and entertainment providers - o generate value for content service providers • To do this I need to - o develop a platform that enables quick and seamless on-boarding	
As a Cloud Provider • I need to - o have an agreed service level objective - o understand the performance of my cloud offering • So that I can - o guarantee Cloud service continuity as required by my business	

- To do this I need to
 - o implement a technical orchestration solution
 - o implement a performance reporting solution.

As a **Content Provider**

- I need to
 - o have a differentiated offering of content
- So that I can
 - o make my content available to a broad spectrum of potential customers
- To do this I need to
 - o establish partnerships with entertainment providers
 - o publish my offerings in entertainment ecosystem catalogs
 - o establish partnerships with advertisement providers to enable offering cheaper viewing options
 - o differentiate my offerings with premium service options in addition to best effort

As an **Advertisement Provider**

- I need to
 - o have an advertisement portfolio for target customer segments
- So that I can
 - o provide value to digital service providers
- To do this I need to
 - o establish partnerships with digital service providers
 - o publish my advertisement services in ecosystem catalogs

As a **Payment Provider**

- I need to

- o provide payment services to a wide variety of business models and scenarios • So that I can - o increase revenue - o enlarge customer base • To do this I need to - o implement a platform that enables many payment methods	
As a Customer • I need to - o purchase sports entertainment services in an easy manner at an acceptable price • So that I can - o enjoy the stay in a sports stadium • To do this I need to - o find easily access to sports entertainment services	

Value Proposition - Stadium App Business Scenario R17.5

What is the value proposition for the ecosystem participants?

For the Sport stadium owner • enable an excellent customer experience for the stadium visitors	
For the Customer • have access to convenience services • have easy access to sports information while in the stadium	
For the Communications service provider • add create value for partners and end customers on top of the connectivity service	

• enable quick and easy on-boarding of partners	
For the Sports entertainment ecosystem manager • enable Sports entertainment services: ◦ connecting stadium owners with content, connectivity and payment service providers ◦ providing an efficient, effective and cost saving supply chain service	
For the Payment provider • to provide simple payment means • to get new revenue	
For the Content provider • to reach a broader audience for my content	
For the Advertisement provider • to provide advertisement service • to enlarge reach of advertisements	

5.4.2. Use Cases - Stadium App Business Scenario R17.5

What are the stadium app business scenario use cases?

We consider following use cases:

- [Stadium owner registers on the sports entertainment ecosystem - Stadium App R17.5](#)
- Stadium owner subscribes to the sports entertainment ecosystem - Stadium App R17.5
- Stadium owner designs and publishes his premium stadium app - Stadium App R17.5
- [Stadium owner on-boards his basic stadium app on the sports entertainment ecosystem - Stadium App R17.5](#)
- [Customer registers on the website of the stadium - Stadium App R17.5](#)
- Customer subscribes to the premium Stadium app service - Stadium App R17.5
- Customer starts using the premium Stadium app service - Stadium App R17.5
- Customer triggers the turbo boost option - Stadium App R17.5

5.4.3. DSRA Support Services overview - Stadium App Business Scenario R17.5

Which DSRA Support Services does the stadium app business scenario use?

Support Service Name	In scope (Y/N)	Comment
Identity Mgmt	Y	To manage Partner and customer identity
Profile Mgmt	N	Out of scope?
Analytics	Y	To provide information about customer and partner behavior
Policy based routing	N	out of scope
Configuration & Activation	Y	To configure the preferences of the digital services, e.g. the option for receiving advertisements or not
Assurance & Traceability	Y	For the Communications service provider to monitor the health of the services and identify potential breaches of the SLA with the Content provider
Charging	Y	To calculate commission fees
Invoicing	Y	To invoice final customer To invoice / payback the content provider, communications service provider, payment service provider
Catalog Lifecycle Mgmt & Federation	Y	To manage the different product offerings in a catalogue
Onboarding	Y	To on-board partners

- [DSRA Support Services use - Stadium App Business Scenario R17.5](#)

DSRA Support Services use - Stadium App Business Scenario R17.5

How do the stadium app business scenario use cases make use of the DSRA support services?

The following table gives an overall view of what support services are related to this Stadium App business scenario.	
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	IDENTITY MANAGEMENT	PROFILE MANAGEMENT	PRIVACY MANAGEMENT	ANALYTICS	POLICY BASED ROUTING	CONFIGURATION & ACTIVATION	ASSURANCE & TRACEABILITY	CHARGING	INVOICING	CATALOG LIFECYCLE MGMT & FEDERATION	ON-BOARDING
Overall	✓	✓				✓				✓	✓

5.4.4. APIs - Stadium App Business Scenario R17.5

What are the relevant APIs for the Stadium App business scenario?

Use case	Actors	API	Touchpoints
Stadium owner registers on the sports entertainment ecosystem	Stadium Owner Identity Provider	Identity API	
Stadium owner subscribes to the sports entertainment ecosystem	Stadium Owner Sports entertainment ecosystem manager	Party Mgmt API Agreement API Party Role API On-boarding (partnership type) API	
Stadium owner on-boards his basic stadium app on the entertainment ecosystem	Stadium Owner	Party Role API On-boarding (partnership type) API	

Use case	Actors	API	Touchpoints
		Product Catalog Mgmt API	
Stadium owner designs and publishes his premium stadium app	Stadium Owner Sports entertainment ecosystem manager Communications Service Provider Content Service Provider Payment Service Provider	Party Role API On-boarding (partnership type) API Product Catalog Mgmt API	
Customer registers on the website of the stadium	Customer Identity Provider	Identity API	
Customer subscribes to the premium stadium service	Customer Stadium Owner Sports entertainment ecosystem manager Communications Service Provider Content Service Provider Payment Service Provider	Digital Service Mgmt API Product Catalog Mgmt API Product Ordering API Activation & Configuration API Product Inventory API	
Customer starts using the premium stadium service	Customer Communications Service Provider Content Service Provider Advertisement Service Provider	Digital Service Mgmt API Usage Mgmt API Performance Mgmt API Product Inventory API	
Customer triggers the turbo boost option	Customer Communications Service Provider	Digital Service Mgmt API Product Catalog Mgmt API Product Ordering API	

Use case	Actors	API	Touchpoints
		Activation & Configuration API Product Inventory API	

5.4.5. Example sequence flows - Stadium App Business Scenario R17.5

What are example sequence flows for the stadium app business scenario?

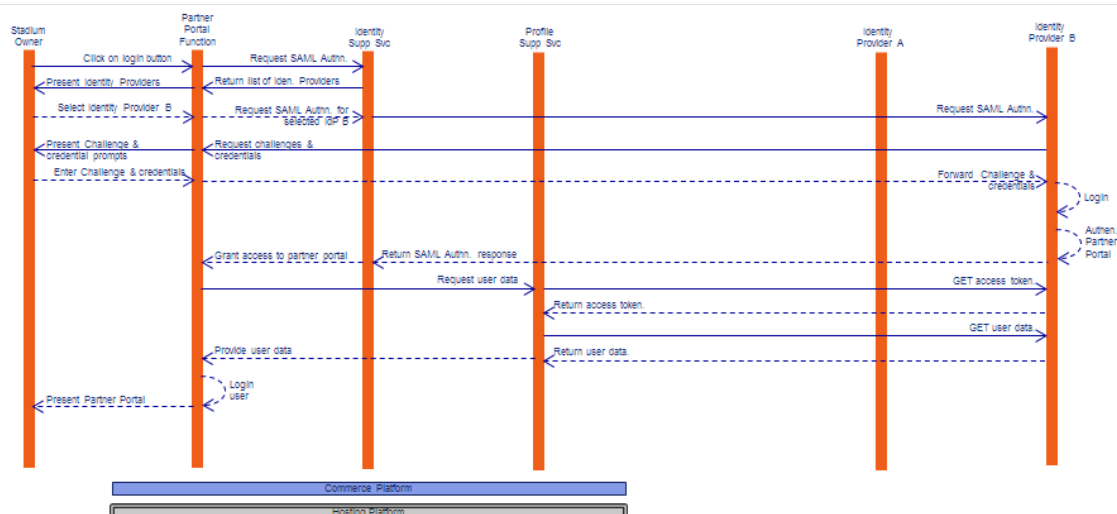
- [Stadium Owner logs in and registers as partner - flow - Stadium App R17.5](#)
- [Stadium Owner on-boards Stadium Apps - flow - Stadium App R17.5](#)
- [Customer logs in and registers for Streaming Stadium App - flow - Stadium App R17.5](#)

Stadium Owner logs in and registers as partner - flow - Stadium App R17.5

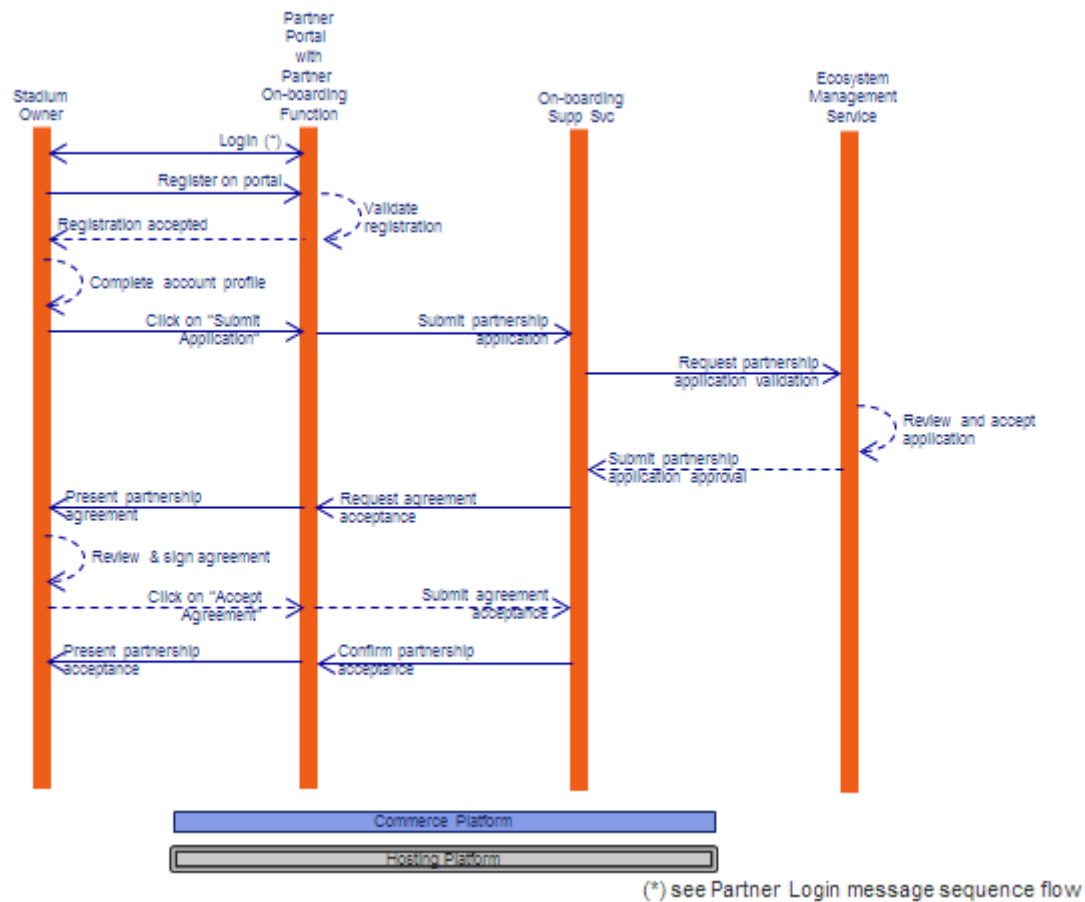
How does the stadium owner log in and register as a partner on the sports entertainment eco-system?

The Stadium Owner registers as a partner and logs in on the Partner Portal of the Sports Entertainment ecosystem. In the scenario that follows, the Stadium Owner selects one of several Identity Providers.

The Stadium Owner registers as a partner and logs in on the Partner Portal of the Sports Entertainment ecosystem.



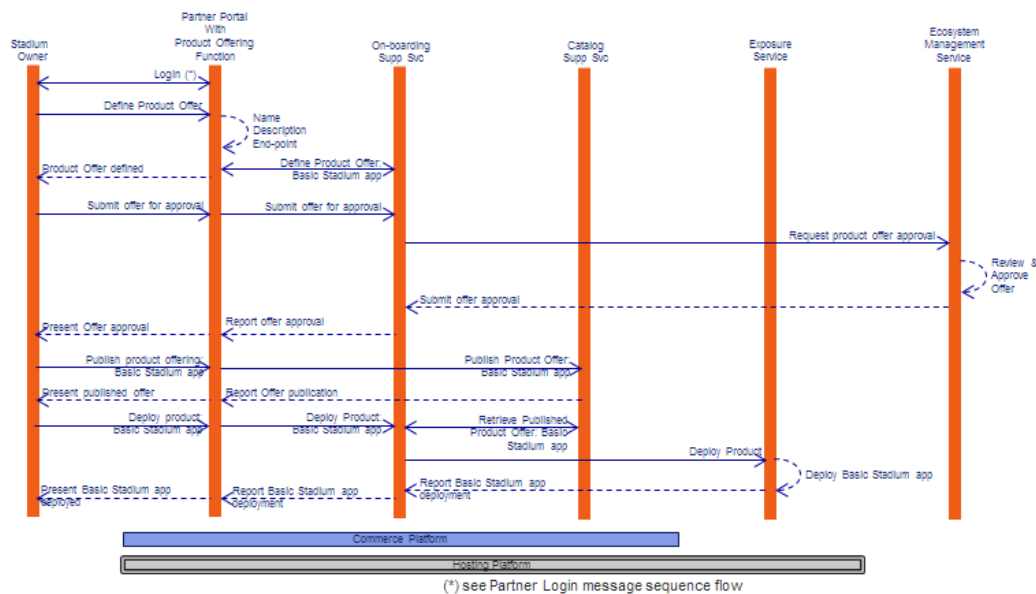
And the Stadium Owner applies for on-boarding as a partner of the eco-system.



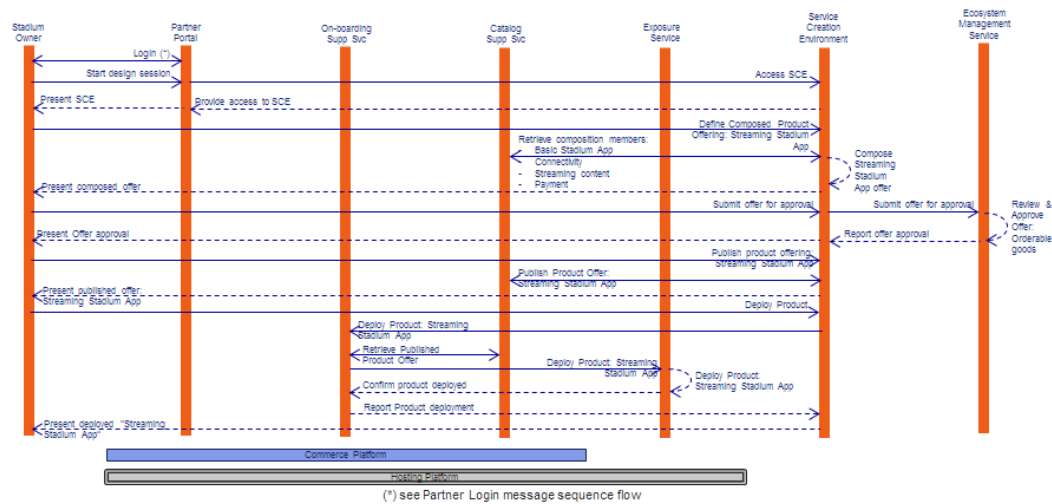
Stadium Owner on-boards Stadium Apps - flow - Stadium App R17.5

How does a stadium owner on-board stadium apps?

The Stadium Owner on-boards his Basic Stadium App.



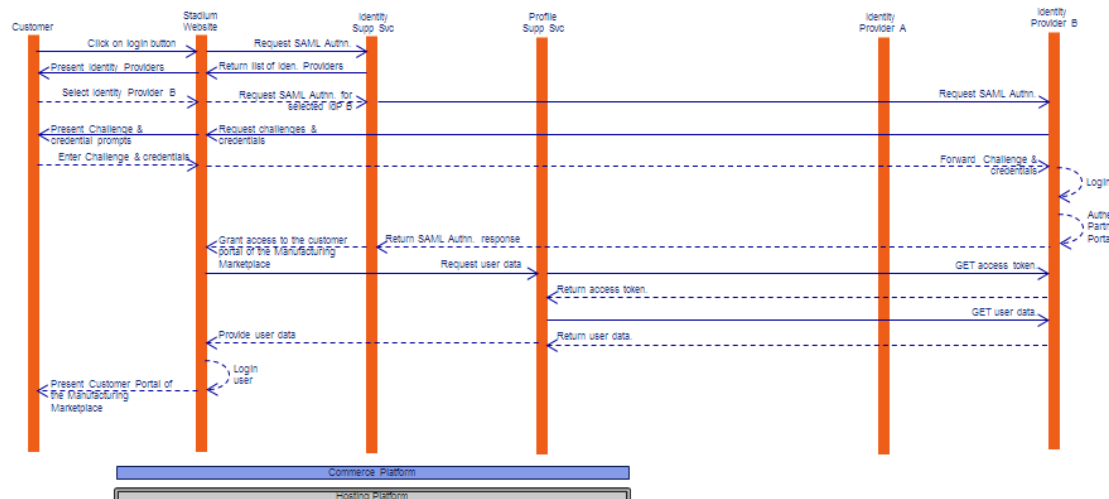
And the Stadium Owner designs the Premium Stadium App



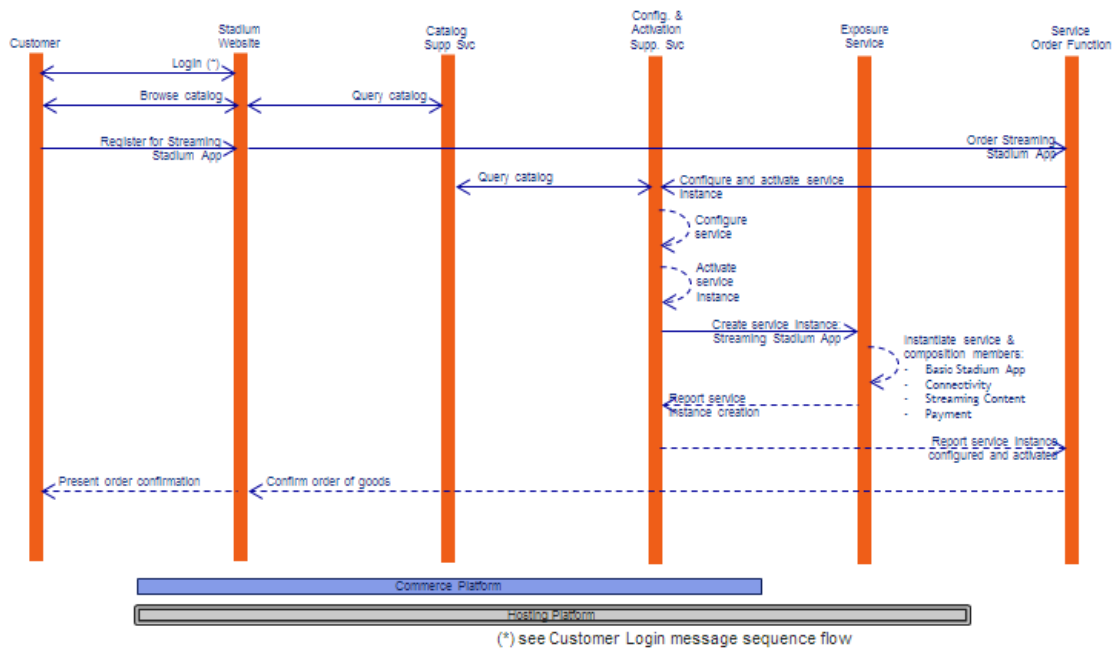
Customer logs in and registers for Streaming Stadium App - flow - Stadium App R17.5

How does a customer log-in and register for the streaming stadium app?

The Customer logs in on the Stadium website.



And the Customer registers for the Streaming Stadium App



5.5. API Wholesale Business Scenario R17.5

How does the API wholesale business scenario implement DSRA concepts?

This chapter illustrates how the API wholesale business scenario implements the DSRA concepts.



- [Description - API Wholesale Business Scenario R17.5](#)
- [Use Cases - API Wholesale Business Scenario R17.5](#)
- [DSRA Platform Capabilities overview - API Wholesale Business Scenario R17.5](#)
- [APIs - API Wholesale Business Scenario R17.5](#)
- [Example sequence flows - API Wholesale Business Scenario R17.5](#)

5.5.1. Description - API Wholesale Business Scenario R17.5

What is the API wholesale business scenario?

This chapter describes the API wholesale business scenario by means of the following:



- [Context - API Wholesale Business Scenario R17.5](#)
- [Drivers - API Wholesale Business Scenario R17.5](#)

- [Key Participants - API Wholesale Business Scenario R17.5](#)
- [Problem Statement - API Wholesale Business Scenario R17.5](#)
- [Value Proposition - API Wholesale Business Scenario R17.5](#)

Context - API Wholesale Business Scenario R17.5

What is the context of the API Wholesale business scenario?

The API wholesale marketplace enables the API wholesaler to expand the variation of providing APIs with resources provided by the Resource Service Provider.



The API wholesaler orders to use resources or functions of the Resource Service Provider as an API and resells that API to the Content Provider. When reselling, the API Wholesaler can add his enhancements or can combine with his original function(s).

The Content Provider provides content (e.g. movie, music, and chat services) for its customers, using the API provided by the API Wholesaler or the Resource Service Provider.

The API Wholesalers and the Content Provider may be the same organization. For example, they belong to a business management division and a development division of the same company, respectively.

Drivers - API Wholesale Business Scenario R17.5

What are the business drivers for the API Wholesale marketplace?

Technological changes: Collaborations using APIs are promoted

Customers are looking for new applications having attractive functions.

Content Providers are looking for new functions or resources which are easy to use, to reach customers.

API Wholesalers are looking for resources or functions as APIs and ways to expose APIs in an easy-to-use form.

Resource Service Providers are looking for chances to offer their resources.

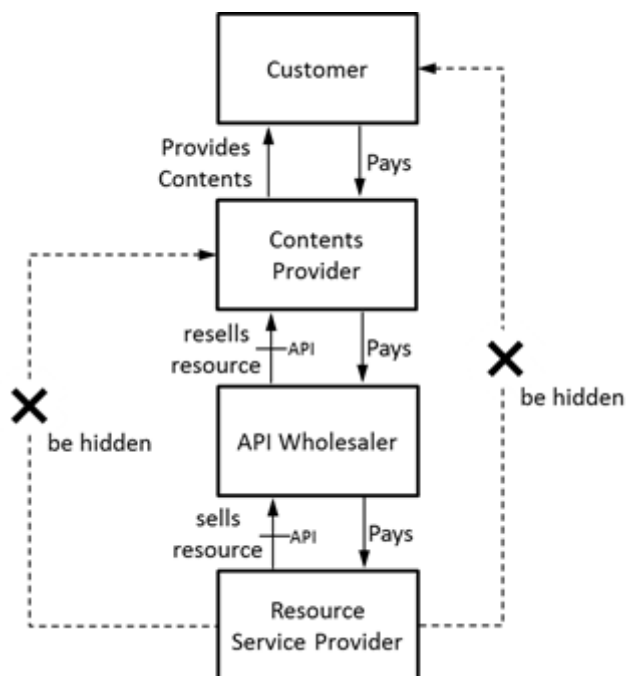


Key Participants - API Wholesale Business Scenario R17.5

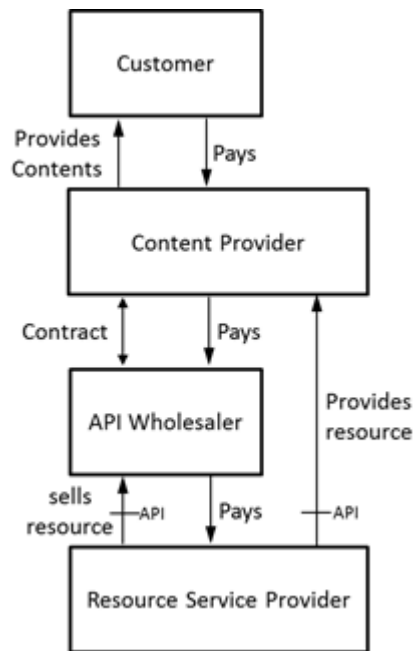
Who are the key participants in the API Wholesale business scenario?

There are two cases of relationships among the key participants.

Firstly, the Content Provider provides content for its customers, using the API provided by the API Wholesaler. The API Wholesaler resells resources which are purchased from the Resource Service Provider. In this model, there may be the requirement that the Resource Service Provider should be hidden from customers and the Content Provider.



Secondly, the Content Provider provides content for its customers, directly using the API provided by the Resource Service Provider. The Content Provider and the API Wholesaler have a contract related to resource usage. The API Wholesaler manages and controls resource usage used by the Content Provider.



In both cases, the API Wholesalers and the Content Provider may belong to the same organization. In that case, there will be no payment between the API wholesaler and the Content Provider.



Problem Statement - API Wholesale Business Scenario R17.5

What challenges are ecosystem participants facing?

• As a Customer, • I need to purchase new and innovative applications (contents services). • So that I can enjoy contents (e.g. watching movie). • To do this, I need to find favorite applications.	
• As a Content Provider • I need to incorporate various functions or resources to my application. • So that I can make my application popular for customers. • To do this, I need to obtain APIs providing above resources.	
• As an API wholesaler	

• I need to resell resources and functions as APIs for contents providers. • So that I can generate revenue by gathering a lot of resource service providers as a “Curator” on the API wholesale business. • To do this, I need to find ways to expose APIs in an easy manner.	
• As a Resource service provider • I need to increase the number of users who use my resource services. • So that I can generate revenue by exposing resource services. • For example, if I’m a communication service provider in “sponsored data service”, I can provide Zero-rating data usage to users.	

Value Proposition - API Wholesale Business Scenario R17.5

What is the value proposition for the API Wholesale eco-system participants?

For the Customer • To be able to purchase and enjoy (multi-)contents applications.	
For the Content Provider • To be able to sell high value applications with reduced lead time because of using external APIs, eliminating the need for internal development.	
For the API wholesaler • To be able to provide APIs with enhancements added or combined with its original function(s).	
For the Resource Service Provider • To be able to expand marketing channels through the API wholesale model.	

5.5.2. Use Cases - API Wholesale Business Scenario R17.5

What are the API Wholesale business scenario use cases?

In this chapter, the following use cases are described:

- [API Authentication Information Transformation and Management - API Wholesale R17.5](#)
- [User Authentication Information Transformation and Management - API Wholesale R17.5](#)
- [API Access Rights Management - API Wholesale R17.5](#)
- [API Usage Log Retrieval - API Wholesale R17.5](#)

API Authentication Information Transformation and Management - API Wholesale R17.5

How is authentication information transformed and managed in the marketplace?

If both of API Providers (API Wholesalers and Resource Service Providers) have API authentication functions, how do API Providers authenticate API users (Content Providers and API Wholesalers), respectively?

The API Wholesaler manages the API authentication information provided by the Resource Service Provider and itself.

Use Case Id		 Example sequence flows
Use Case Name	API Authentication Information Transformation and Management	
Summary	<u>The API Wholesaler manages the API authentication information provided by the Resource Service Provider and itself.</u>	
Actor(s)	Content Provider API Wholesaler Resource Service Provider	
Lifecycle Phases		
Begins When	The Content Provider sends an authentication request to the API Wholesaler.	

Description	See Example sequence flows .	
Ends When	The Contents Provider receives an API response from the API Wholesaler.	

User Authentication Information Transformation and Management - API Wholesale R17.5

How is user authentication information transformed and managed in the marketplace?

If there is the requirement that a backend service provider should be hidden from end users and Contents Providers in the API Wholesale model, how do Service Providers authenticate end-users?

The API Wholesaler manages the User authentication information provided by the Resource Service Provider and itself.

Use Case Id		 Example sequence flows
Use Case Name	User Authentication Information Transformation and Management	
Summary	<u>The API Wholesaler manages the User authentication information provided by the Resource Service Provider and itself.</u>	
Actor(s)	Customer Content Provider API Wholesaler Resource Service Provider	
Lifecycle Phases		
Begins When	The customer accesses the application provided by the Content Provider.	
Description	See Example sequence flows .	
Ends When	The customer receives content on the application from the Content Provider.	

API Access Rights Management - API Wholesale R17.5

How does the API wholesaler give API access rights to the content provider?

The content provider may access the API resources of the resource service provider to develop their application. The API wholesaler wants to manage API access rights of the content provider to those limited resources. The API wholesaler sets roles of the content provider on the resource service provider's platform. **The API wholesaler**

defines and grants access to the roles of the content provider, on the resource service provider's platform.

Use Case Id		 Example sequence flows
Use Case Name	API Access Rights Management	
Summary	<u>The API wholesaler defines and grants access to the roles of the content provider, on the resource service provider's platform.</u>	
Actor(s)	Content Provider API Wholesaler Resource Service Provider	
Lifecycle Phases		
Begins When	The API wholesaler allows the content provider to access API resources of the resource service provider.	
Description	See Example sequence flows .	
Ends When	The content provider has been enabled to use API provided by the resource service provider.	

API Usage Log Retrieval - API Wholesale R17.5

How does the API wholesaler retrieve the usage of API resources accessed by the content providers?

The API wholesaler wants to analyze the usage of API resources accessed by the content providers for marketing or to know irregular usage and trace security attacks in later reaction (auditing). The API wholesaler retrieves API log information accessed by the content providers from the resource service provider's platform.

The API wholesaler retrieves API log information accessed by the content providers from the resource service provider's platform.

Use Case Id		 Example sequence flows
Use Case Name	API Usage Log Retrieval	

Summary	The API wholesaler retrieves API log information accessed by the content providers from the resource service provider's platform.	
Actor(s)	Content Provider API Wholesaler Resource Service Provider	
Lifecycle Phases		
Begins When	The content provider uses an API provided by the resource service provider.	
Description	See Example sequence flows .	
Ends When	The resource service provider provides API usage log information, which the content provider accessed, to the API wholesaler.	

5.5.3. DSRA Platform Capabilities overview - API Wholesale Business Scenario R17.5

Which DSRA platform capabilities does the API wholesale business scenario use?

Platform Capability Name	In scope (Y/N)	Comment
Identity Mgmt	Y	To manage Partner and customer identity To authenticate API users
Intelligence & Analytics	N	To analyze API log information
Policy based routing	N	Out of scope
Fulfillment	N	To define and grant access to user roles
Assurance	N	Out of scope
Charging	N	Out of scope (flow focuses before charging)
Catalog Lifecycle Mgmt & Federation	Y	To onboard partners and their offerings
Monitoring & Management	Y	To collect API usage logs

- [DSRA Platform Capability use - API Wholesale Business Scenario R17.5](#)

DSRA Platform Capability use - API Wholesale Business Scenario R17.5

How does the API wholesale business scenario use cases make use of the DSRA platform capabilities?

Scenario	IDENTITY MANAGEMENT	PROFILE MANAGEMENT	PRIVACY MANAGEMENT	ANALYTICS	POLICY BASED ROUTING	CONFIGURATION & ACTIVATION	ASSURANCE & TRACEABILITY	CHARGING	INVOICING	CATALOG/LIFECYCLE MGMT & FEDERATION	ON-BOARDING
API Authentication Information Transformation and Management	✓										✓
User Authentication Information Transformation and Management	✓										✓
API Access Rights Management						✓					✓
API Usage Log Retrieval				✓							

5.5.4. APIs - API Wholesale Business Scenario R17.5

What are the relevant APIs for the API Wholesale business scenario?

Use case	Actors	API	Touchpoints
API Authentication Information Transformation and Management	Content Provider API Wholesaler Resource Service Provider	On-boarding Identity	
API Authentication Information Transformation and Management	Customer Content Provider	On-boarding Identity	

Use case	Actors	API	Touchpoints
	API Wholesaler Resource Service Provider		
API Access Rights Management	Content Provider API Wholesaler Resource Service Provider	User Role and Permission	
API Usage Log Retrieval	Content Provider API Wholesaler Resource Service Provider	Usage Mgmt	

5.5.5. Example sequence flows - API Wholesale Business Scenario R17.5

What are example sequence flows for the API wholesale business scenario?

- [API Authentication Information Transformation and Management - flow - API Wholesale R17.5](#)
- [User Authentication Information Transformation and Management - flow - API Wholesale R17.5](#)
- [API Access Rights Management - flow - API Wholesale R17.5](#)
- [API Usage Log Retrieval - flow - API Wholesale R17.5](#)

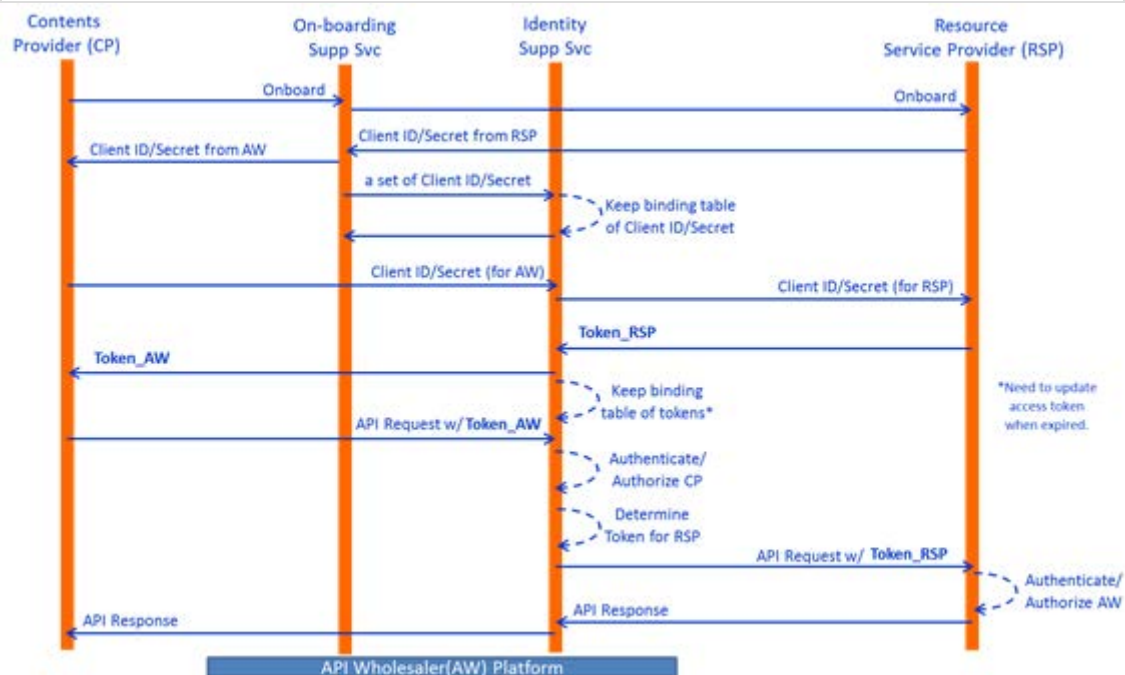
API Authentication Information Transformation and Management - flow - API Wholesale R17.5

If both API Providers - API Wholesalers and Resource Service Providers - have API authentication functions, how do API Providers authenticate API users - Content Providers and API Wholesalers - respectively?

To limit access to an API, the Resource Service Providers and the API Wholesaler would provide an access token to API Wholesaler and the Content service developer, respectively.

To limit access to an API, the Resource Service Providers and the API Wholesaler would provide an access token to API Wholesaler and the Content service developer, respectively.

The API Wholesaler needs to manage authentication information by binding the access tokens provided by the Resource Service Provider and itself.



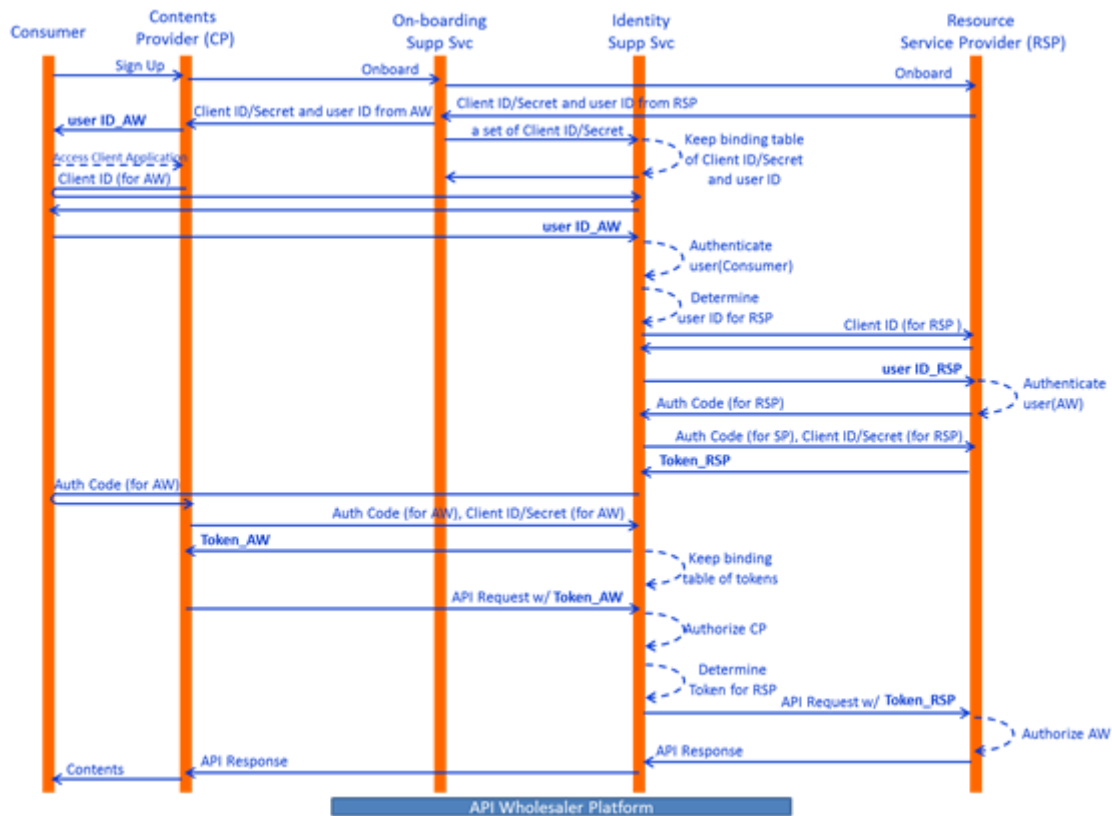
User Authentication Information Transformation and Management - flow - API Wholesale R17.5

If there is the requirement that a "backend" Service Provider should be hidden from end users and Content Providers in the API Wholesale model, how do Service Providers authenticate end-users?

When the Resource Service Provider provides private resources of end users (e.g. phone call, chat), the Resource Service Provider and the API Wholesaler would provide the accounts to the API Wholesaler and the Content service developer, respectively.

When the Resource Service Provider provides private resources of end users (e.g. phone call, chat), the Resource Service Provider and the API Wholesaler would provide the accounts to the API Wholesaler and the Content service developer, respectively.

The API Wholesaler needs to manage the accounts by binding the user ID provided by the Resource Service Provider and itself.



API Access Rights Management - flow - API Wholesaler R17.5

The Content Provider may access the API resources of the Resource Service Provider to develop their application. The API Wholesaler wants to manage API access rights of the Content Provider to those limited resources.



Under a previously established contract between the API Wholesaler and the Content Provider, the Content Provider applies for on-boarding the specific API, which the Content Provider chose from catalog, to the API Wholesaler.

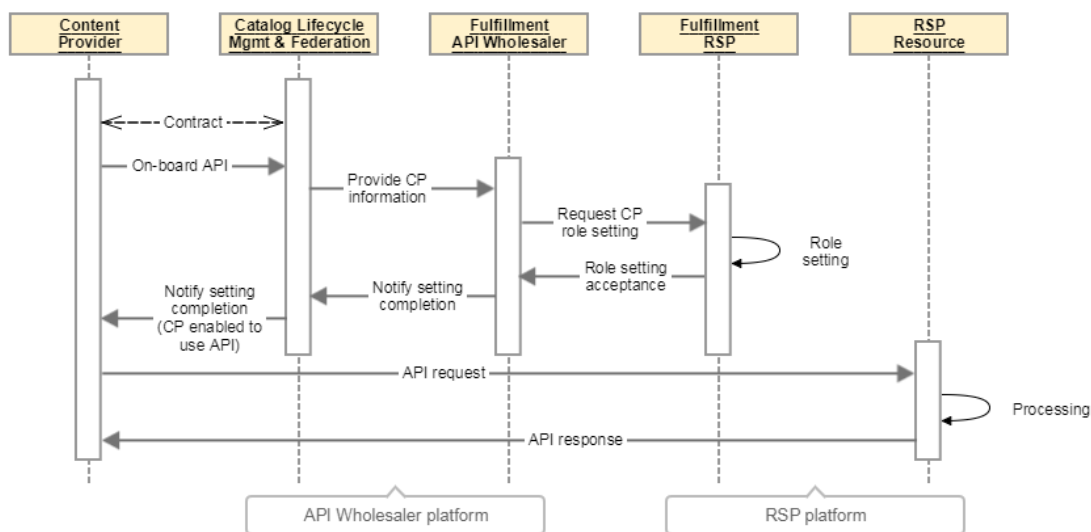


Specifically, the Content Providers send application information such as the name of API desired for use and the period of use.

After this on-boarding transaction, the API Wholesaler sets roles of the Content Provider on the Resource Service Provider's platform.

After role setting, the Content Provider can access API resources of the Resource Service Provider.

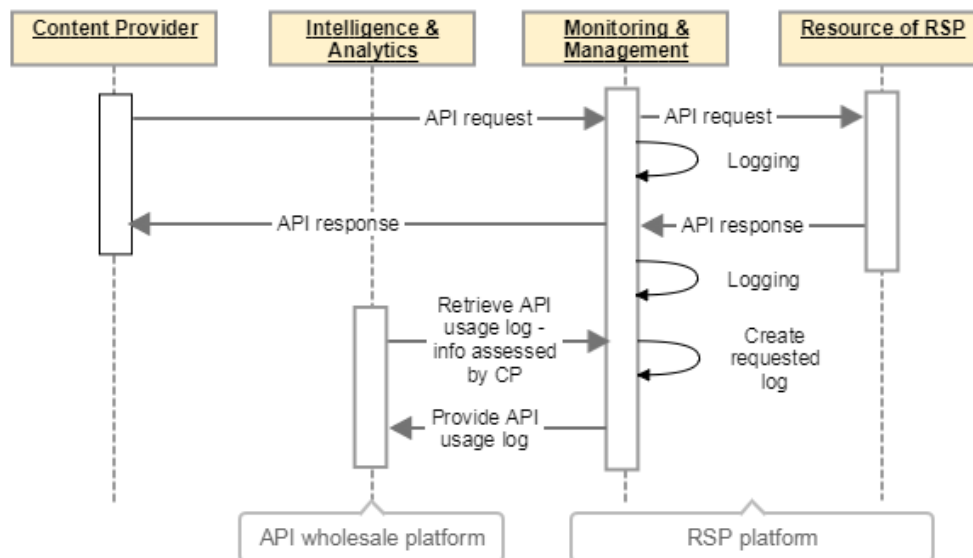
Sequence Diagram

**API Usage Log Retrieval - flow - API Wholesale R17.5**

The API wholesaler retrieves API log information accessed by the content providers from the resource service provider's platform.



API usage log retrieval Sequence Diagram



6. Definition of Terms R17.5

What are the key terms used in the DSRA Guide?

Term	Definition
Component Service	An individual service (product) that is registered on the DSRA platform with the intention that developers will use it as a part of a larger E2E product
Composite Service	An assembly of one or (usually) more component services into an E2E service. The service providers and the E2E product developer will often be (but need not be) different. See also product distinction below. Note that service composition may be recursive so a composite service may become a component of yet another service.
Developer	The creator of an E2E composite product
Digital Ecosystem Platform (DEP)	A platform adhering to the DSRA, designed to enable / sustain one or more digital ecosystems.
DPRA	Digital Platform Reference Architecture
Digital Service (DS)	- Any process, activity or value that is accessible via network based interfaces (public or private). - A DS may or may not generate revenue, or may generate revenue indirectly - A DS may or may not have an end user interface - Infrastructure abstraction - Portability - Agnostic of technology (access, transmission, ...), location, delivery model, hosting technology, ...
Digital Service Ecosystem (DSE)	A Digital Service Ecosystem (Wikipedia): A digital ecosystem is a distributed, adaptive, open socio-technical system with properties of self-organization, scalability and sustainability inspired from natural ecosystems.
DSM API	Digital Service Management API (TMF665) enabling the management of digital service lifecycle
DSRA	Digital Services Reference Architecture.

Term	Definition
	The DSRA maps to the Actualization Platform view of the DPRA.
DSRA Platform Capability	A Digital Ecosystem Platform capability that aids in the establishment and ongoing operation of a digital ecosystem. Example of a DSRA Platform Capability, is an API Management. This is a critical definition. The entire concept of the DSRA and the rationale for TM Forum's participation in this domain, since there is support (e.g.: Management) capabilities that are required to facilitate future business models and mission critical services when assembled from across multiple service providers. The key focus of DSRA is on management services that allows for FCAPS functions to take place in this complex, distributed ecosystem – charging for the video, end-to-end assurance of the multi-party product, etc. The focus is not on the functional service that is part of the product delivered to an end user: e.g.: it is not the movie content, nor the transport service nor the program guide that a consumer interacts with. This relationship of functional-to-management services was first clearly articulated in the Software Enabled Services project which defined each building block as having a functional component and one or more management components. For the DSRA platform, we define the set of shared management services as “platform capabilities”. The word "management" has many facets of interpretation whether from a process perspective (configuration &) or from a functional perspective (monitoring), and this in addition to organizational, every day activity, governance or audit interpretations. It was advised to the DSRA team, to follow the best practice from the Frameworx community in avoiding the use of the word "management", but rather find collective words that describe the set of associated functions and processes. This will be considered for the next iteration of the DSRA Guide. Since there are some existing references to the term "management" in the current version of the DSRA Guide, until the next iteration - when a more descriptive collective name will be introduced - one needs to understand the meaning of "management" in this DSRA Guide. Let's look at an example. The DSRA Support Services allow for example monitoring the health of Composite Products composed of Service Components from multiple ecosystem partners, on multiple technology domains and verticals, end-to-end. The DSRA Support Services consume the Management API operations

Term	Definition
	implemented by a digital service to retrieve health reports, notifications, or to configure the digital service with user data. In previous versions of the DSRA Guide, platform capabilities were referred to as Support Services.
Functional API	A Functional API reflects the service provided and used alone or bundled to deliver directly the service or to build an associated application. It exposes assets and encompasses a wide range of service / resources. See also Types of APIs managed in the DSRA .
Management API	A Management API allows to perform all management operations before, during and after the use of a service. Through the management API, a digital service exposes capability to allow DSRA Support Services to: for example, get health reports, subscribe to notifications or configure the digital service with user data. Please note that the distinction between a digital services Functional API and Management API is only a semantic difference. See also Types of APIs managed in the DSRA .
Online Charging System (OCS)	3GPP defines an online charging system as performing: 1. Usage data collection (by records or real-time protocol) 2. Rating and various levels of discounting 3. Account balance management, and 4. Transfer of required data to the next stage of billing and / or accounting systems
Product	Any component or composite service that is sold and has commercial terms associated with it – typically a price and possibly an SLA, as defined by TM Forum Catalog Management.
Service	Refers to either component service or composite service
Service Provider	The owner of a component service.
Support Service	A Support Service in the context of the DSRA is the legacy term for what is now referred to as DSRA Platform Capability or DPRA Technical Platform Capability. It is a <u>function</u> (or a consistent set of functions) provided by a DSRA compliant platform to actors involved in an ecosystem. A platform capability is not an API, but may use an API to exchange data

Term	Definition
	with the « rest of the world » (i.e. other systems or ecosystems).

7. Reading Guide R17.5

How to read and use this DSRA Guide?

The Digital Services Reference Architecture is providing best practices, methods and tools for the enablement of managing open, interoperable virtualized digital services focusing on maximizing re-use, agility of operations and scalability to meet future demands.

As more services are being developed using an agile composition of existing capabilities as well as products in the industry, a consistent framework for enabling end to end management of these services across the eco-system is required with attention paid to the integration of these capabilities to reduce customization tax while ensuring delivery of excellent end customer experience.

As more services are being developed using an agile composition of existing capabilities as well as products in the industry, a consistent framework for enabling end to end management of these services across the eco-system is required

- [DSRA Guide Structure R17.5](#)
- [DSRA Guide Usage R17.5](#)
- [DSRA Guide Pictograms R17.5](#)

7.1. DSRA Guide Structure R17.5

What are we trying to achieve?

In the document there are some General sections, the DSRA Guide is structured in 3 parts.


[Introduction to DSRA](#)

Basics illustrates the fundamental concepts behind the DSRA, including underlying principles, standards and digital services management enablement vision.


[DSRA Business Alignment Architecture](#)

Practices describes best practices and current initiatives that can be leveraged to move towards the vision. Besides


 Practices

a general section, there are also dedicated sections for many Vertical Industries. (Work in Progress)	
Stories provide User Stories from which functional requirements can be deduced. (Work in Progress)	 Stories
Finally, there is another General part that contains all the rest.	 DSRA Guide Structure

The flow of the document contained in the above-mentioned parts is as follows. After this introduction, the key architectural principles that need to be followed to be successful, are being described, as well as the Software Enabled Services design pattern.

This is followed by a set of user stories that reveal functional requirements, an overview of what can be leveraged from existing investment, systems and services to cope with the digital disruption in this new world.

This is followed by an overview of what happens in many industry verticals.

Finally, an overview is given of relevant artifacts and activities in other TM Forum teams and/or SDOs

7.2. DSRA Guide Usage

R17.5

How should you be using this DSRA Guide?

The structure, layout and Reading Assists used in this DSRA Guide are inspired by thisisservicedesignthinking	 thisisservicedesignthinking
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Colors

As mentioned before, this DSRA Guide consists of 3 parts identified by Colors. **Blue** refers to **Basics**, **Green** to **Practices** that refer to the Best Practices, Methods, Tools and Standards, **Red** to **Cases**. Sections that relate to each of these parts, are decorated with the corresponding colors. Additional sections are decorated in **Grey**.

Reference Pane

The layout of the sections in this DSRA Guide have a Body Pane and a Reference Pane. The Reference Pane can hold quotes, pointers to other texts of this DSRA Guide, external pointers to reference artifacts, Pictograms and Navigation Assists.

What?

Nearly all sections in this guide are entitled with a question in the upper left corner. These repeat the topic you are going to get an answer to in the current section of the DSRA Guide. These answers are giving in breakout sentences in the body of the text. In the body text, those sentences are underlined so that you can quickly identify the answers.

Navigation Aids

The [Reference Pane](#) of a section that has underlying child sections displays a Navigation Tree pointing to the underlying sections.

To enable easy navigation through the DSRA Guide, a pointer to the start of the current part of the DSRA Guide is in the upper right corner (Reference Pane).

Pointers Internal pointers are proceeded by an internal pointer pictogram in the associated color of the DSRA Guide part it refers to (Blue , Green , Red or Grey)	 internal link
External pointers are preceded by a dedicated pictogram.	 external link

Overview Page

An overview page with all the icons used in this DSRA Guide. As you will start to remember the meaning of the icons you will be able to relate to the associated concepts without having to read the details.

Pictograms

A pictogram is associated with each of the best practices and standards that are described in this DSRA guide. When reference is made to a best practice, method, tool or standard in other texts, the associated icon appears in the right pane. The overview page in this guide provides with all icons used in this guide and their meaning.

Notes

Notes are highlighted with a yellow background

The **Practices** and **Cases** Sections are still to be developed and will be part of future publications.

7.3. DSRA Guide Pictograms R17.5

How should you be using the DSRA Guide?

			
general	basics	practices	cases
			
hyperlink	scope	objective	business model
			
business	customer	quality	revenue
			
scenario	use case	user story	role
			
partnering	interaction	control	lifecycle
			
architecture	process	data	API
			
management	orchestration	analytics	digital
			
internet	application		

8. Administrative Appendix DSRA R17.5

This Appendix provides additional background material about the TM Forum and this document. In general, sections may be included or omitted as desired; however, a Document History must always be included.

8.1. Document History

8.1.1. Version History

Created by TM Forum Publishing Staff

Version Number	Date Modified	Modified by:	Description of changes
v.6.0.0	4/JUN/2016	Johan Vandenberghe	Updates for Fx16
v6.0.1	16/JUN/2016	Alicja Kawecki	Minor cosmetic edits prior to publication for Fx16
v6.0.2	7/JUL/2016	Johan Vandenberghe	- Corrected numerous hyperlinks (some were referring to pages or documents that were no longer available, or were pointing to the work spaces of the publication spaces on confluence) - Correctly ordered child pages (in transferring from the work space to the publication space, for some pages, the order of the child pages was not respected) - Corrected a small number of typos.
v6.0.3	29/SEP/2016	Alicja Kawecki	Updated cover page and Notice to reflect TM Forum Approved status
v7.0.0	19/JUN/2017	Johan Vandenberghe	Updates for Fx17
v7.0.1	27/JUL/2017	Alicja Kawecki	Minor formatting/style edits prior to publication for Fx17

Version Number	Date Modified	Modified by:	Description of changes
v7.0.2	28/NOV/2017	Adrienne Walcott	Updated to reflect TM Forum Approved Status
v8.0.0	28 Dec 2017	Johan Vandenberghe	Updates for Release 17.5; Reformatting for Word; new diagrams;
8.0.1	12 Jan 2018	Adrienne Walcott	Formatting/style edits prior to publishing
8.0.2	25-May-2018	Adrienne Walcott	Updated to reflect TM Forum Approved Status

8.1.2. Release History

Release Number	Date Modified	Modified by:	Description of changes
17.0.0	June 2017	Nancy Lyness	Initial Release of this Introductory Guide
17.0.1	28 November 2017	Adrienne Walcott	Updated to reflect TM Forum Approved Status
17.5.0	December 2017	Nancy Lyness	Release 17.5 Updates;
17.5.1	25-May-2018	Adrienne Walcott	Updated to reflect TM Forum Approved Status

8.2. Company Contact Details

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8.3. Acknowledgments

This document was prepared by the members of the TM Forum Digital Services Reference Architecture (DSRA) team that defines enabling technologies within the Open Digital Project.

The Digital Services Reference Architecture is a genuinely collaborative effort. The TM Forum would like to thank the following people for contributing their time and expertise to the production of this document. It is just not possible to recognize all the organizations and individuals that have contributed or influenced the introduction. We apologize to any person or organization we inadvertently missed in these acknowledgments.

Key individuals that reviewed, provided input, managed, and determined how to utilize inputs coming from all over the world, and really made this document happen were:

Name	Affiliation
John Frame	Sigma Systems
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Eric Troup	Microsoft
Jean-Luc Tymen	Orange
Johan Vandenberghe	Bell Labs Consulting
John Wilmes	TM Forum